# Chapter 10
# Graph connectivity

*Discrete Structures for Computing* on June 16, 2024

Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai
Faculty of Computer Science and Engineering
University of Technology - VNUHCM
trtanh@hcmut.edu.vn



Contents

Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Acknowledgement

Some slides about Euler and Hamilton circuits are created by Chung Ki-hong and Hur Joon-seok from KAIST.



Contents

Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Course outcomes

| | Course learning outcomes |
|---|---|
| L.O.1 | Understanding of logic and discrete structures |
| | L.O.1.1 – Describe definition of propositional and predicate logic |
| | L.O.1.2 – Define basic discrete structures: set, mapping, graphs |
| L.O.2 | Represent and model practical problems with discrete structures |
| | L.O.2.1 – Logically describe some problems arising in Computing |
| | L.O.2.2 – Use proving methods: direct, contrapositive, induction |
| | L.O.2.3 – Explain problem modeling using discrete structures |
| L.O.3 | Understanding of basic probability and random variables |
| | L.O.3.1 – Define basic probability theory |
| | L.O.3.2 – Explain discrete random variables |
| L.O.4 | Compute quantities of discrete structures and probabilities |
| | L.O.4.1 – Operate (compute/ optimize) on discrete structures |
| | L.O.4.2 – Compute probabilities of various events, conditional ones, Bayes theorem |



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

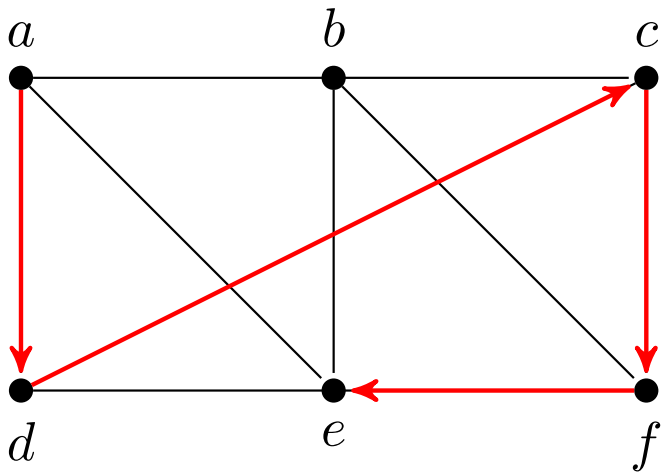
# Contents


1. **Connectivity**
   - Paths and Circuits
2. **Euler and Hamilton Paths**
   - Euler Paths and Circuits
   - Hamilton Paths and Circuits
3. **Shortest Path Problem**
   - Dijkstra's Algorithm
   - Bellman-Ford Algorithm
   - Floyd-Warshall Algorithm
   - Ford's algorithm
   - Others
4. **Graph Coloring**

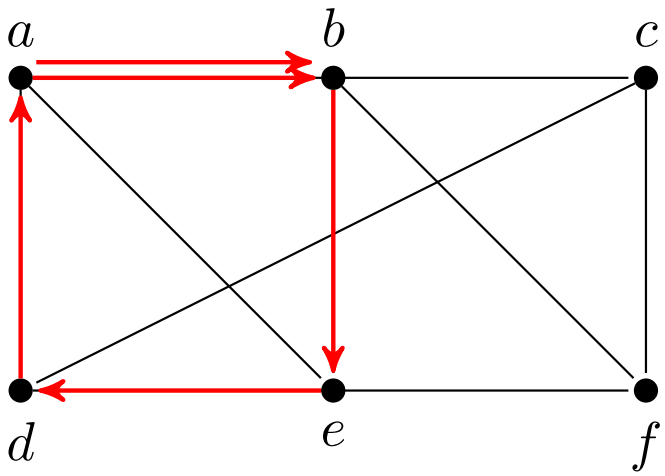
# Path and Circuits

## Definition (in undirected graph)

- Path (*đường đi*) of length $n$ from $u$ to $v$: a sequence of $n$ edges $\{x_0, x_1\}, \{x_1, x_2\}, \ldots, \{x_{n-1}, x_n\}$, where $x_0 = u$ and $x_n = v$.
- A path is a circuit (*chu trình*) if it begins and ends at the same vertex, $u = v$.
- A path or circuit is simple (*đơn*) if it does not contain the same edge more than once.




Simple path




Not simple path




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

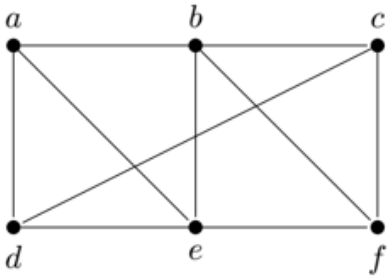
Floyd-Warshall Algorithm

Ford's algorithm

Others

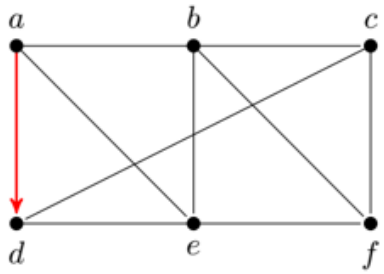
Graph Coloring

# Paths and Circuits

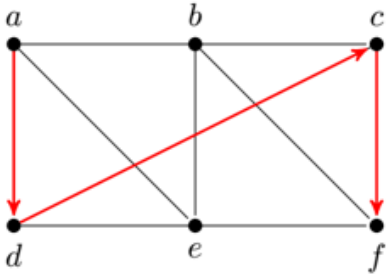


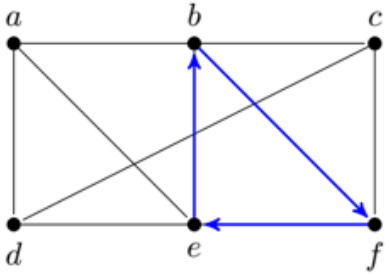

Original Graph

Simple path (length 1)



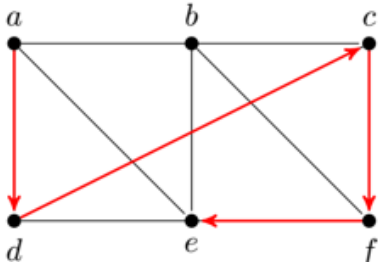

Simple path (length 3)

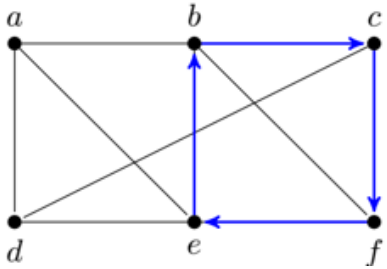



Simple Circuit (length 3)

Simple path (length 4)




Simple Circuit (length 4)



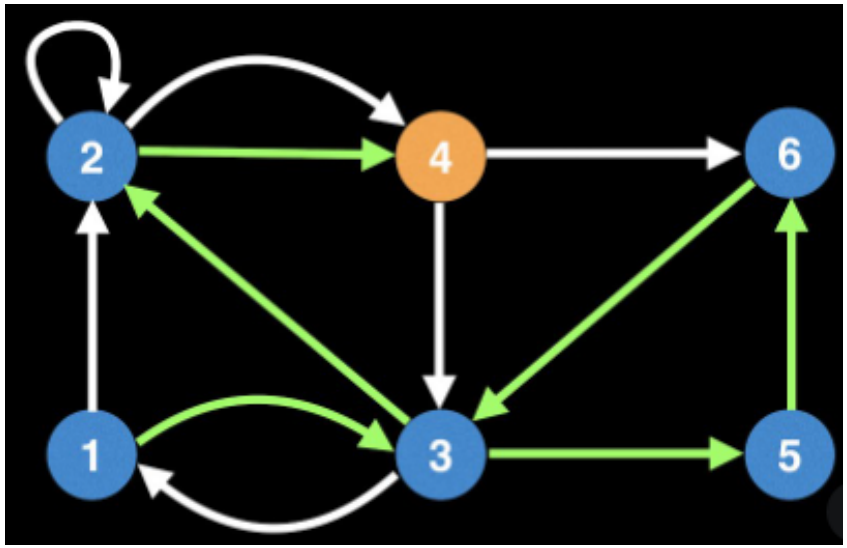
Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Path and Circuits

## Definition (in directed graphs)

Path is a sequence of $(x_0, x_1), (x_1, x_2), \ldots, (x_{n-1}, x_n)$, where $x_0 = u$ and $x_n = v$.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

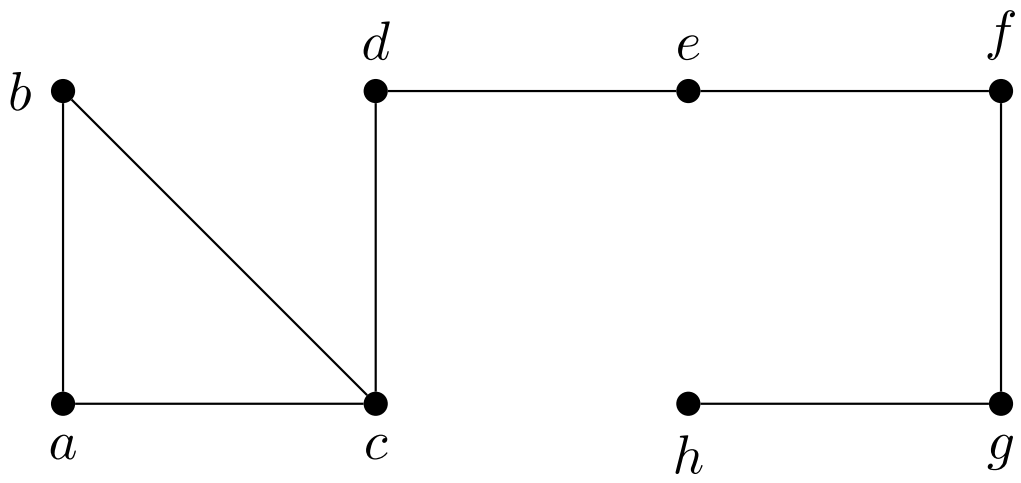
Others

Graph Coloring

# Connectedness in Undirected Graphs

## Definition

- An undirected graph is called connected (*liên thông*) if there is a path between every pair of distinct vertices of the graph.
- There is a simple path between every pair of distinct vertices of a connected undirected graph.




Connected graph




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

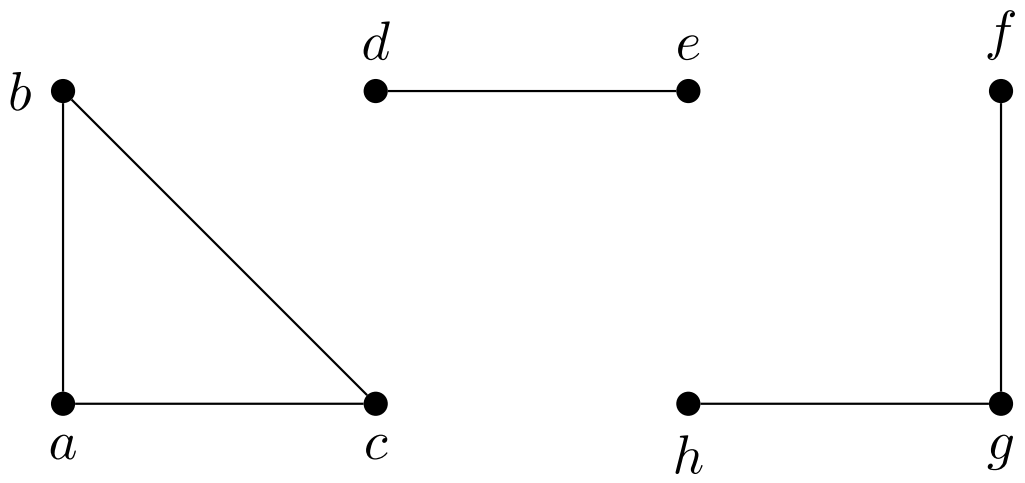
Others

Graph Coloring

# Connectedness in Undirected Graphs

## Definition

- An undirected graph is called connected (*liên thông*) if there is a path between every pair of distinct vertices of the graph.
- There is a simple path between every pair of distinct vertices of a connected undirected graph.




Disconnected graph




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

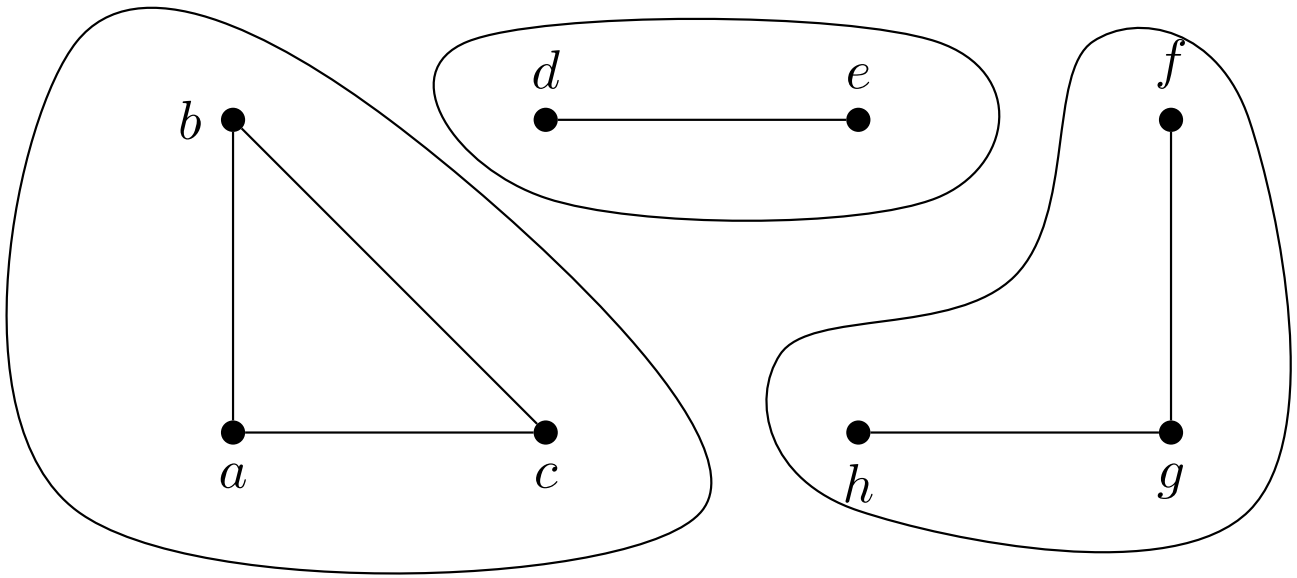
Others

Graph Coloring

# Connectedness in Undirected Graphs

## Definition

- An undirected graph is called connected (*liên thông*) if there is a path between every pair of distinct vertices of the graph.
- There is a simple path between every pair of distinct vertices of a connected undirected graph.




Connected components (*thành phần liên thông*)




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

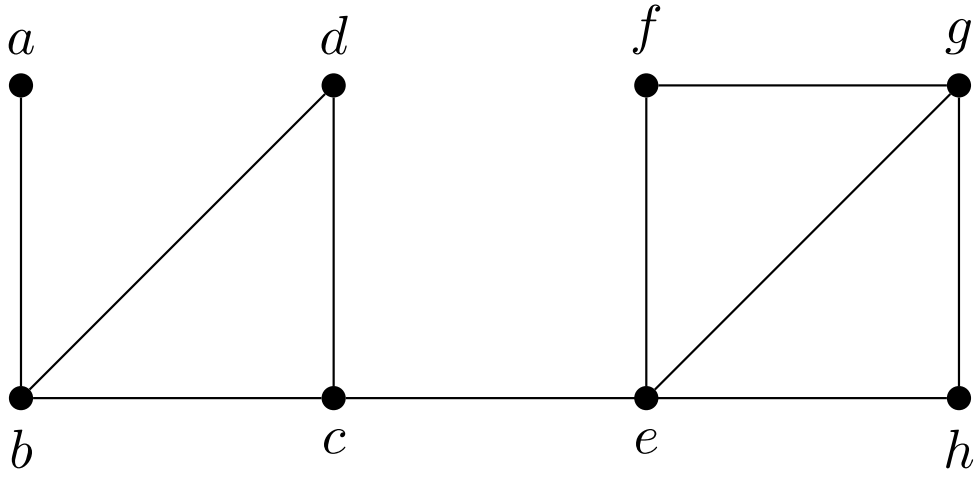
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*).



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

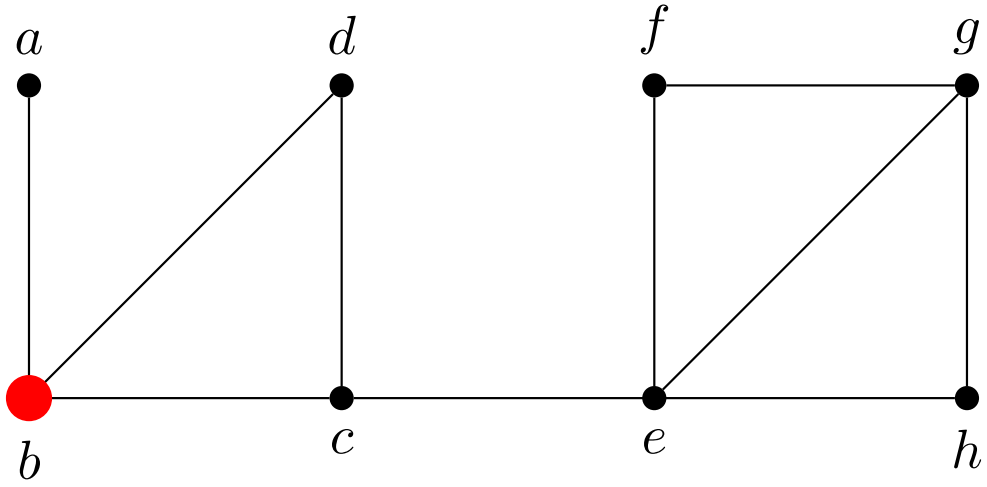
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*).



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

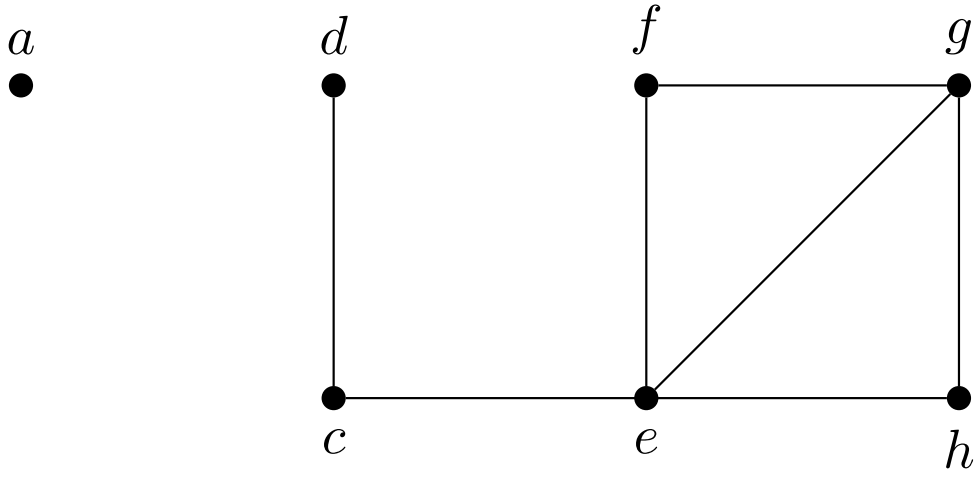
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*).



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

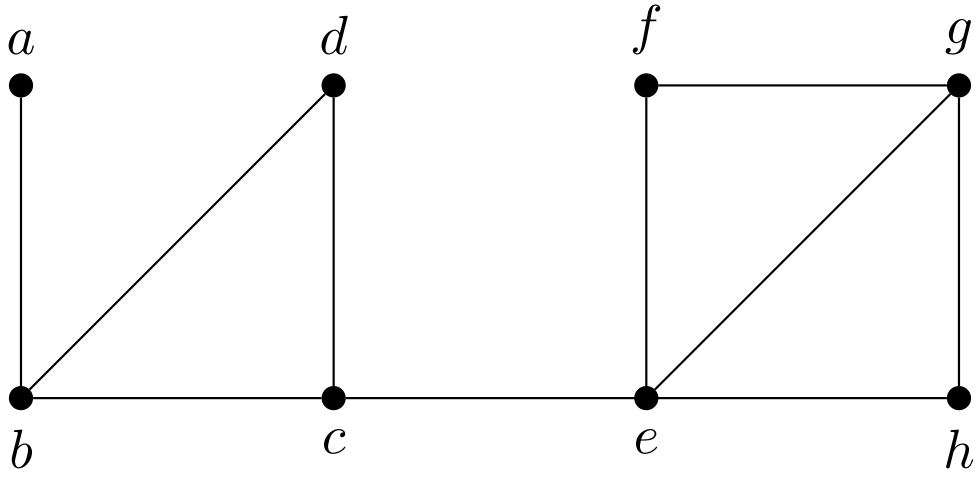
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*). What else?



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

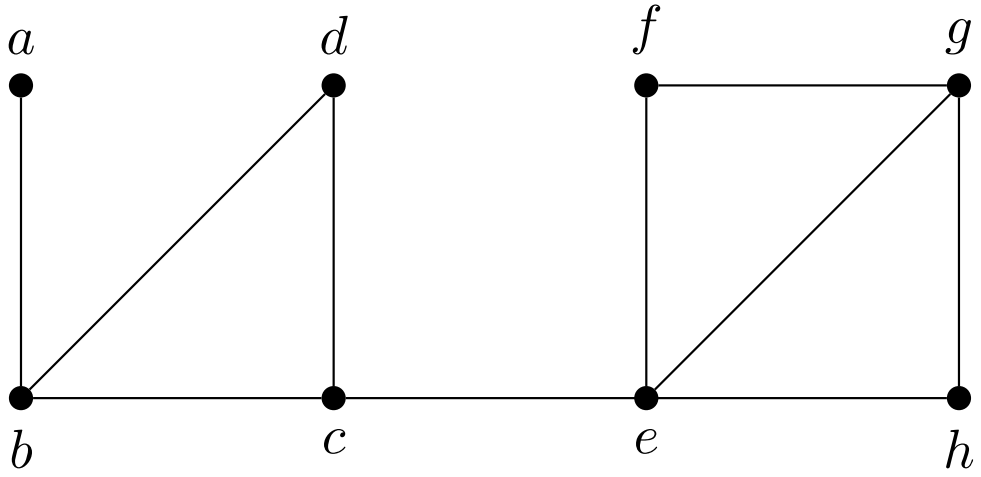
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*). What else?
- $\{a, b\}$ is a cut edge (*cạnh cắt*) or bridge (*cầu*).




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

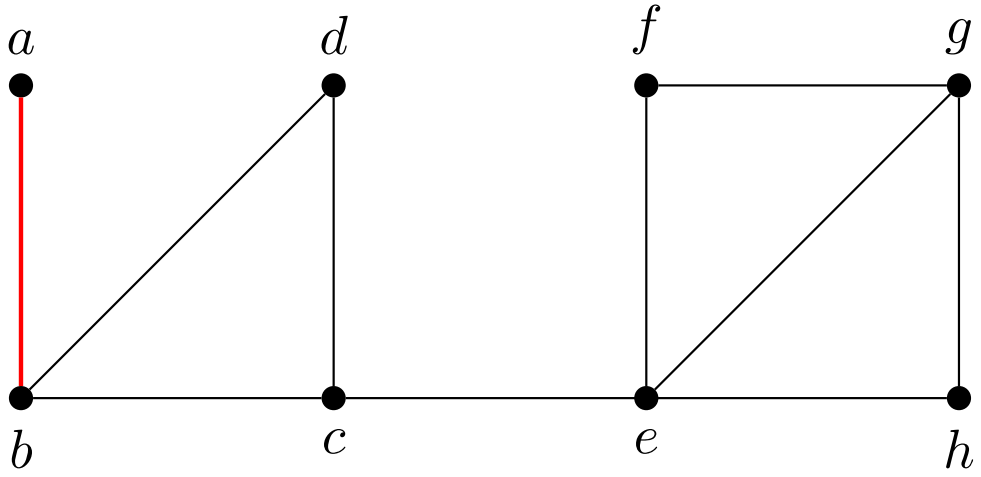
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*). What else?
- $\{a, b\}$ is a cut edge (*cạnh cắt*) or bridge (*cầu*).




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

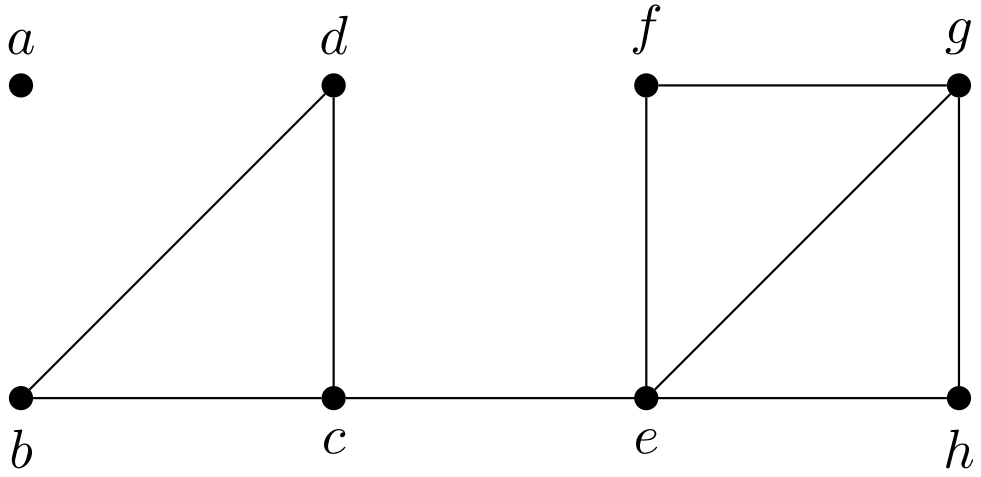
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*). What else?
- $\{a, b\}$ is a cut edge (*cạnh cắt*) or bridge (*cầu*).




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

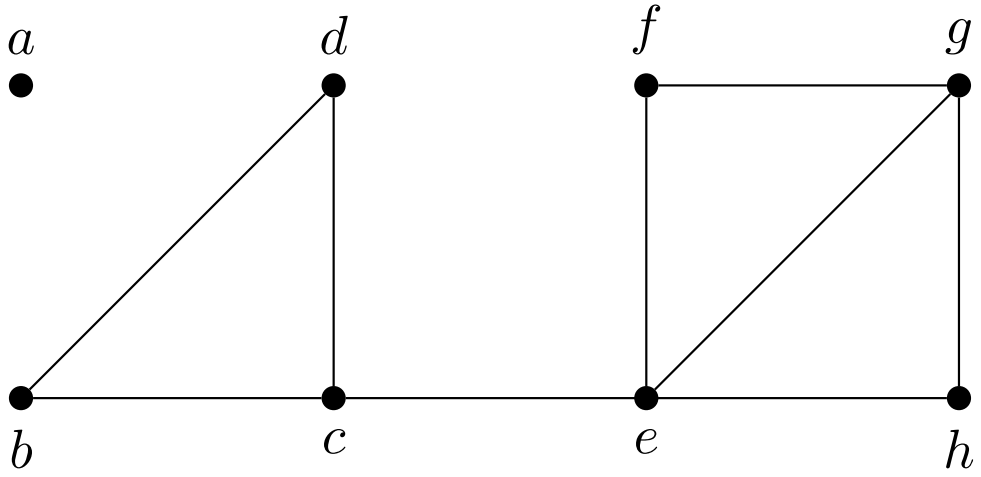
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- $b$ is a cut vertex (*đỉnh cắt*) or articulation point (*điểm khớp*). What else?
- $\{a, b\}$ is a cut edge (*cạnh cắt*) or bridge (*cầu*). What else?




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

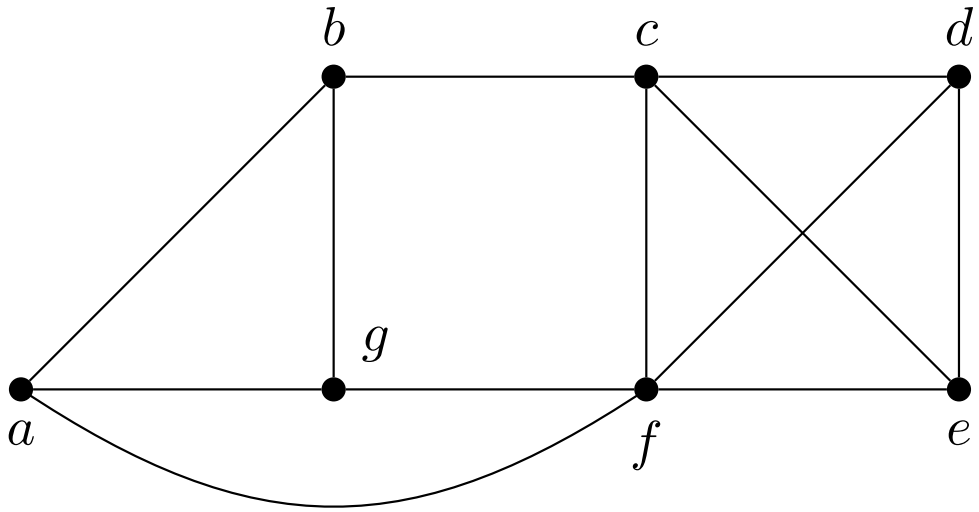
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

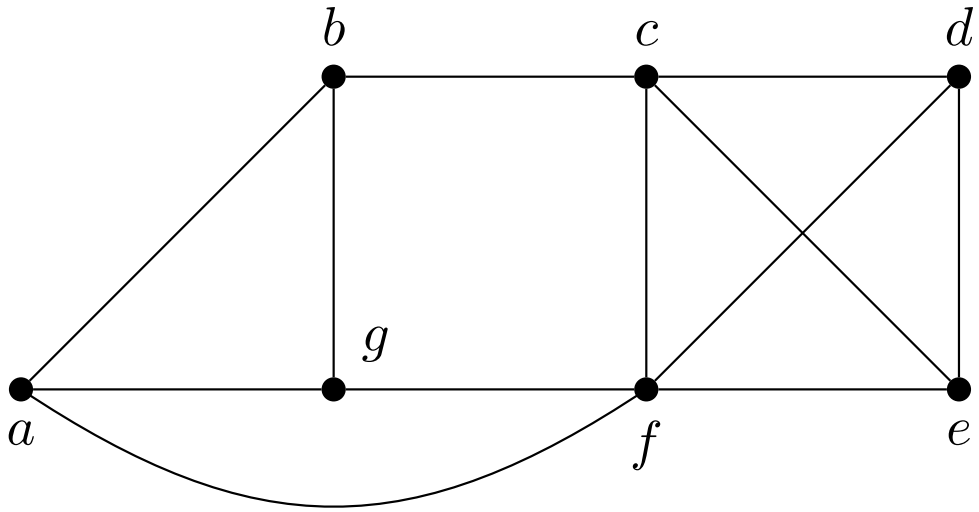
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

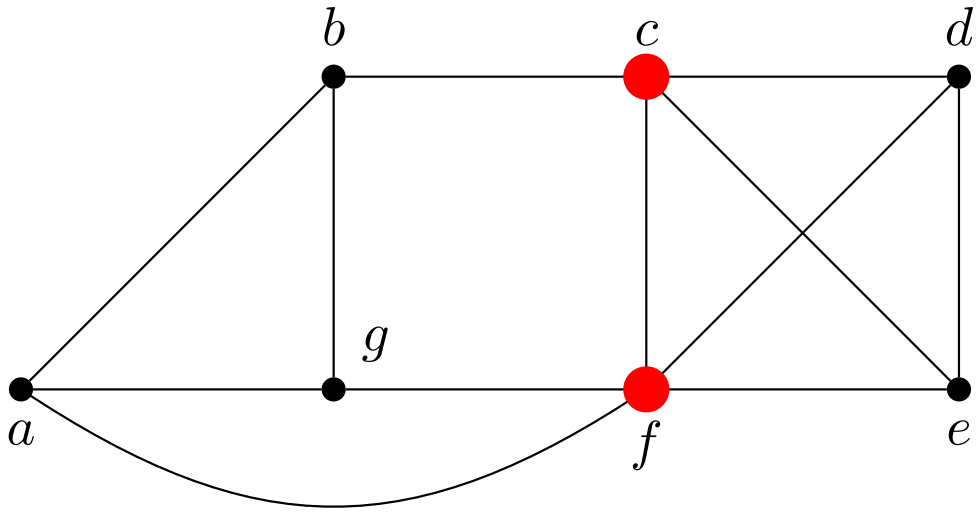
- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

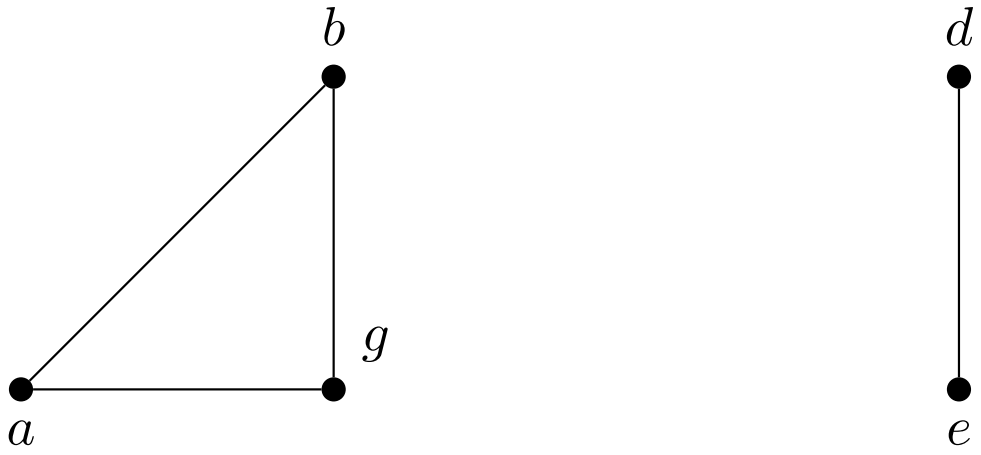
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

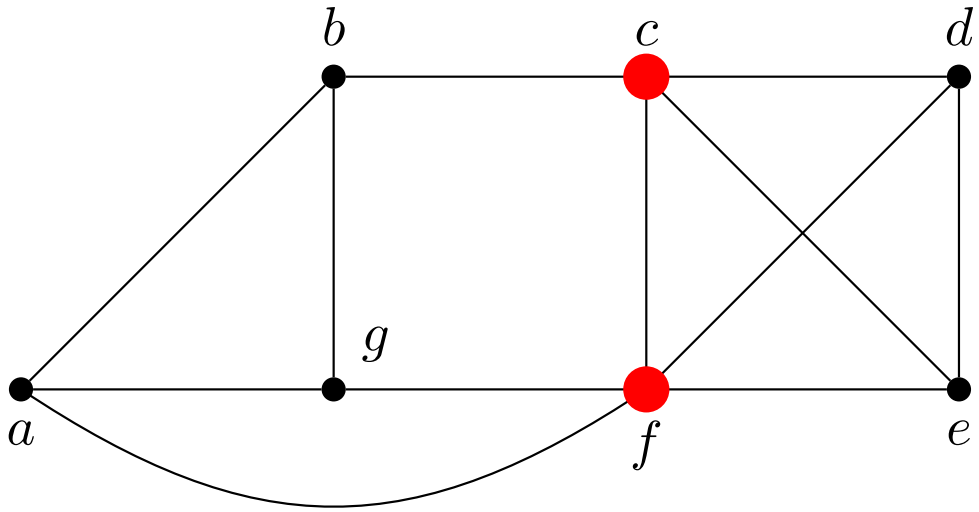
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)
- The vertex cut is $\{c, f\}$, so the minimum number of vertices in a vertex cut, vertex connectivity (*liên thông đỉnh*) $\kappa(G) = 2$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

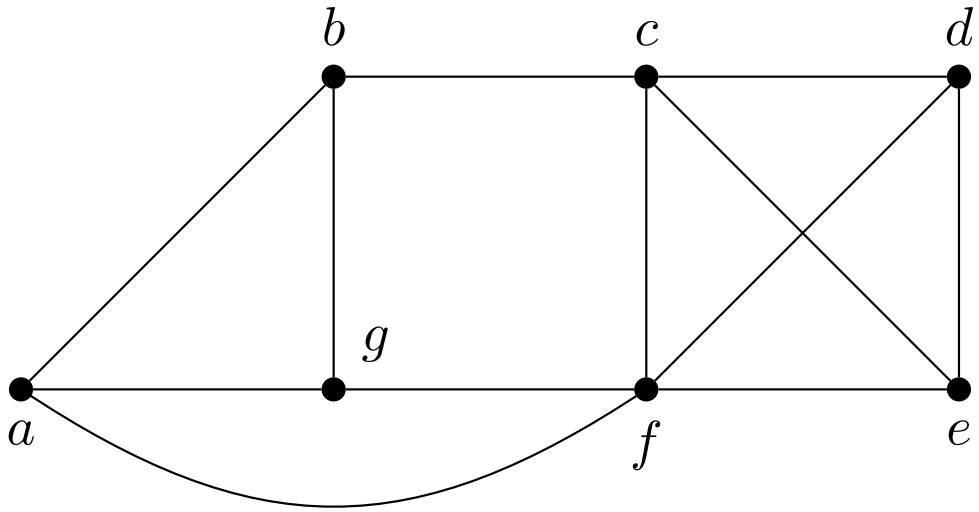
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)
- The vertex cut is $\{c, f\}$, so the minimum number of vertices in a vertex cut, vertex connectivity (*liên thông đỉnh*) $\kappa(G) = 2$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

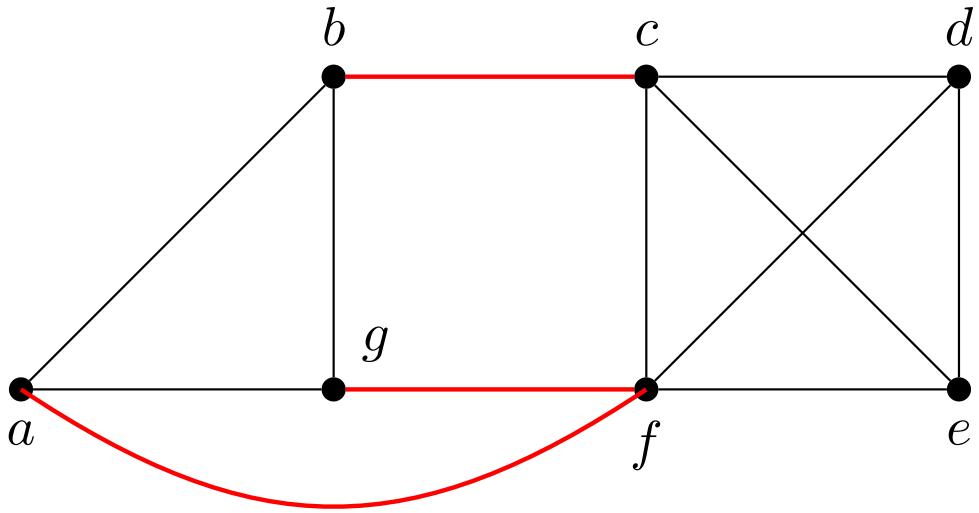
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)
- The vertex cut is $\{c, f\}$, so the minimum number of vertices in a vertex cut, vertex connectivity (*liên thông đỉnh*) $\kappa(G) = 2$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

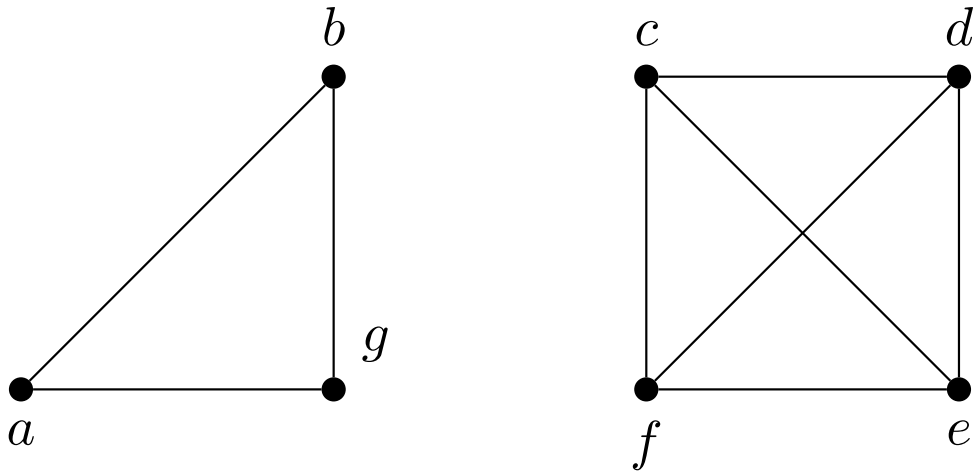
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)
- The vertex cut is $\{c, f\}$, so the minimum number of vertices in a vertex cut, vertex connectivity (*liên thông đỉnh*) $\kappa(G) = 2$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

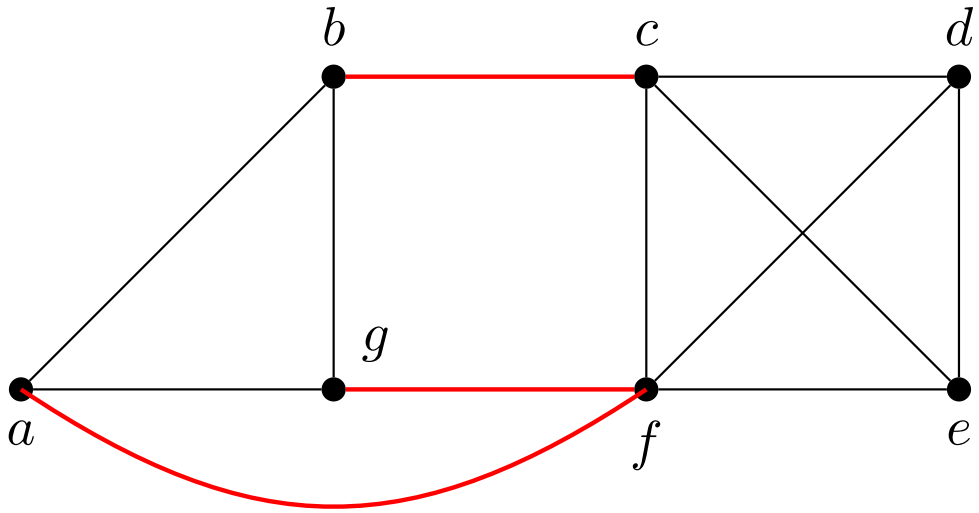
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# How Connected is a Graph?




## Definition

- This graph doesn't have cut vertices: nonseparable graph (*đồ thị không thể phân tách*)
- The vertex cut is $\{c, f\}$, so the minimum number of vertices in a vertex cut, vertex connectivity (*liên thông đỉnh*) $\kappa(G) = 2$.
- The edge cut is $\{\{b, c\}, \{a, f\}, \{f, g\}\}$, the minimum number of edges in an edge cut, edge connectivity (*liên thông cạnh*) $\lambda(G) = 3$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Applications of Vertex and Edge Connectivity

- Reliability of networks
  - Minimum number of routers that disconnect the network
  - Minimum number of fiber optic links that can be down to disconnect the network
- Highway network
  - Minimum number of intersections that can be closed
  - Minimum number of roads that can be closed



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

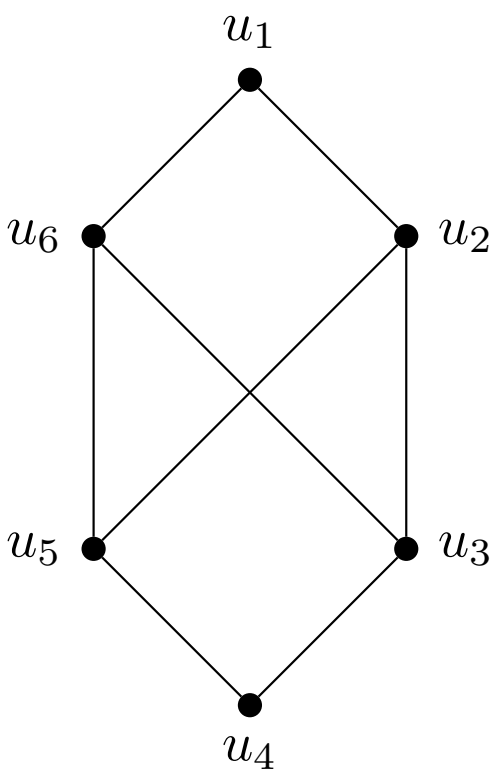
Others

Graph Coloring

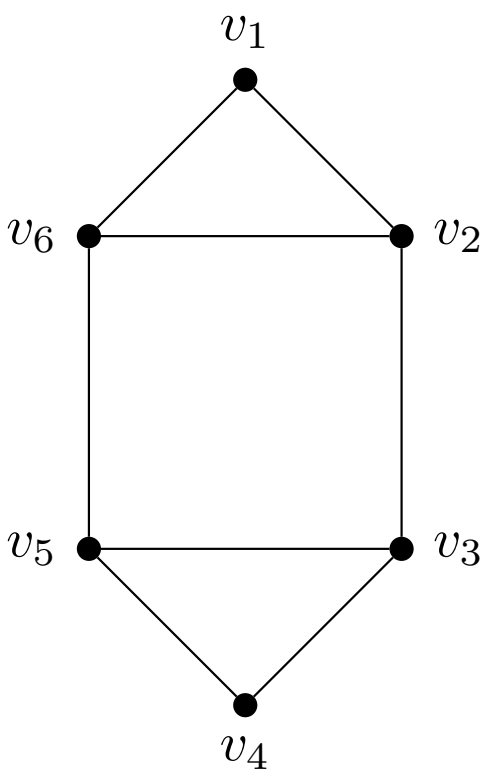
# Applications

## Example

Determine whether the graphs below are isomorphic.




$G$




$H$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

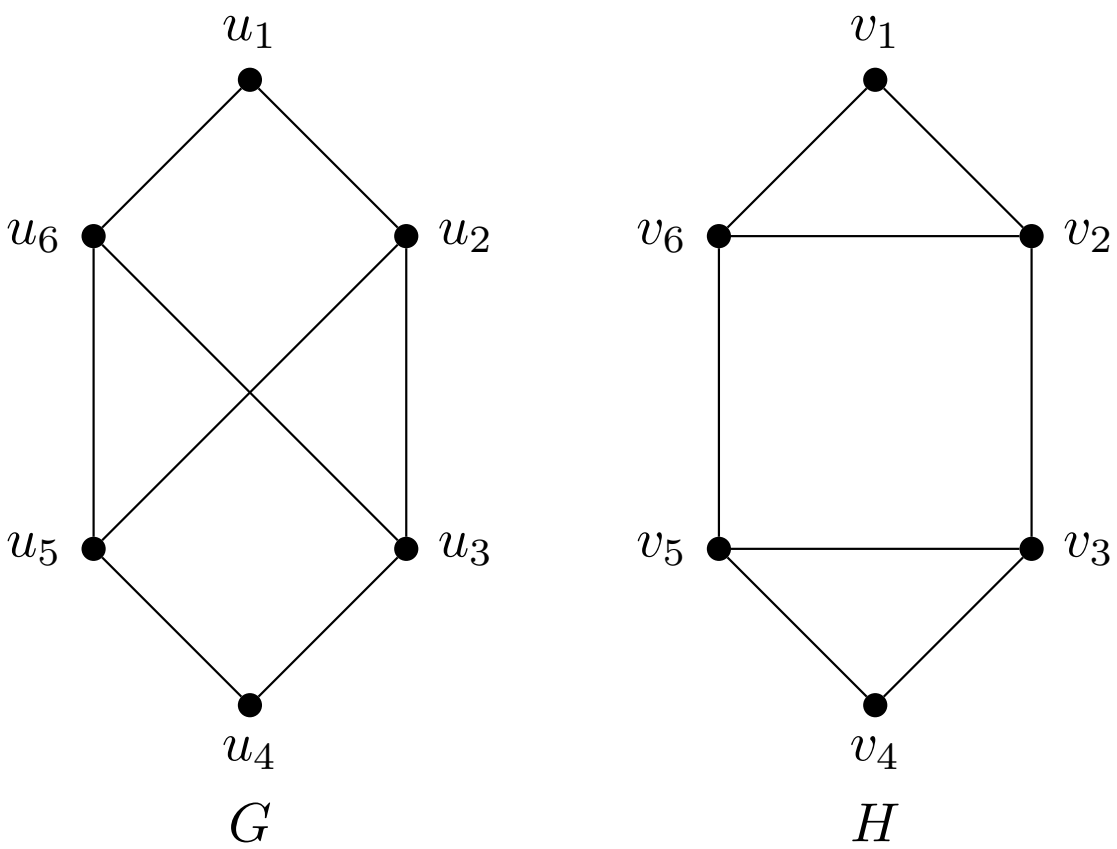
Others

Graph Coloring

# Applications

## Example

Determine whether the graphs below are isomorphic.




## Solution

*$H$ has a simple circuit of length three, not $G$.*




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

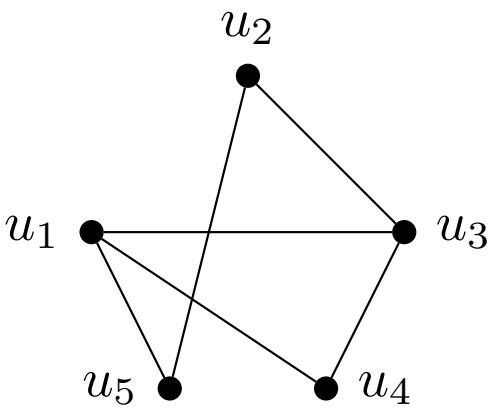
Others

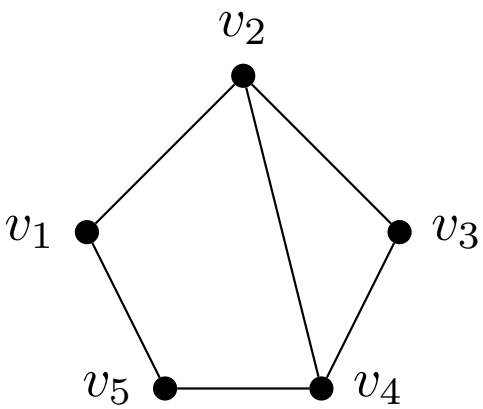
Graph Coloring

# Applications

## Example

Determine whether the graphs below are isomorphic.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

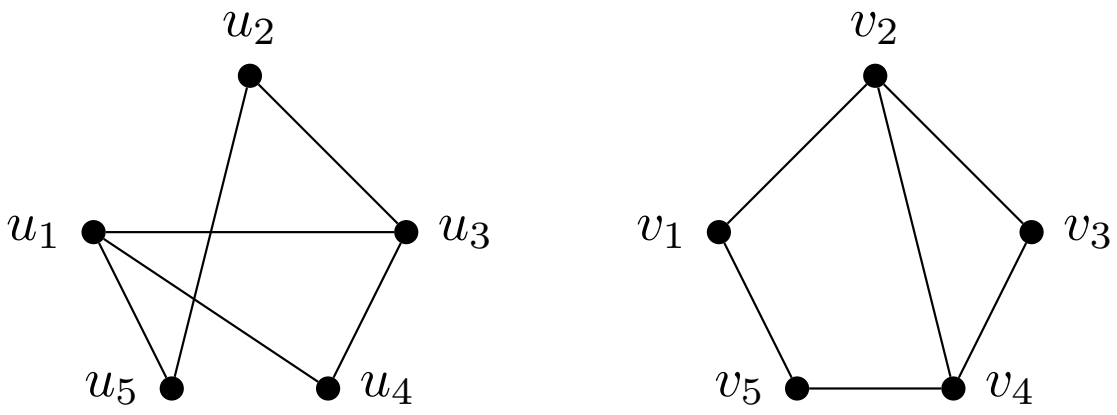
Others

Graph Coloring

# Applications

## Example

Determine whether the graphs below are isomorphic.




## Solution

*Both graphs have the same vertices, edges, degrees, circuits. They may be isomorphic.*
*To find a possible isomorphism, we can follow paths that go through all vertices so that the corresponding vertices in the two graphs have the same degrees.*




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

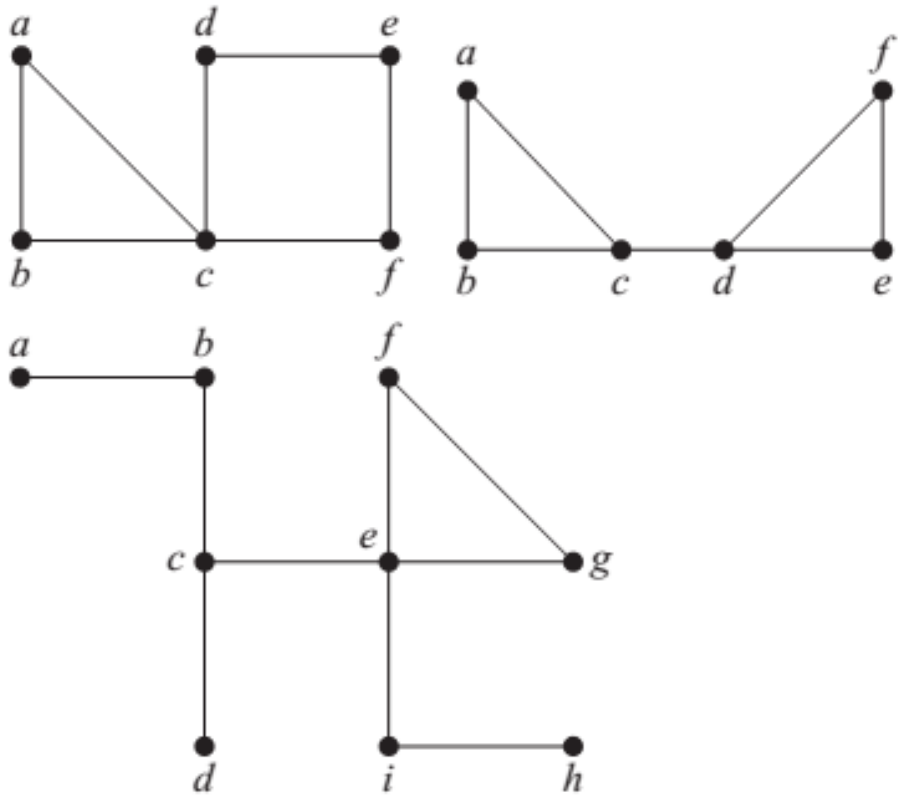
# Exercise

Find all the cut vertices, cut edges of the graphs

a) $C_n$, where $n \geq 3$

b) $W_n$ where $n \geq 3$

c) $K_{m,n}$ where $m \geq 2, n \geq 2$






Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

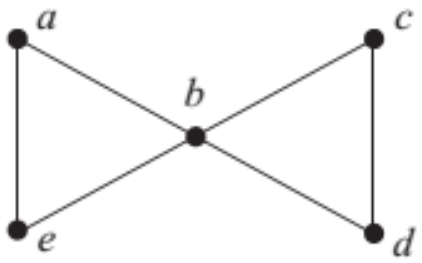
Others

Graph Coloring

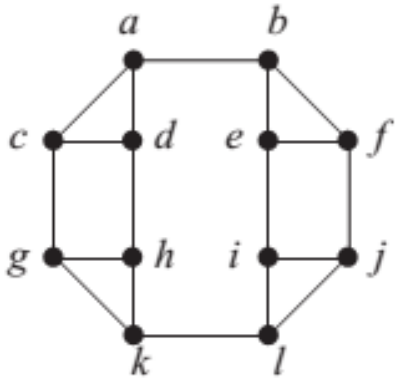
# Exercise

For each of these graphs, find $\kappa(G), \lambda(G)$

a)




b)







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

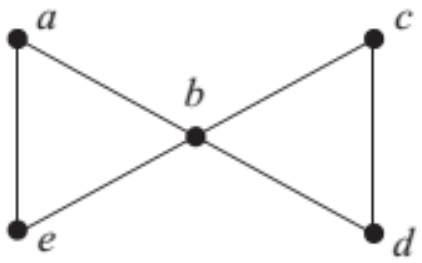
Others

Graph Coloring

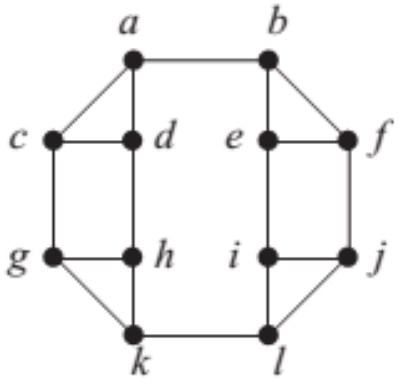
# Exercise

For each of these graphs, find $\kappa(G), \lambda(G)$

a)




b)




Construct a graph $G$ with $\kappa(G) = 1, \lambda(G) = 2$, and $\min_{v \in V} deg(v) = 3$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

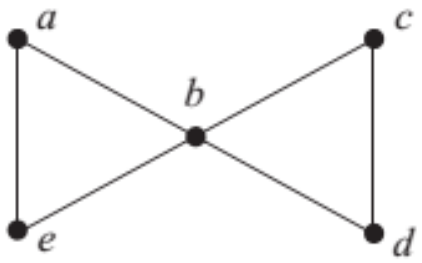
Others

Graph Coloring

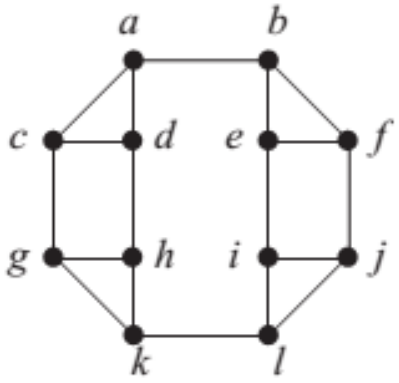
# Exercise

For each of these graphs, find $\kappa(G), \lambda(G)$

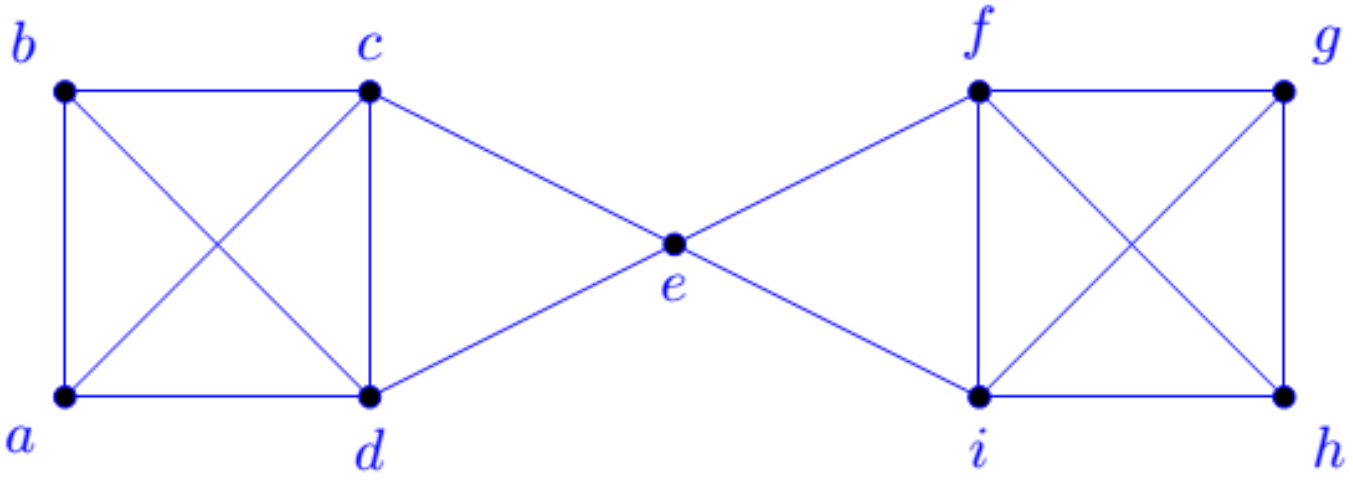
a)




b)




Construct a graph $G$ with $\kappa(G) = 1, \lambda(G) = 2$, and $\min_{v \in V} deg(v) = 3$.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

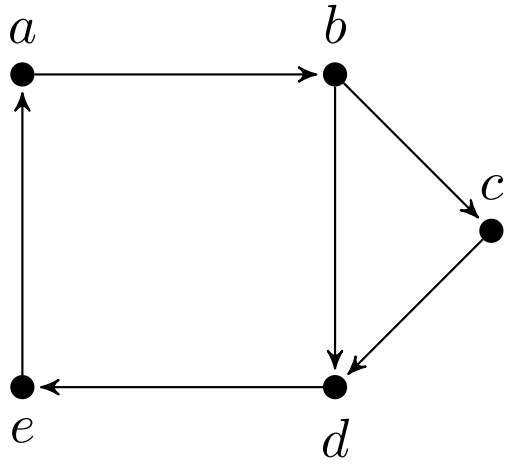
Others

Graph Coloring

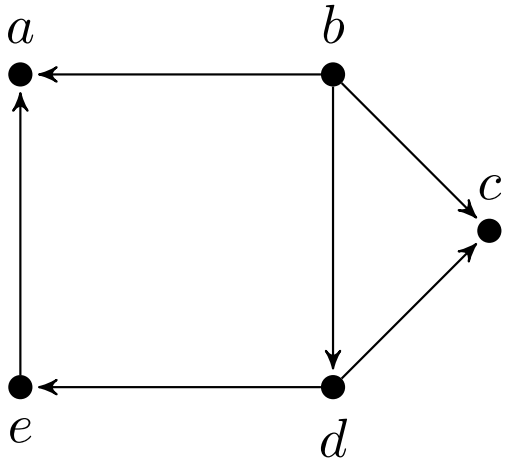
# Connectedness in Directed Graphs

## Definition

- An directed graph is strongly connected (*liên thông mạnh*) if there is a path between any two vertices in the graph (for both directions).
- An directed graph is weakly connected (*liên thông yếu*) if there is a path between any two vertices in the underlying undirected graph.




Strongly connected




Weakly connected




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

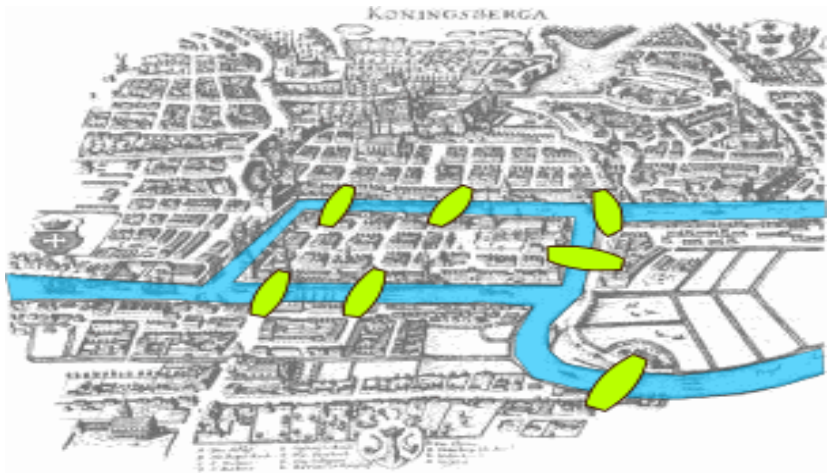
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# The Famous Problem of Seven Bridges of Königsberg




- Is there a route that a person crosses all the seven bridges once?



Contents

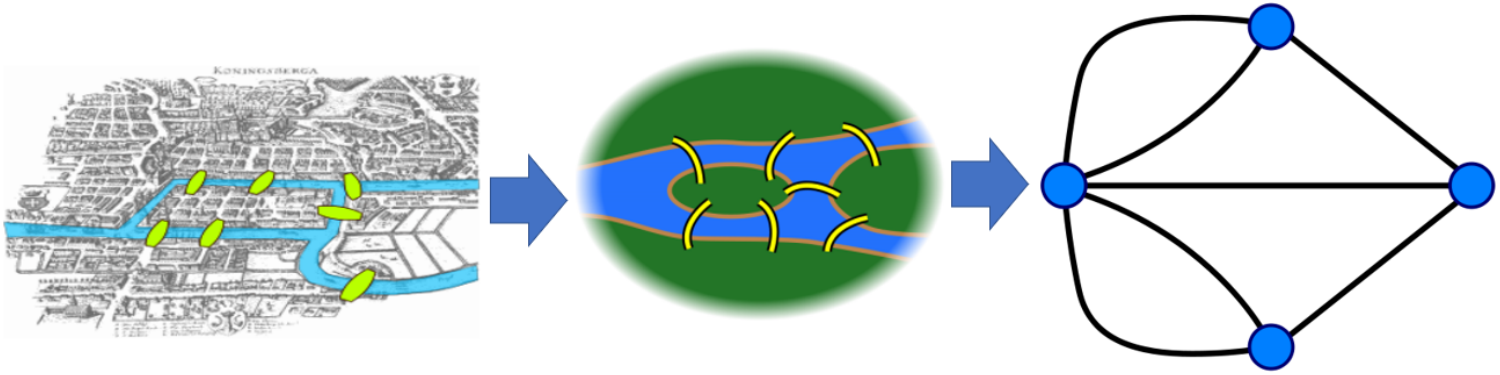
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Euler Solution



- Euler gave the solution: It is not possible to cross all the bridges exactly once.



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# What is Euler Path and Circuit?

- **Euler Path** (*đường đi Euler*) is a path in the graph that passes each edge only once.
  The problem of Seven Bridges of Königsberg can be also stated: Does Euler Path exist in the graph?
- **Euler Circuit** (*chu trình Euler*) is a path in the graph that passes each edge only once and return back to its original position.
  *From Definition, Euler Circuit is a subset of Euler Path.*



Contents

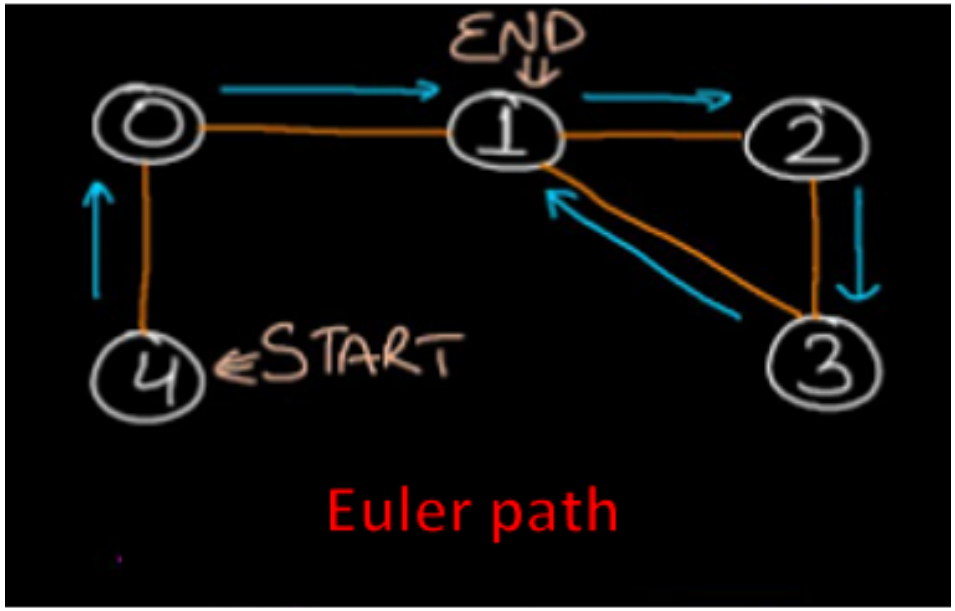
Connectivity
Paths and Circuits

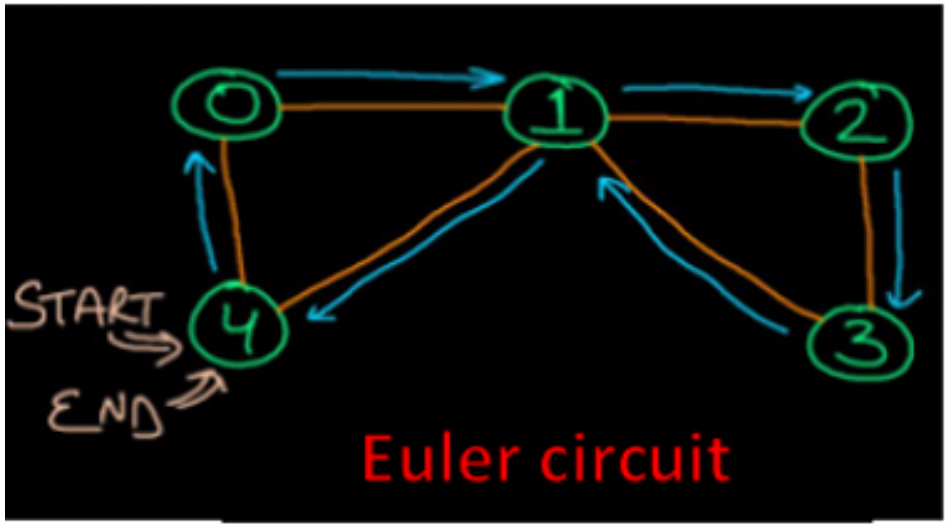
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Examples of Euler Path and Circuit







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Conditions for Existence

In a connected multigraph,

- Euler Circuit existence: **no odd-degree nodes exist** in the graph.
- Euler Path existence: **2 or no odd-degree nodes exist** in the graph.



Contents

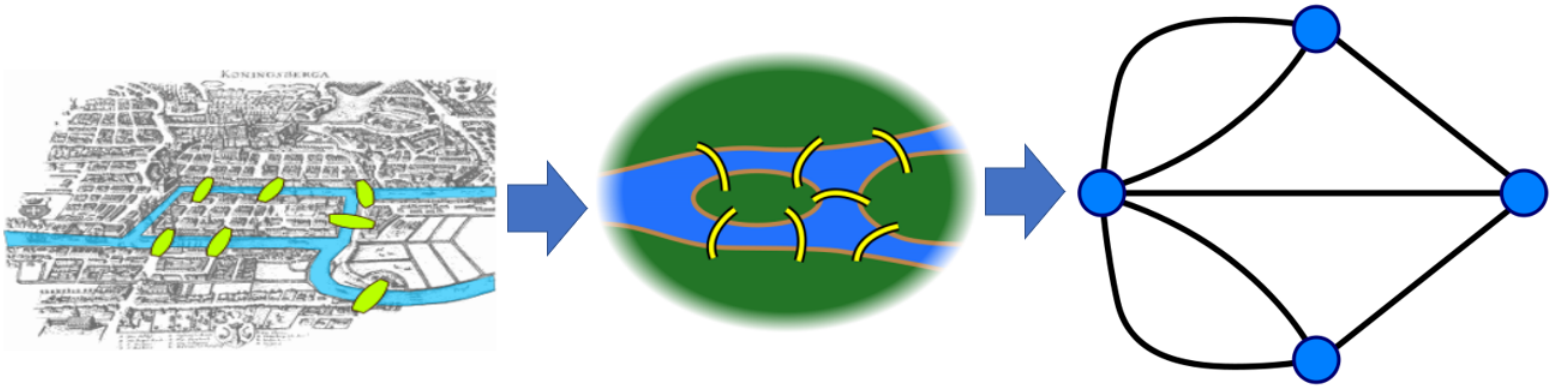
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Back to the Seven Bridges Problem



- Four vertices of odd degree
- No Euler circuit $\rightarrow$ cannot cross each bridge exactly once, and return to starting point
- No Euler path, either



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Searching Euler Circuits and Paths – Fleury's Algorithm

- Choose a random vertex (if circuit) or an odd degree vertex (if path)
- Pick an edge joined to another vertex so that it is not a cut edge unless there is no alternative.
- Remove the chosen edge. The above procedure is repeated until all edges are covered.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Searching Euler Circuits and Paths – Hierholzer's Algorithm

- Choose a starting vertex and find a circuit
- As long as there exists a vertex $v$ that belongs to the current tour but that has adjacent edges not part of the tour, start another circuit from $v$

More efficient algorithm, $O(n)$.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

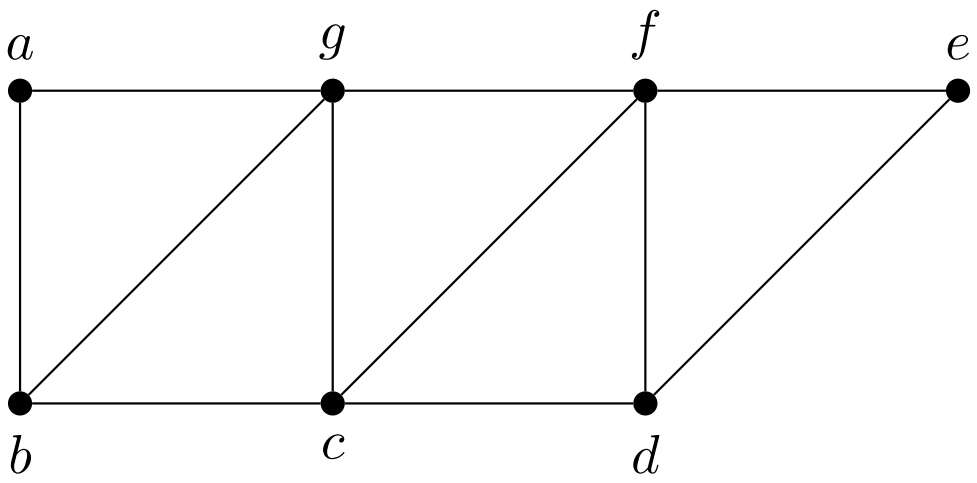
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

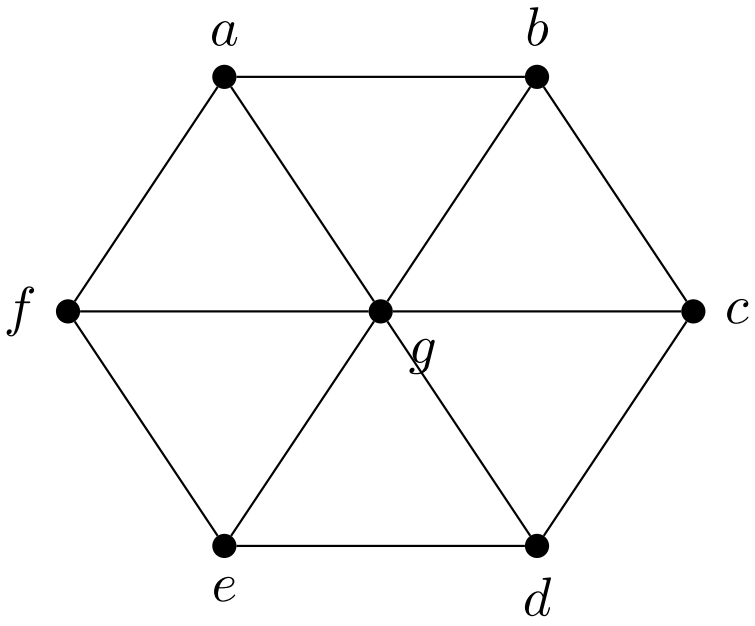
Graph Coloring

# Exercise

## Example

Are these following graph Euler path (circuit)? If yes, find one.






Contents

Connectivity
Paths and Circuits

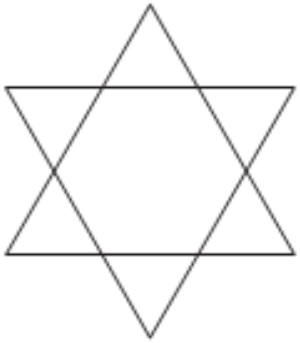
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

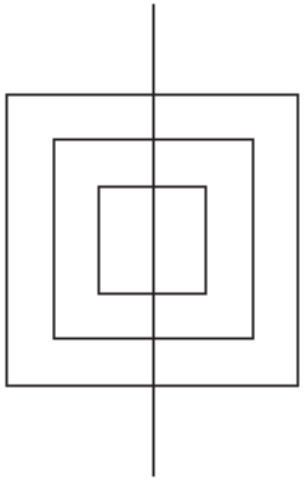
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Exercise

Determine whether the picture shown can be drawn with a pencil in a continuous motion without lifting the pencil or retracing part of the picture.

a) 

b) 



Contents

Connectivity
Paths and Circuits

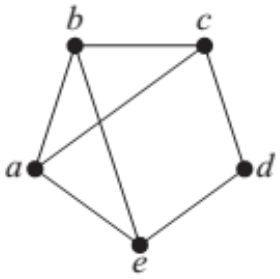
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

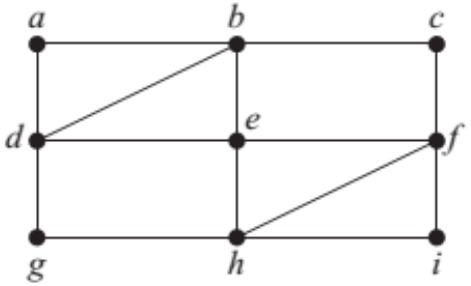
Graph Coloring

# Exercise - Euler path & circuit

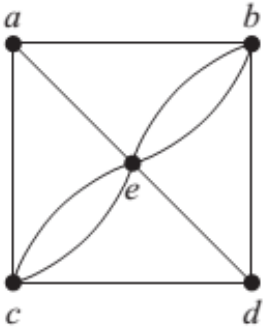
a)



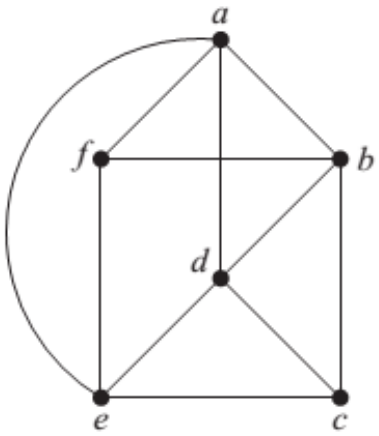

b)



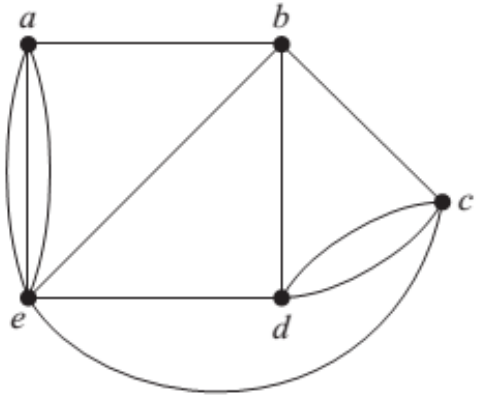

c)



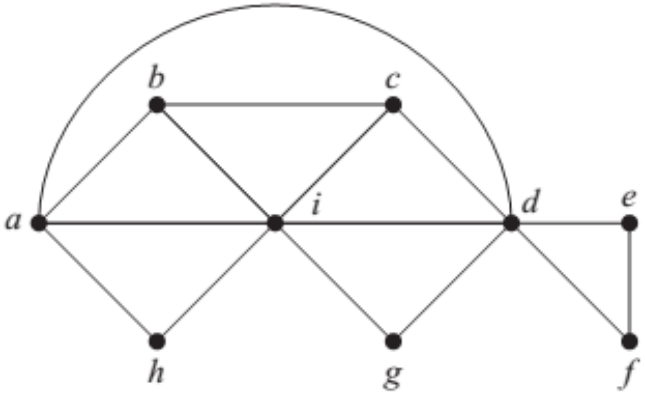

d)




e)




f)

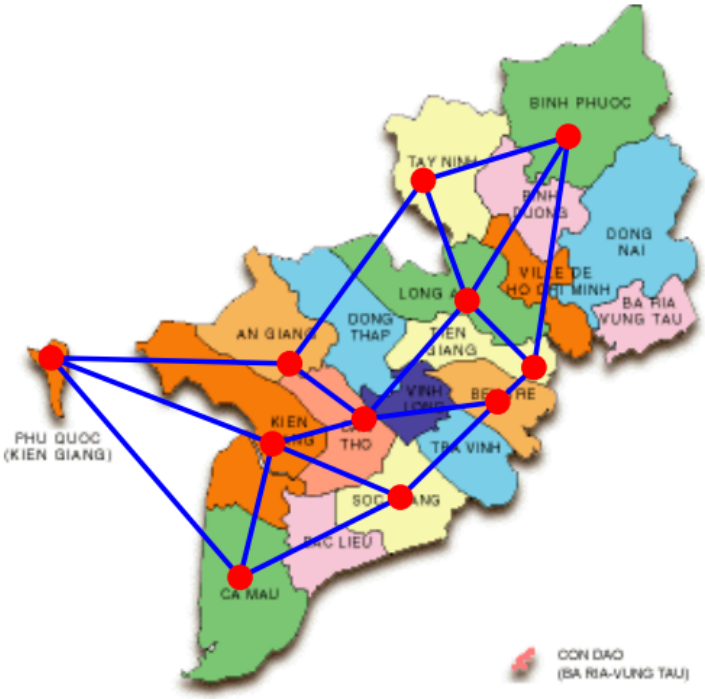





Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Traveling Salesman Problem




*Is there the possible tour that visits each city exactly once?*




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

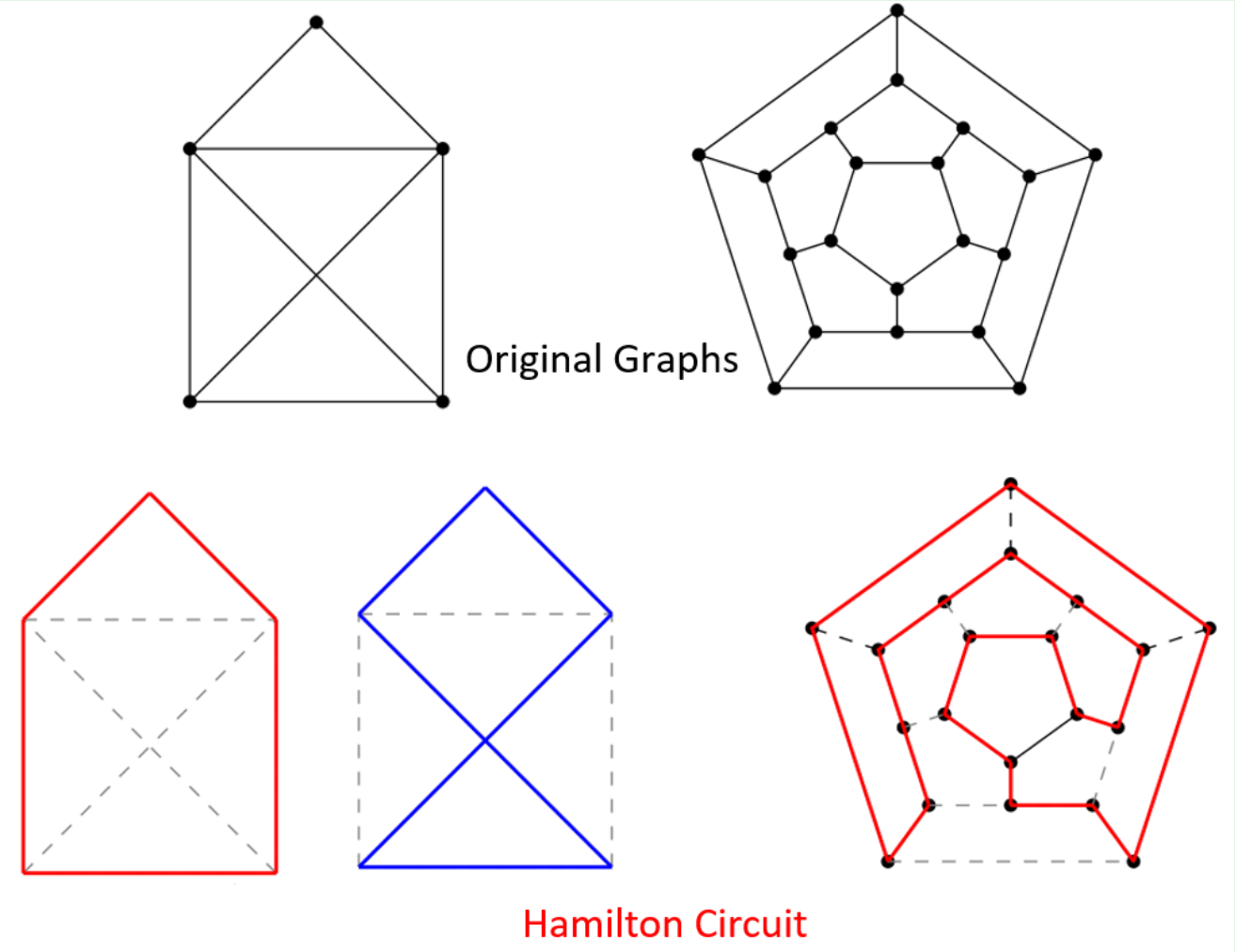
Graph Coloring

# What Is A Hamilton Circuit?

## Definition

The circuit that visit each vertex in a graph once.

## Example






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

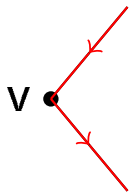
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Rules of Hamilton Circuits

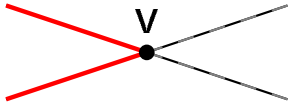
$\deg(v) = 2$ for $\forall v$ in Hamilton circuit!

Rule 1 if $\deg(v) = 2$, both edge must be used.




Rule 2 No subcircuit (*chu trình con*) can be formed.

Rule 3 Once two edges at a vertex $v$ is determined, all other edges incident at $v$ must be removed.






Contents

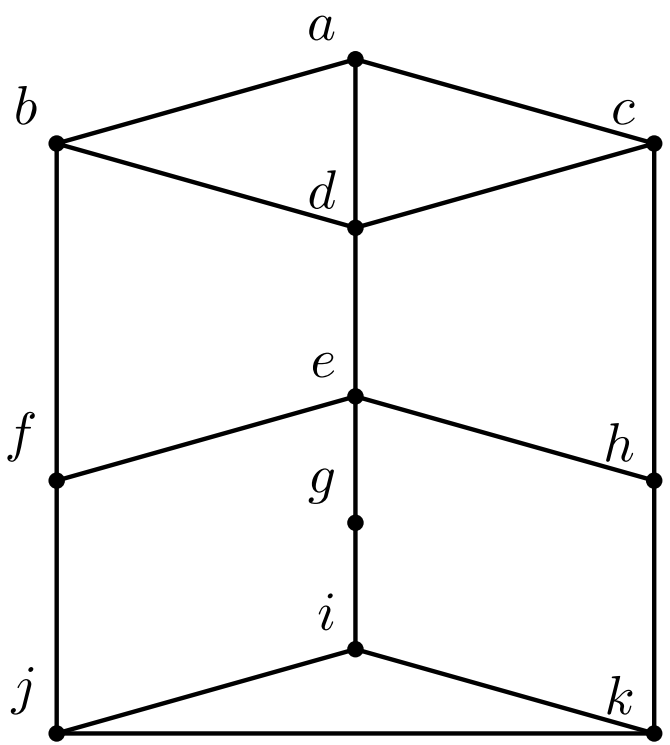
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

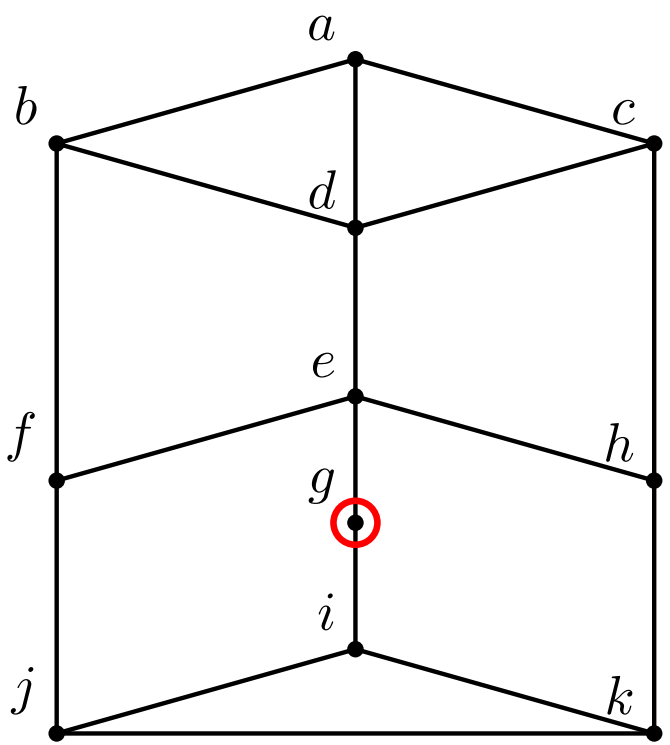
Edges : possible routes



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

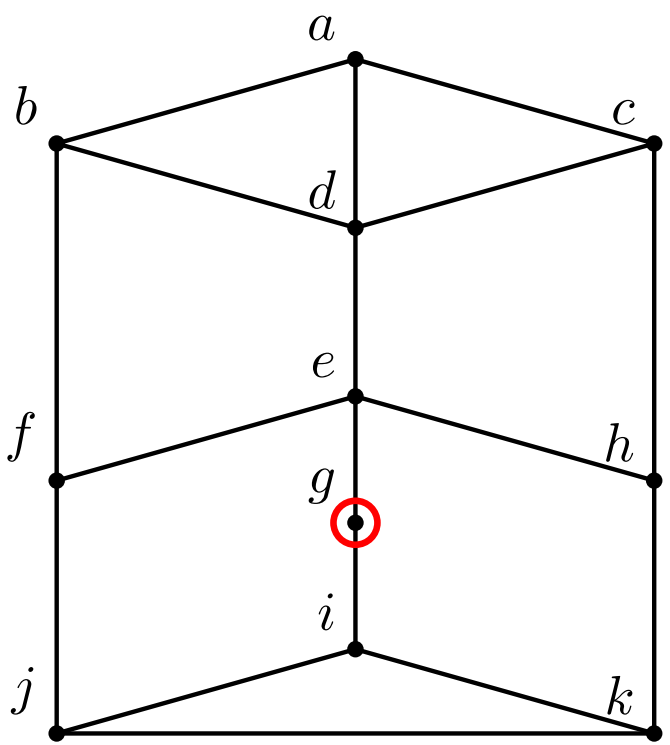
Edges : possible routes



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

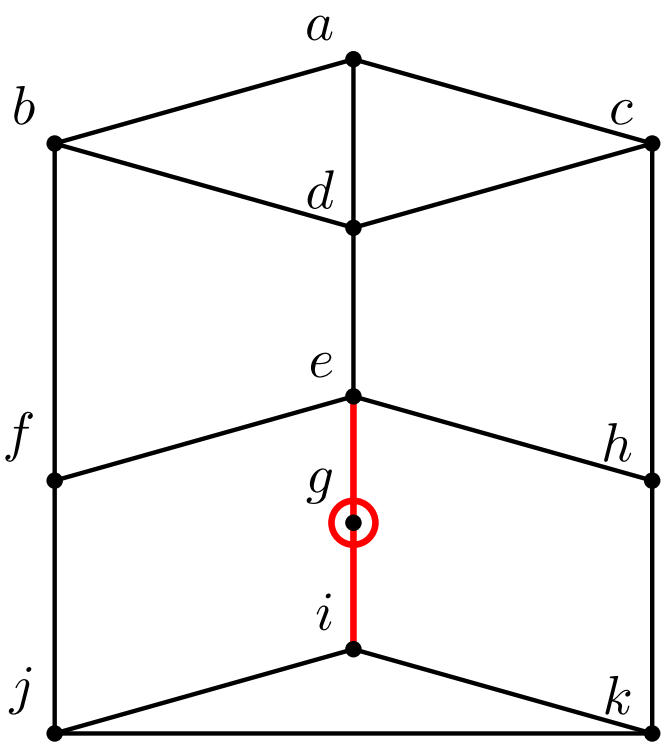
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

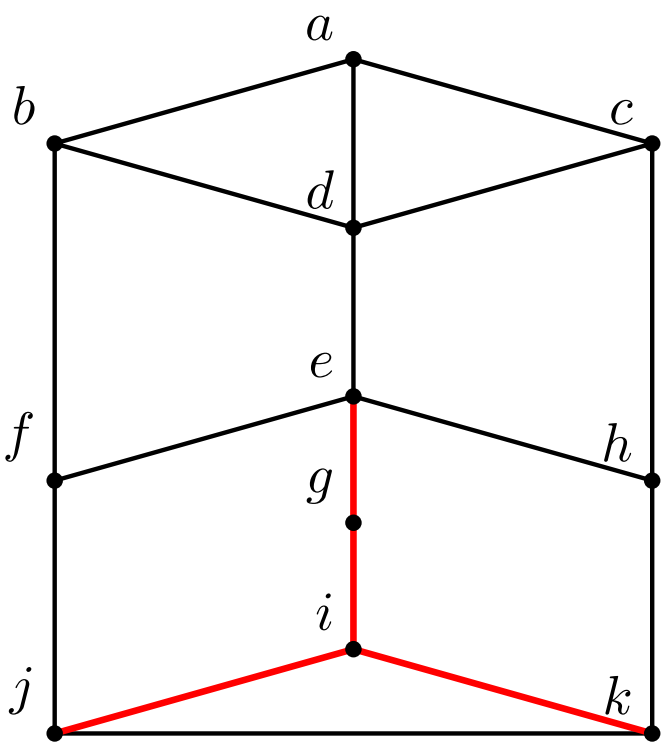
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

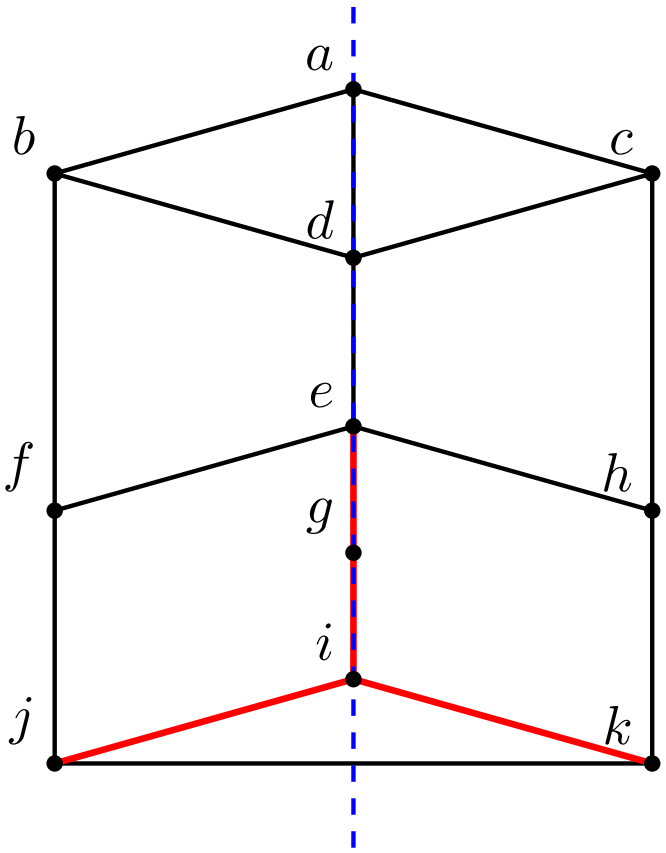
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

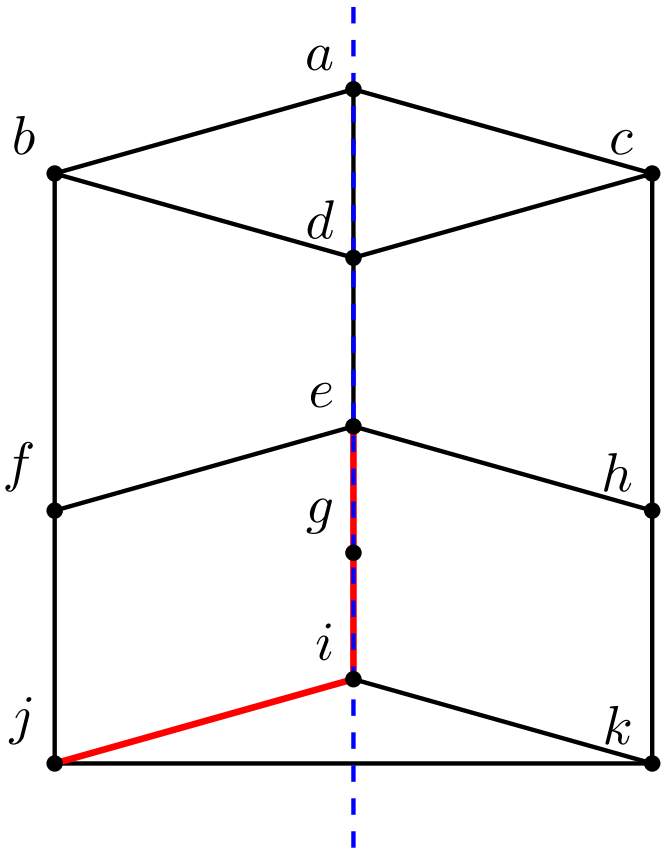
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

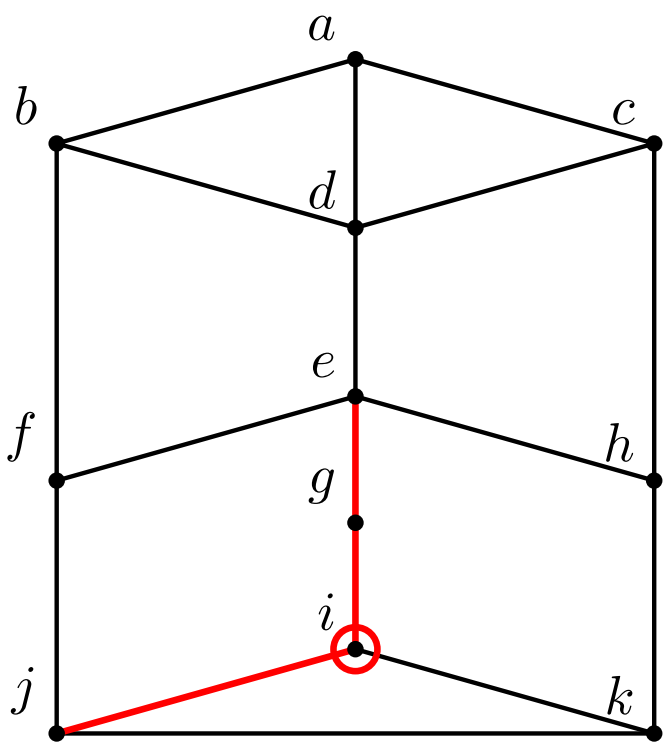
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

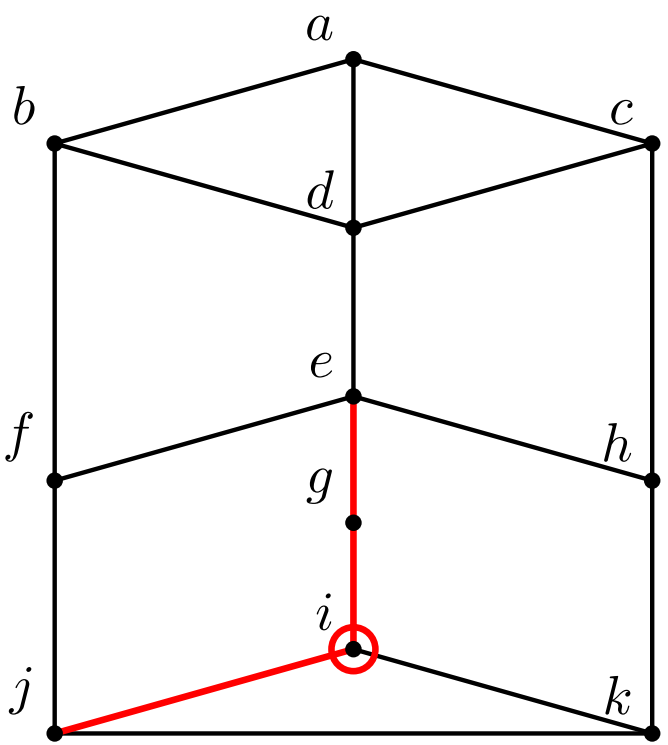
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$

Rule 3

Once two edges are determined, other edges must be removed



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

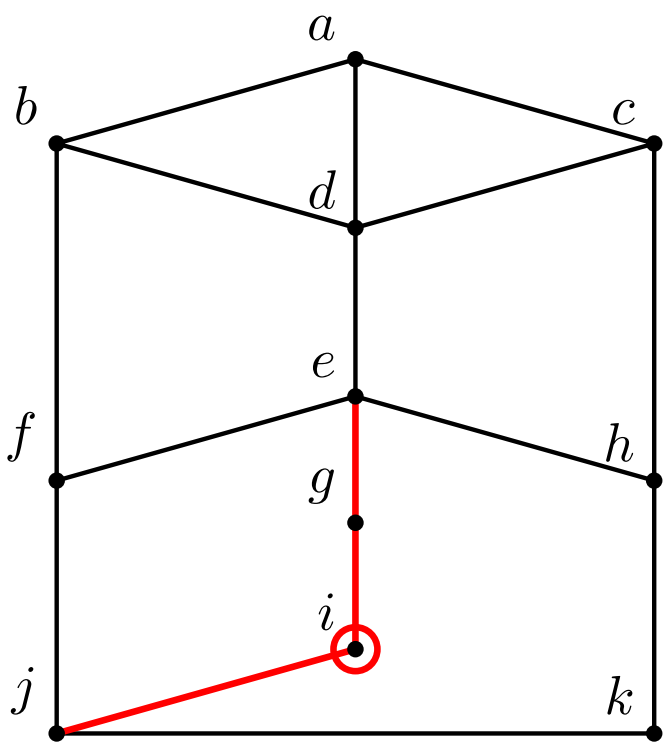
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$

Rule 3

Once two edges are determined, other edges must be removed



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

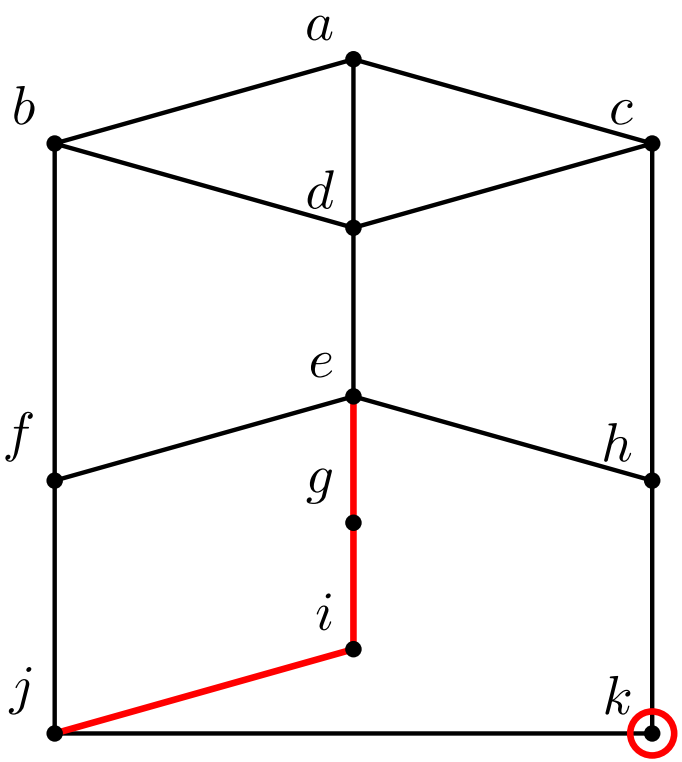
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$

Rule 3

Once two edges are determined, other edges must be removed



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

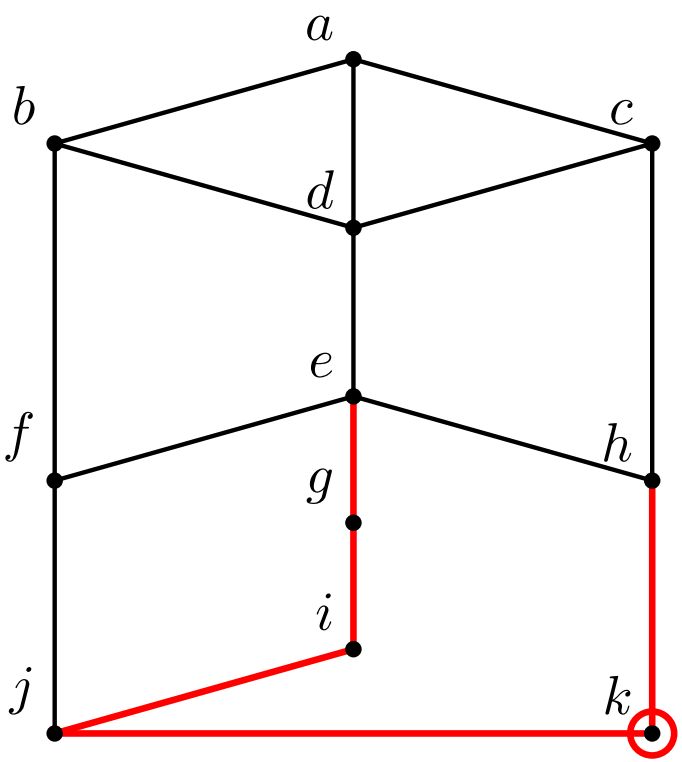
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

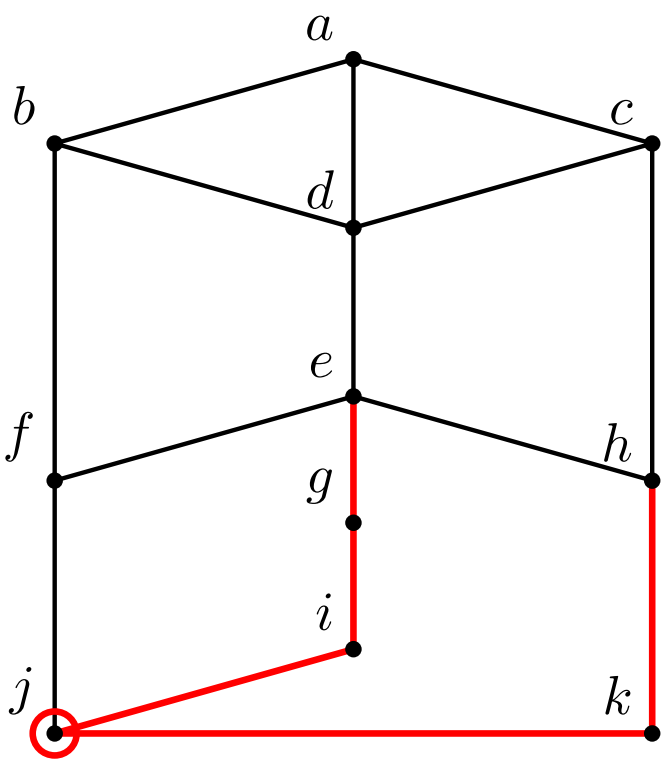
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

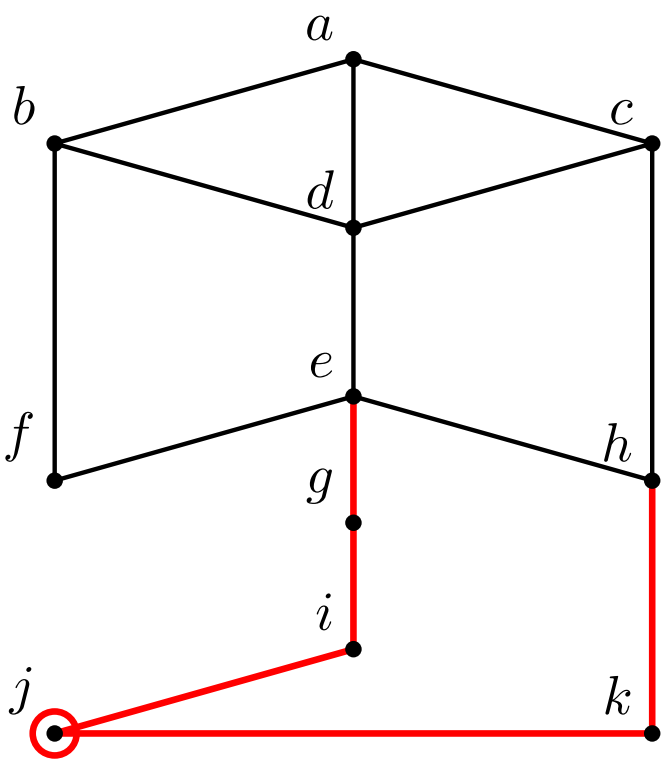
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

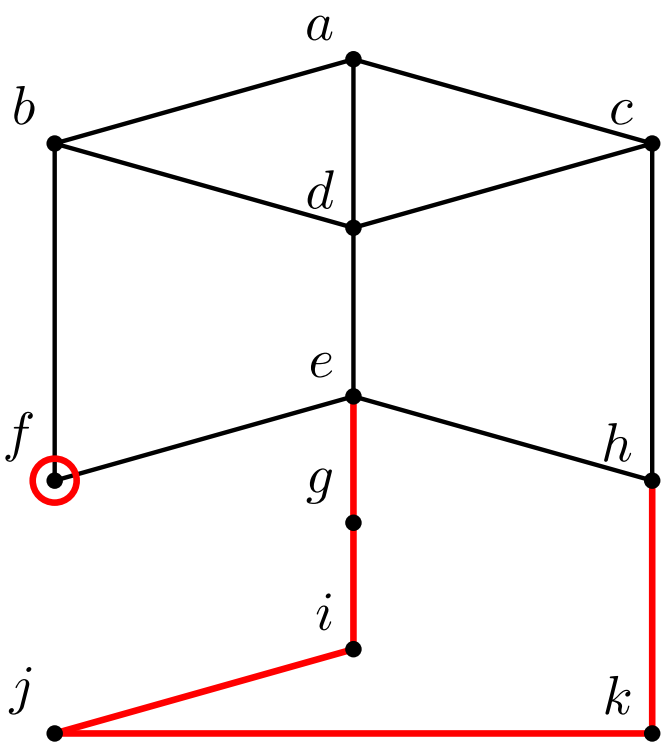
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

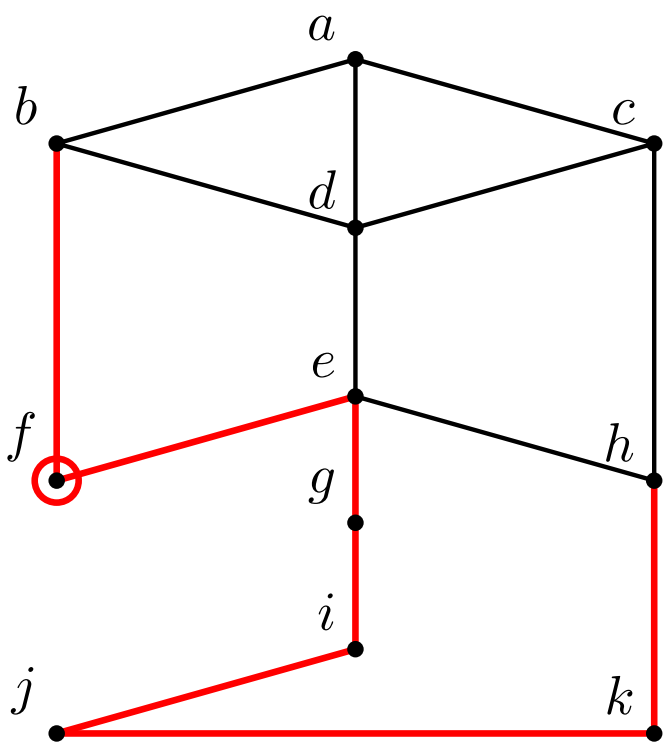
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

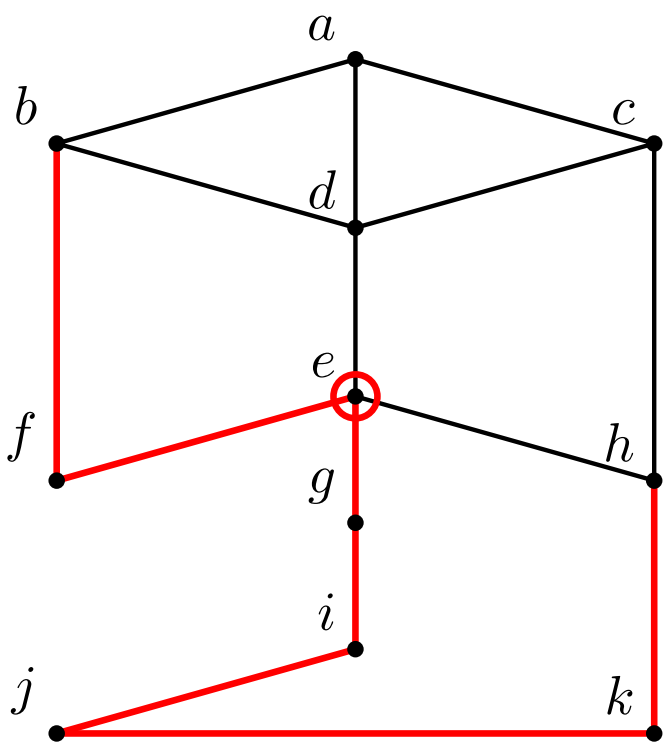
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

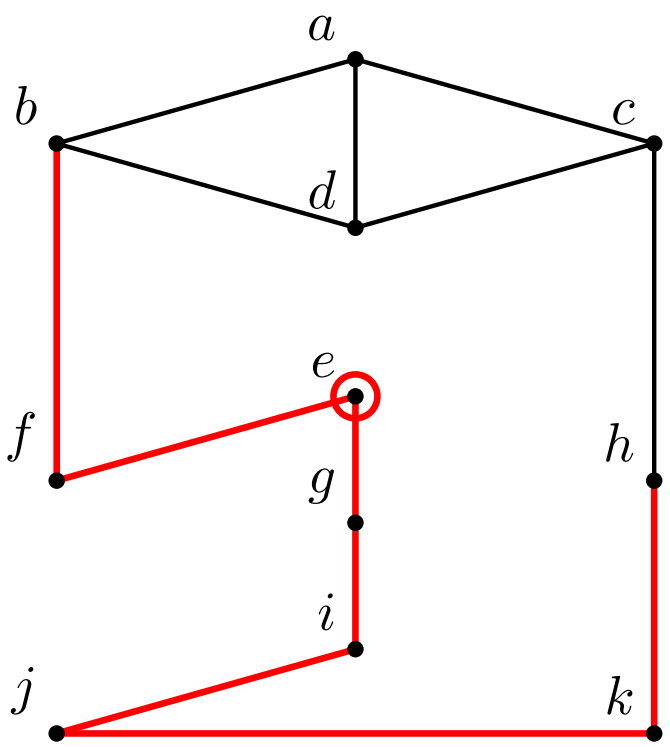
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

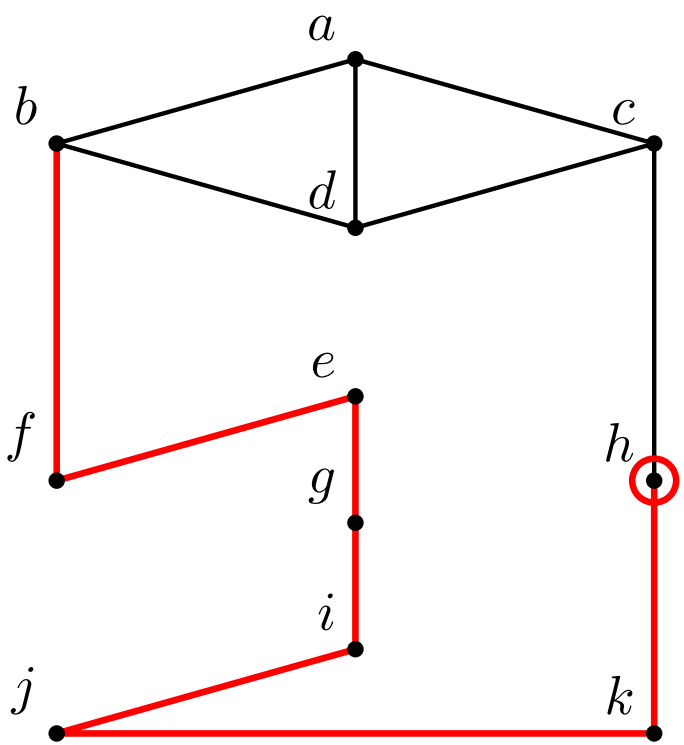
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

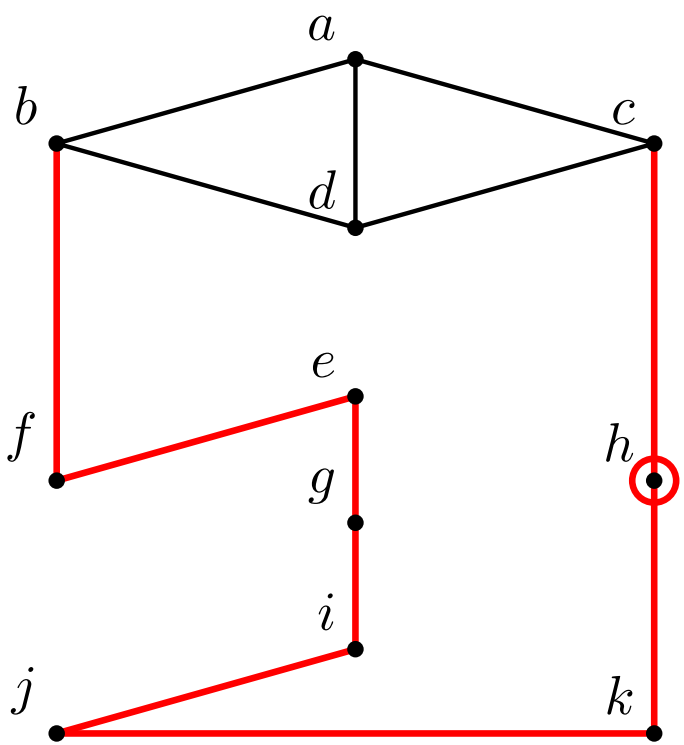
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

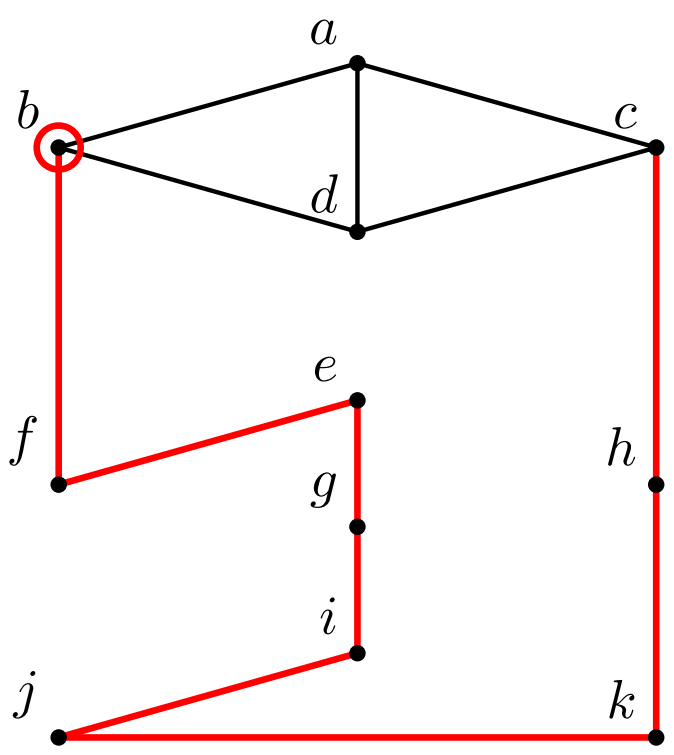
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

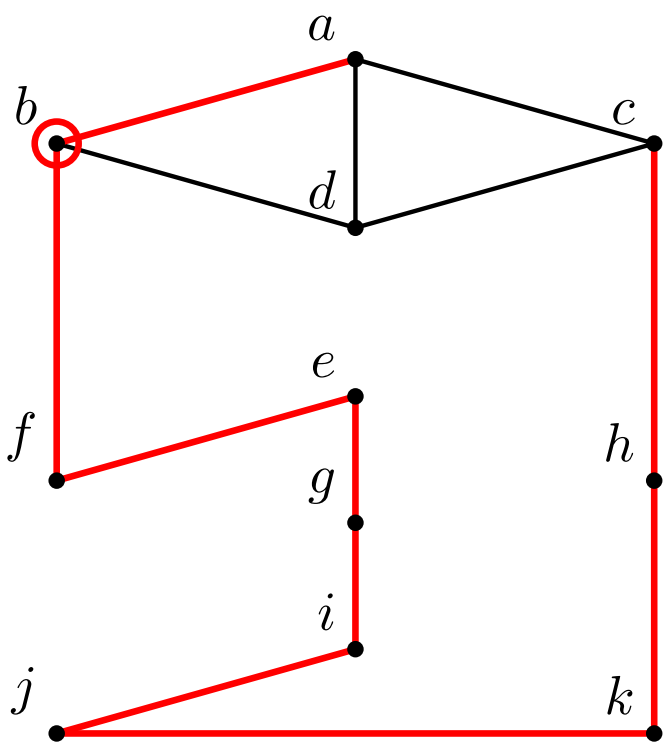
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

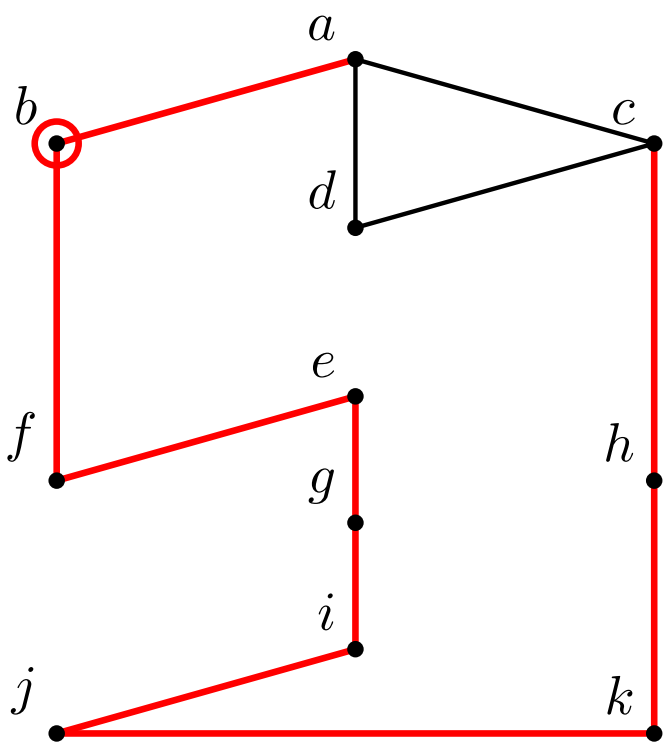
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

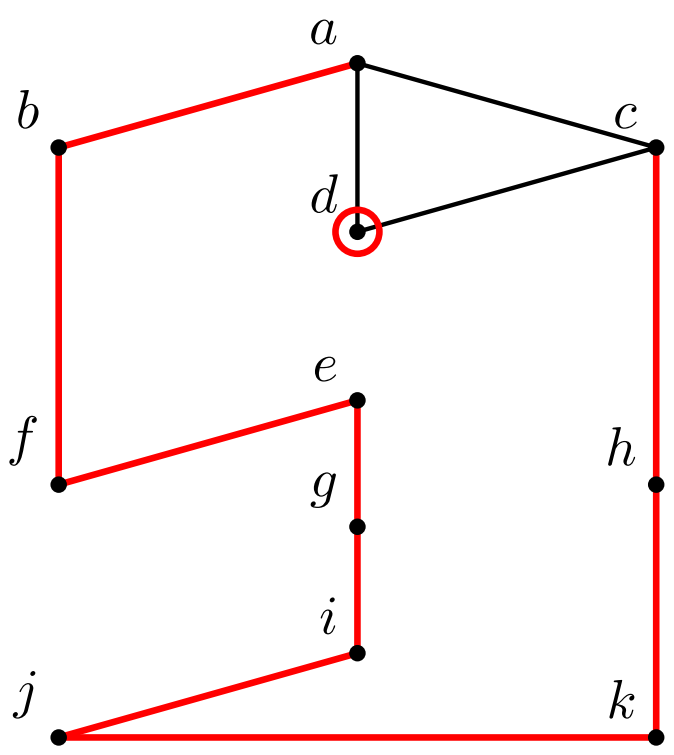
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

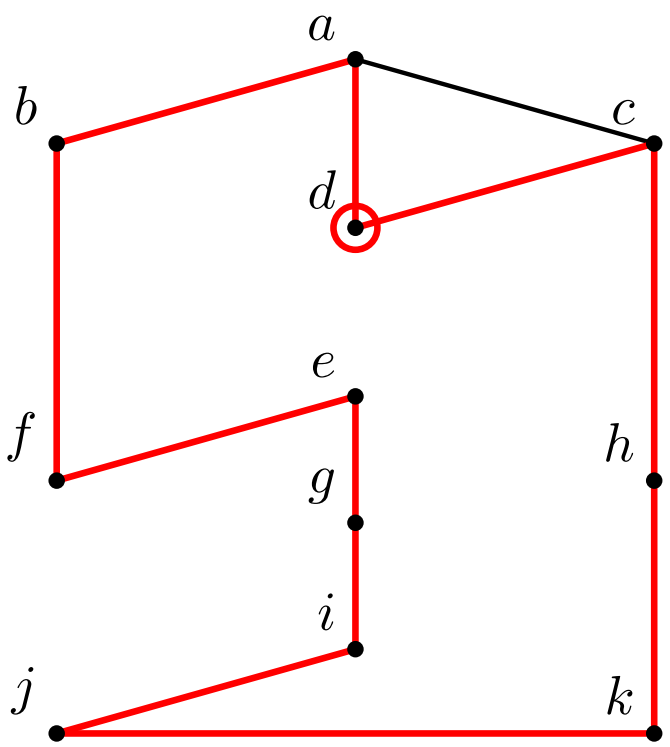
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

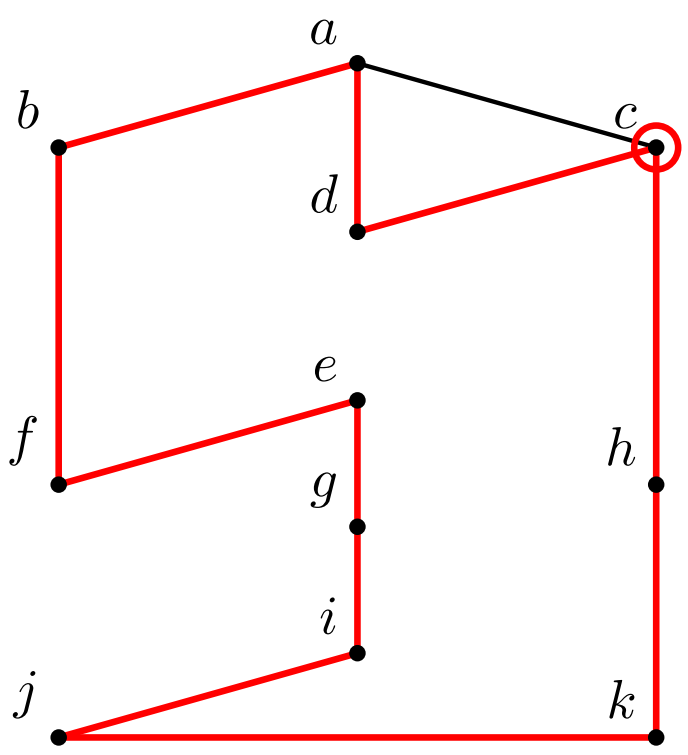
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

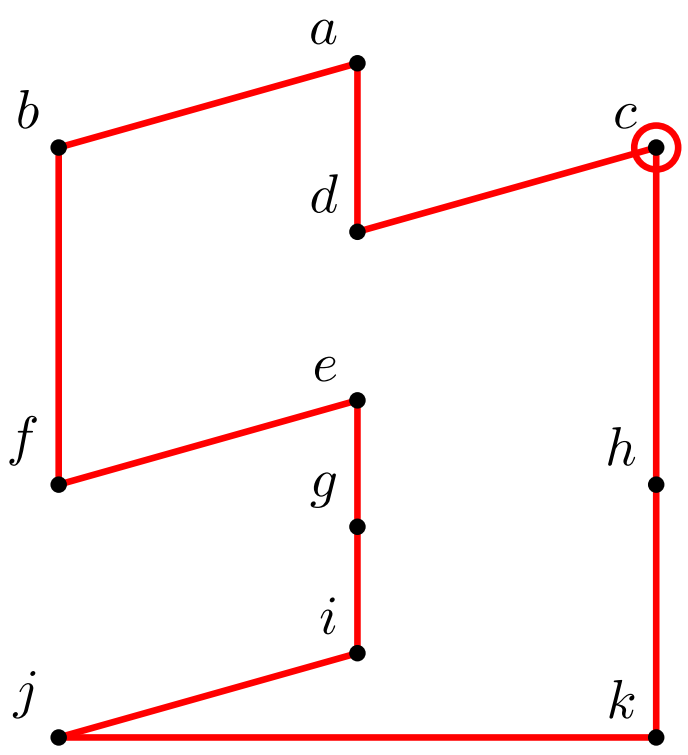
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

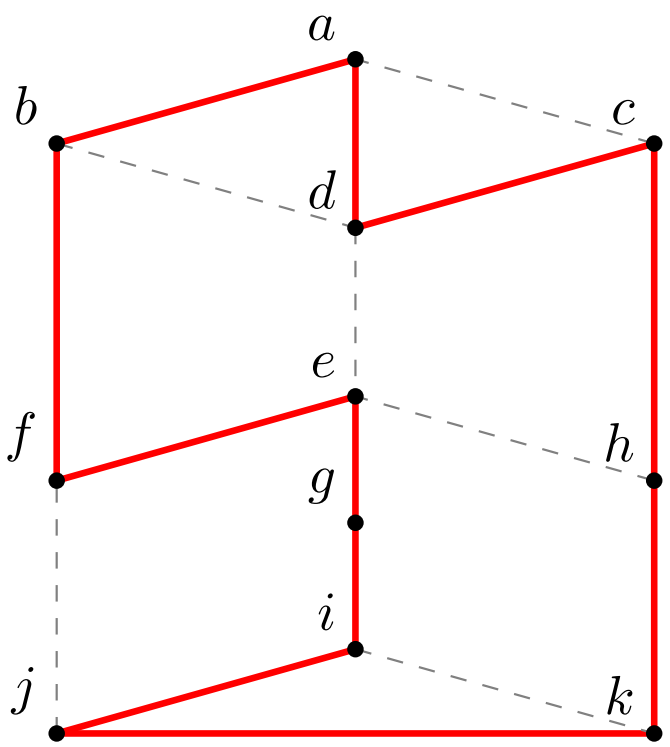
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




We get Hamilton circuit!

Vertices : cities

Edges : possible routes

Rule 1

$\deg(v) = 2$

Rule 3

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

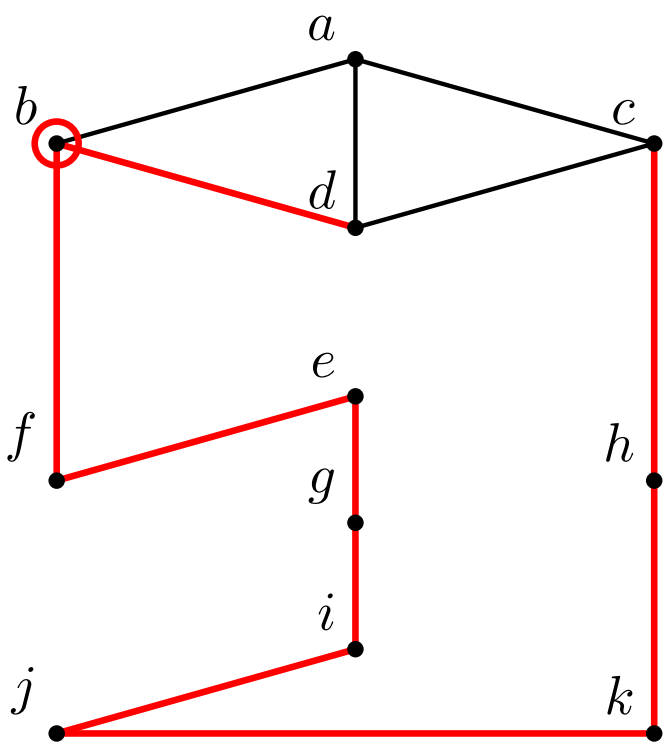
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

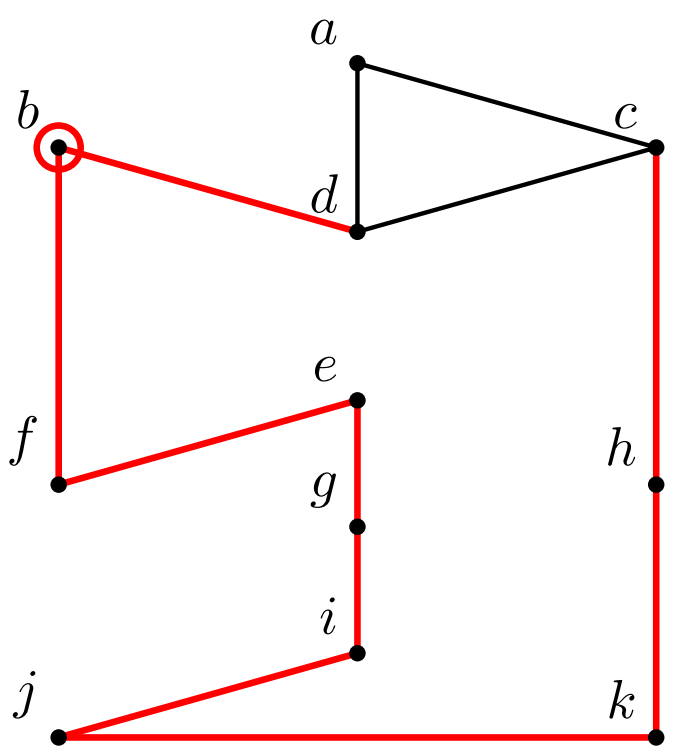
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

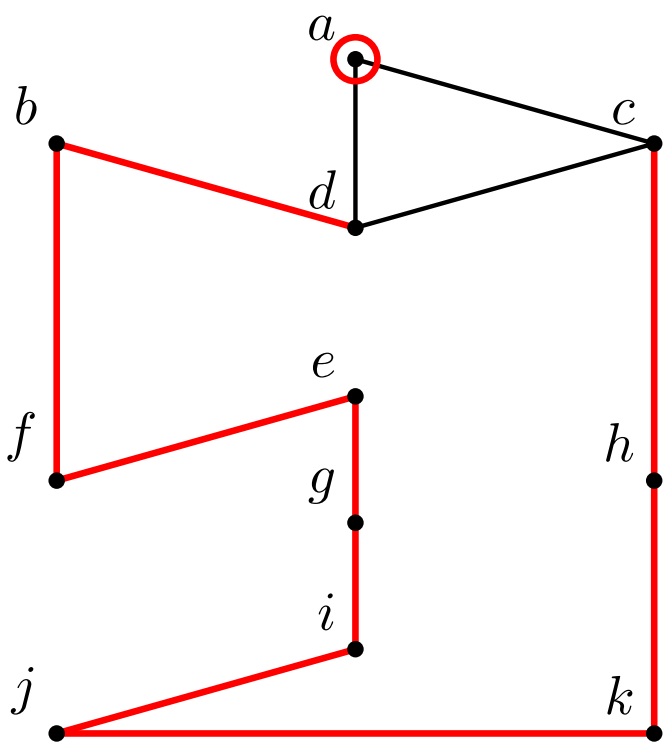
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

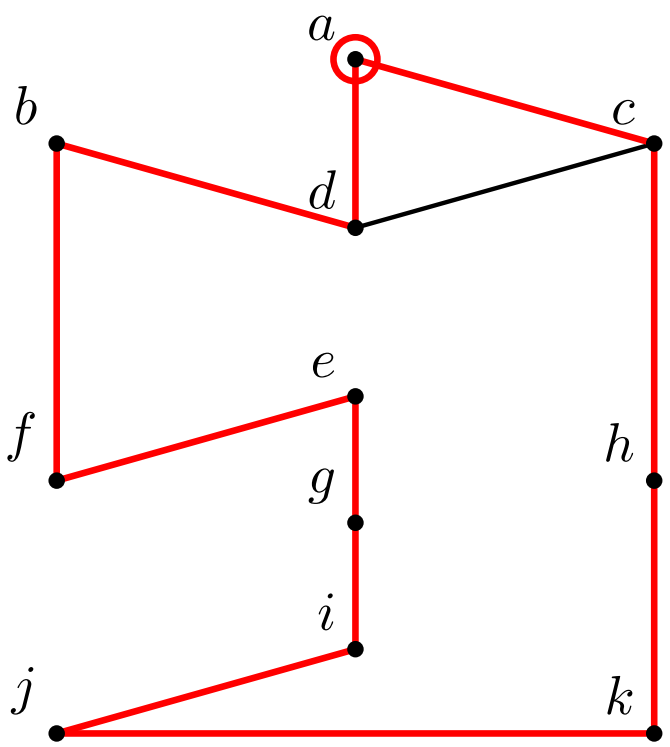
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

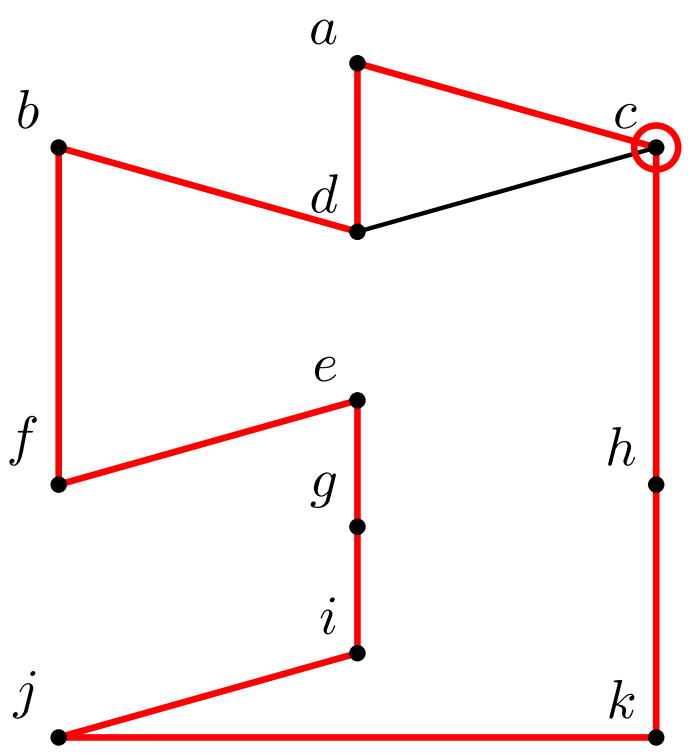
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

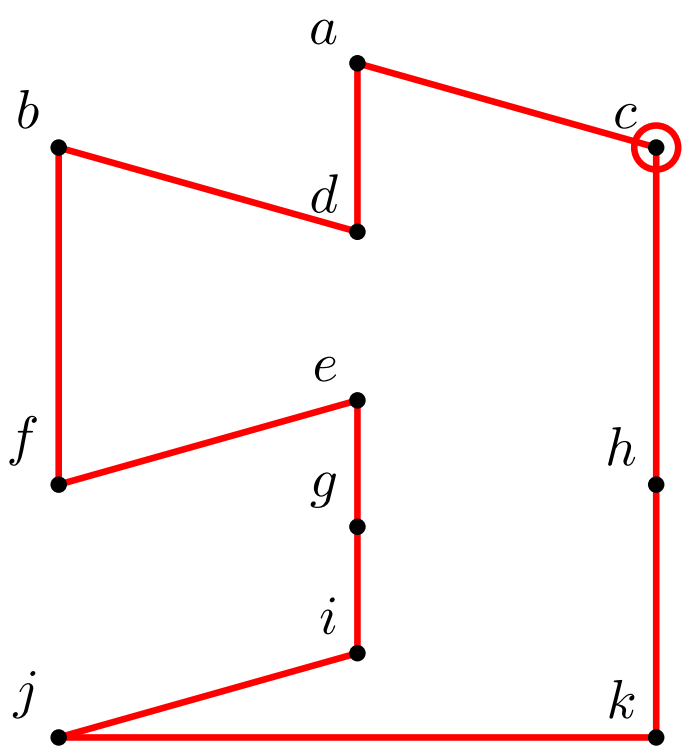
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

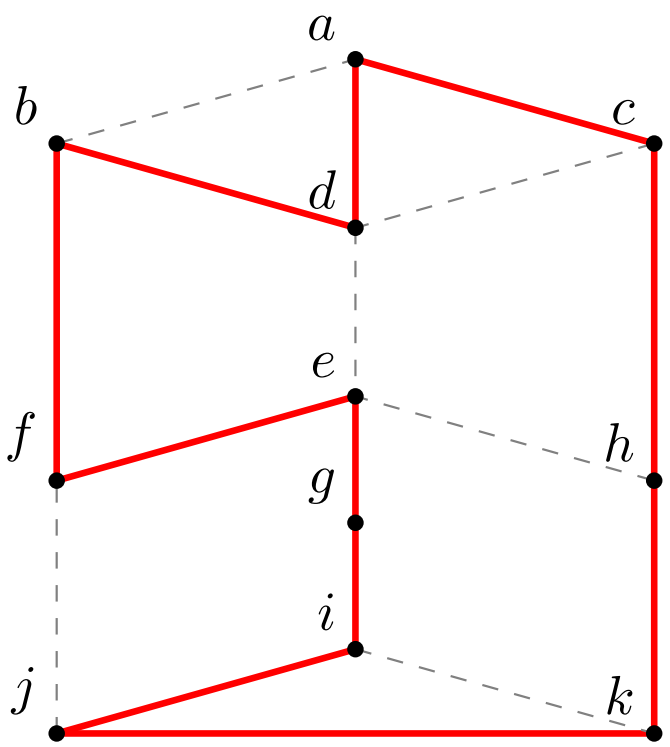
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Finding Hamilton Circuits




We get Hamilton circuit!

Vertices : cities

Edges : possible routes

**Rule 1**

$\deg(v) = 2$

**Rule 3**

Once two edges are determined, other edges must be removed




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

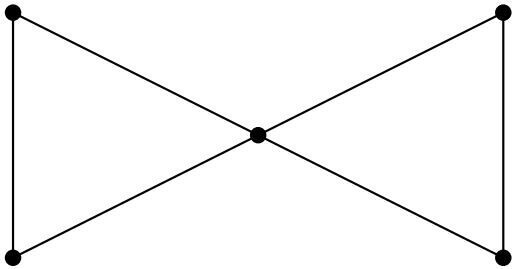
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit





Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

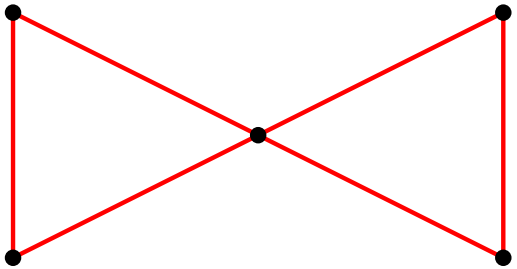
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit





Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

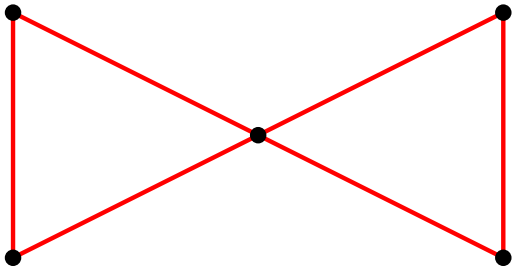
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Existence of Hamilton Circuit

Hamilton circuit does not exist for all graph. But, there is no specific way to find whether Hamilton circuit exists or not.

Simple check by rules of Hamilton circuit



Violates Rule 2! (No subcircuit)



Contents

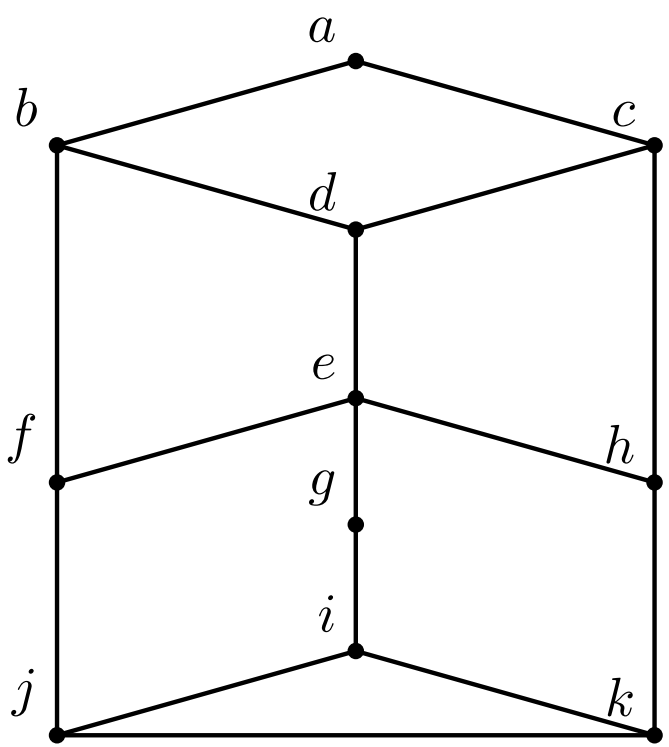
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

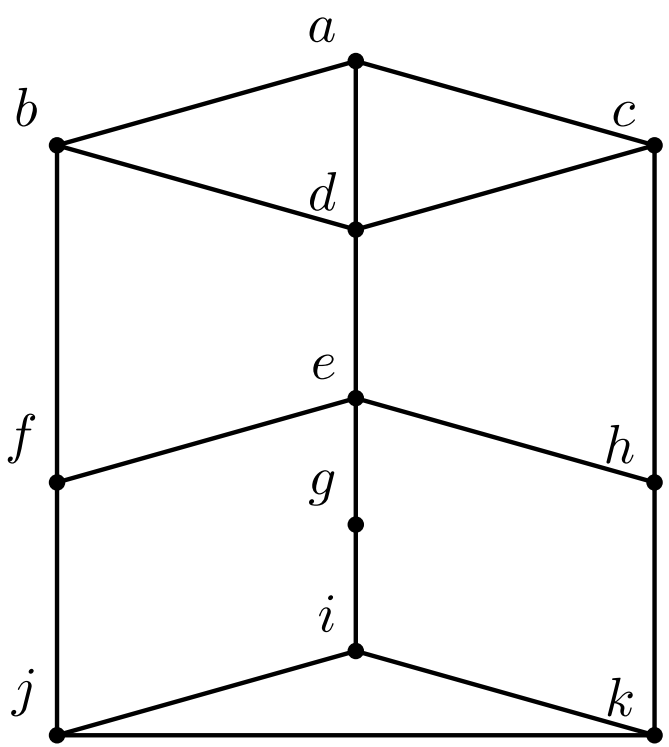
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

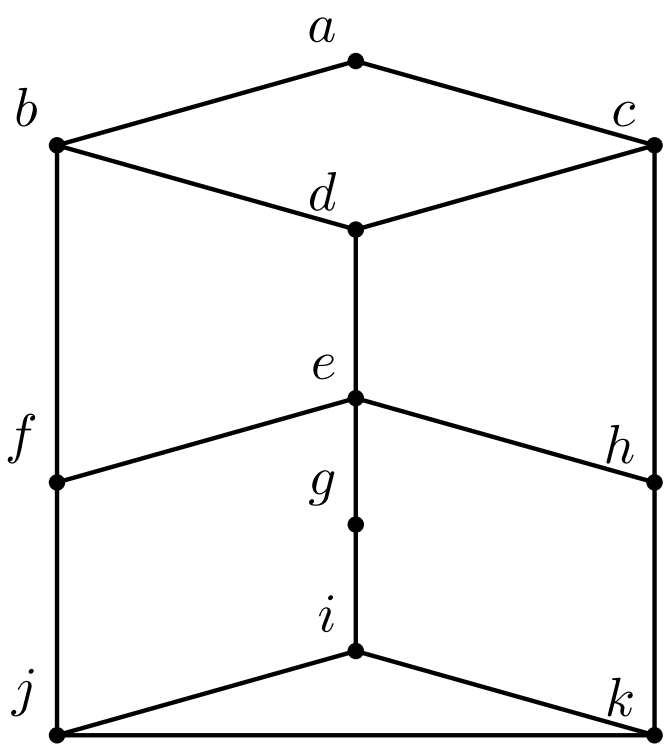
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

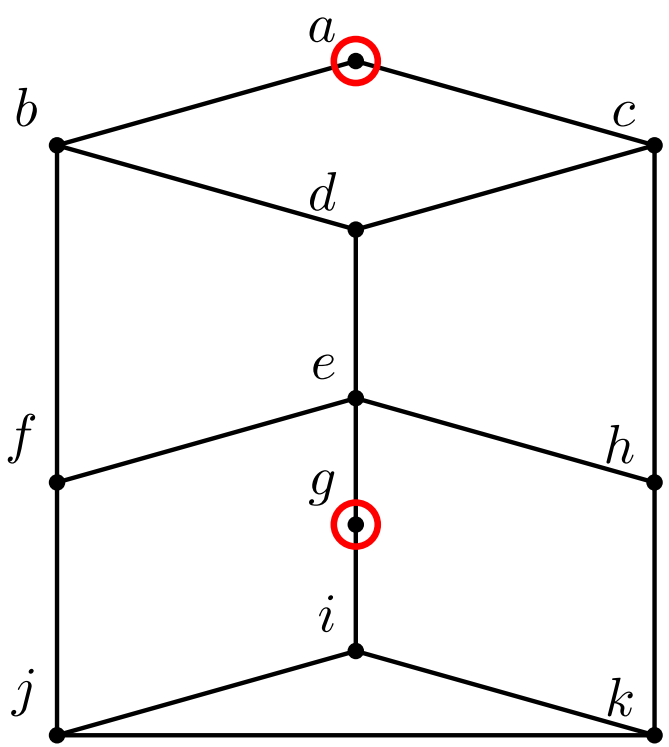
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

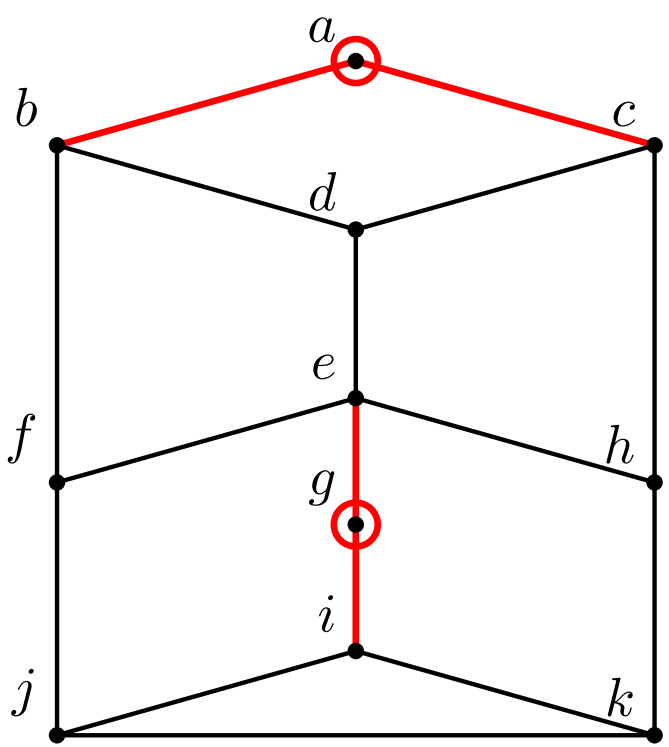
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

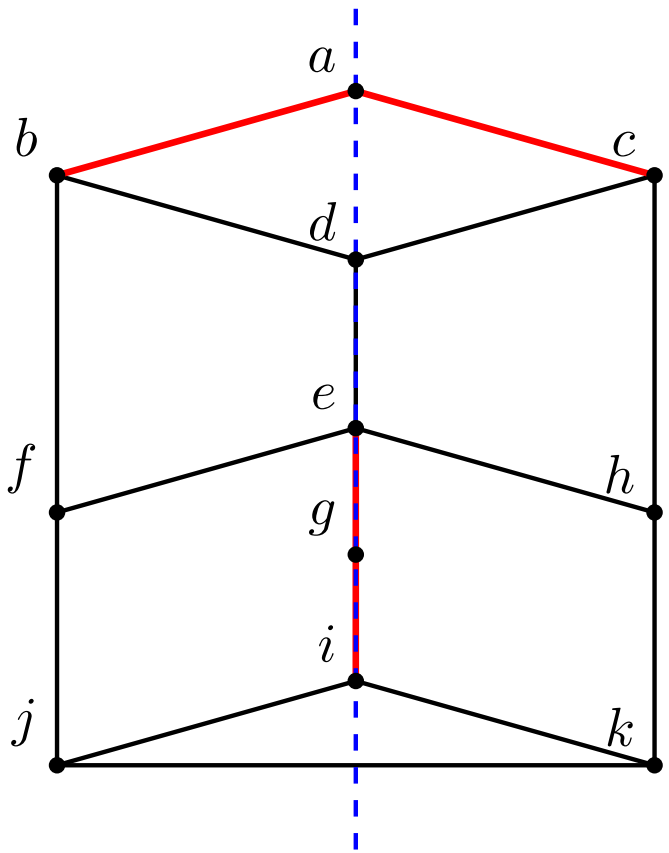
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

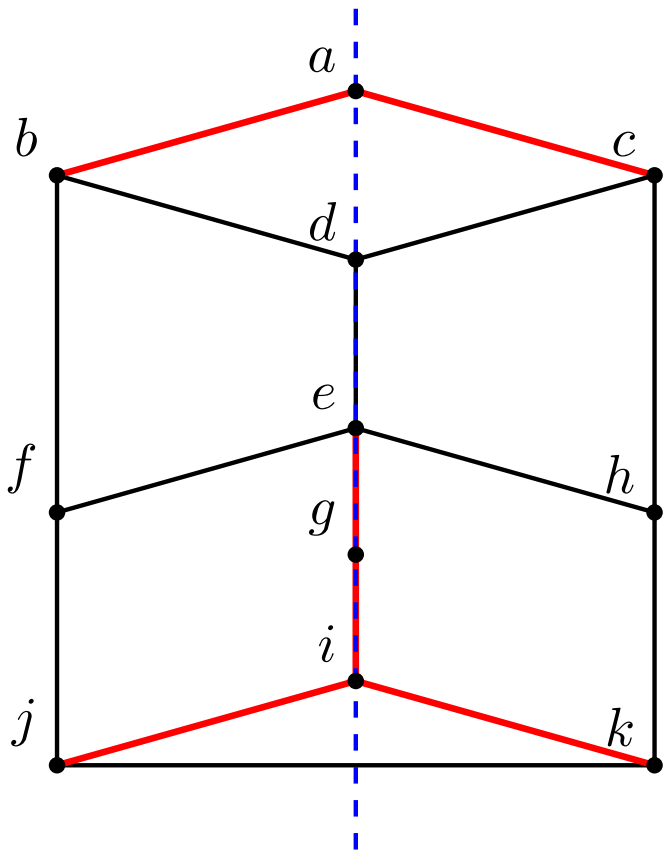
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

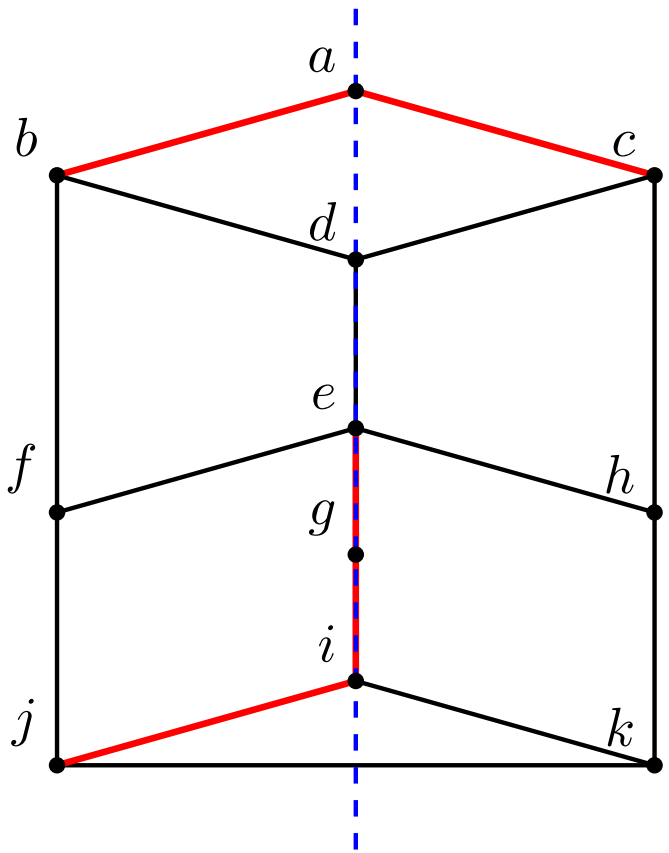
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

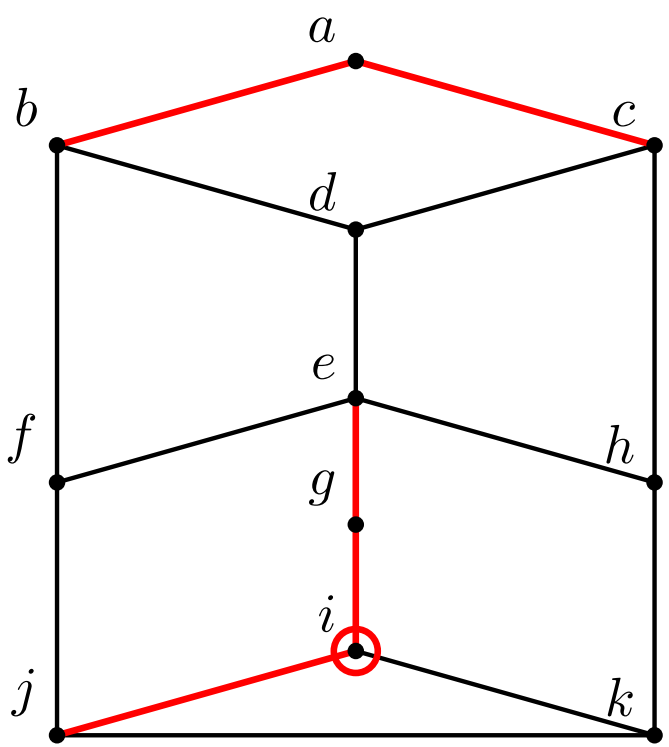
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

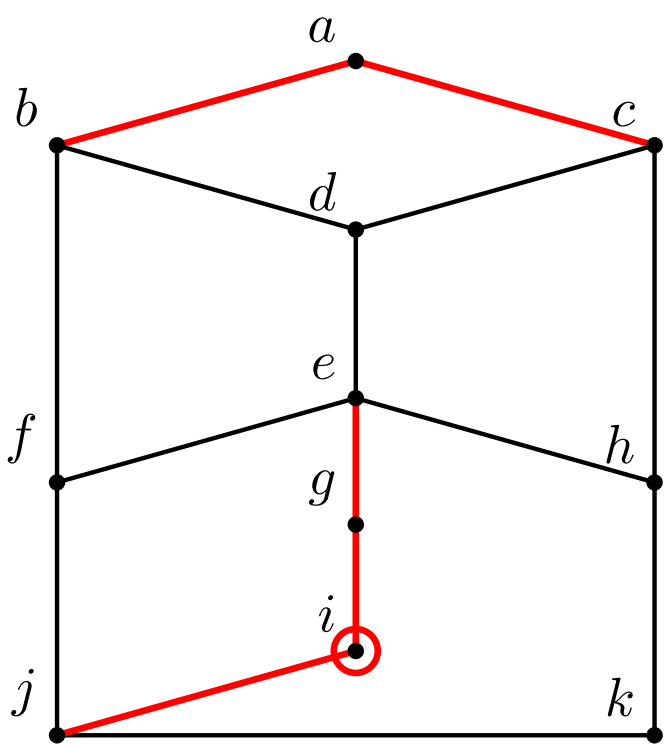
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

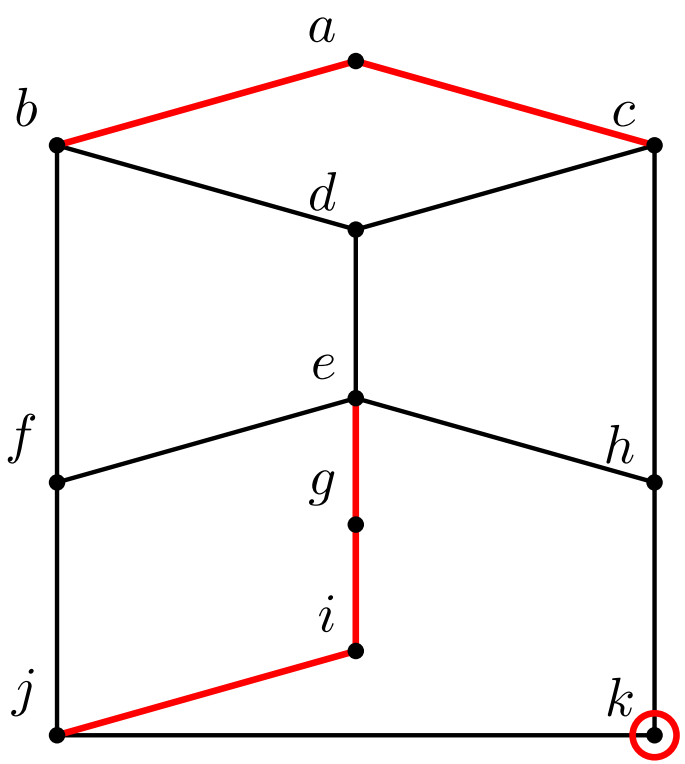
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

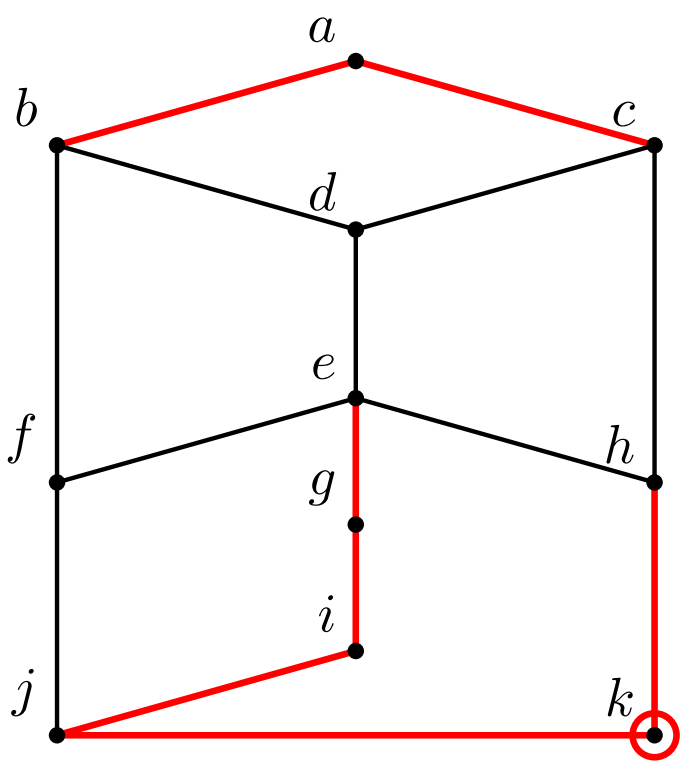
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

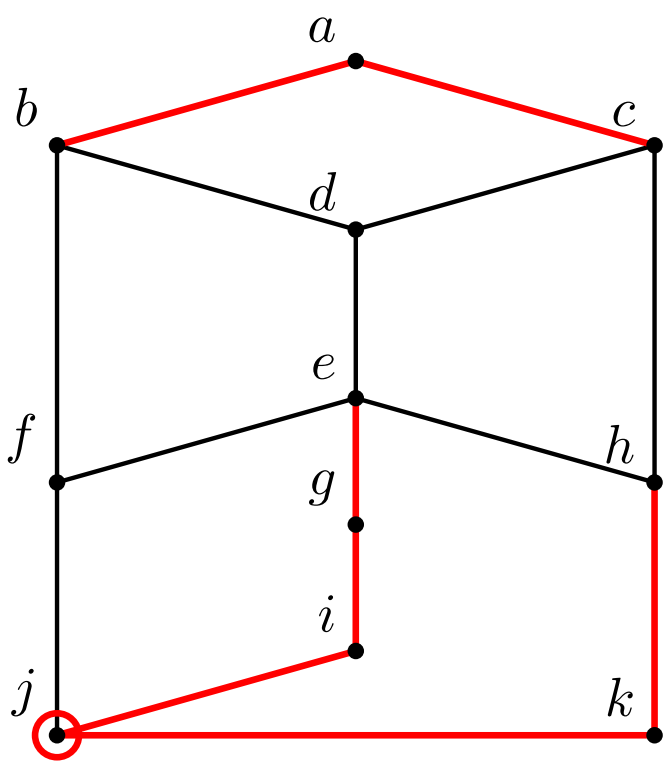
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

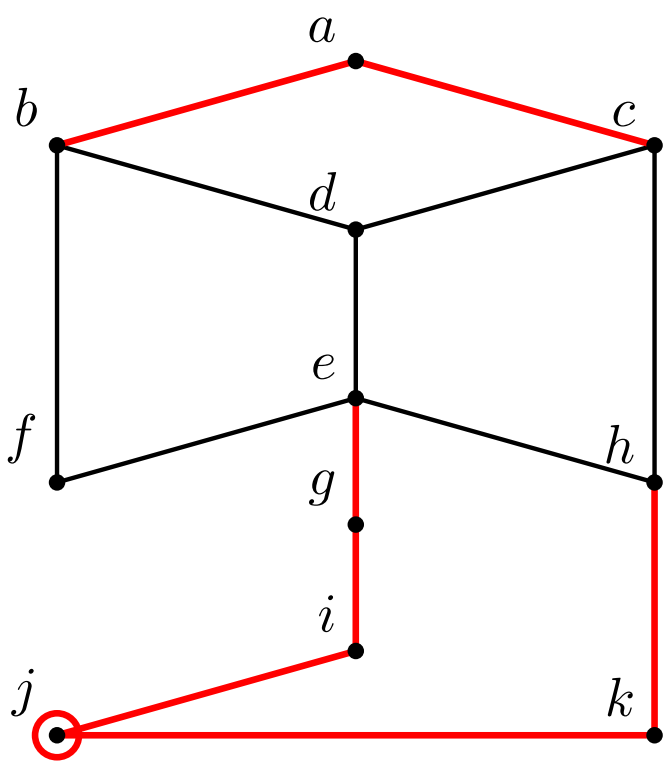
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

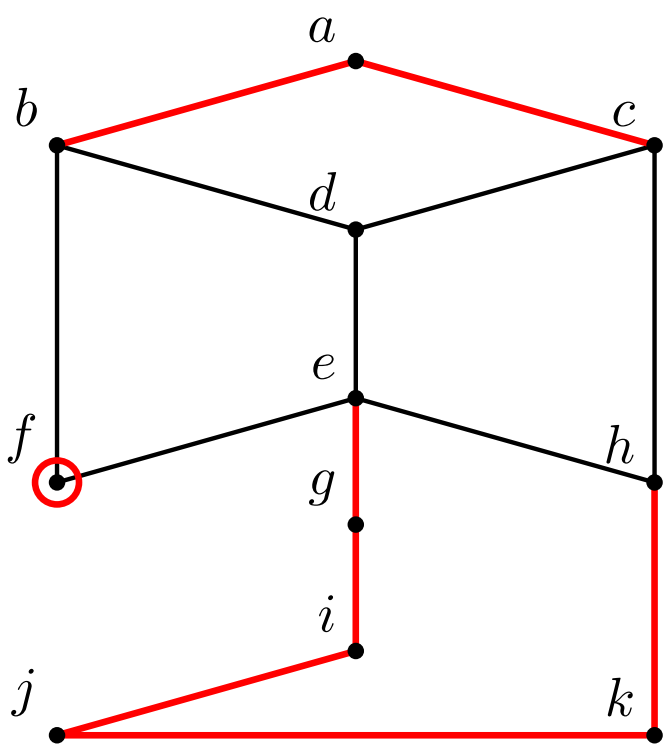
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

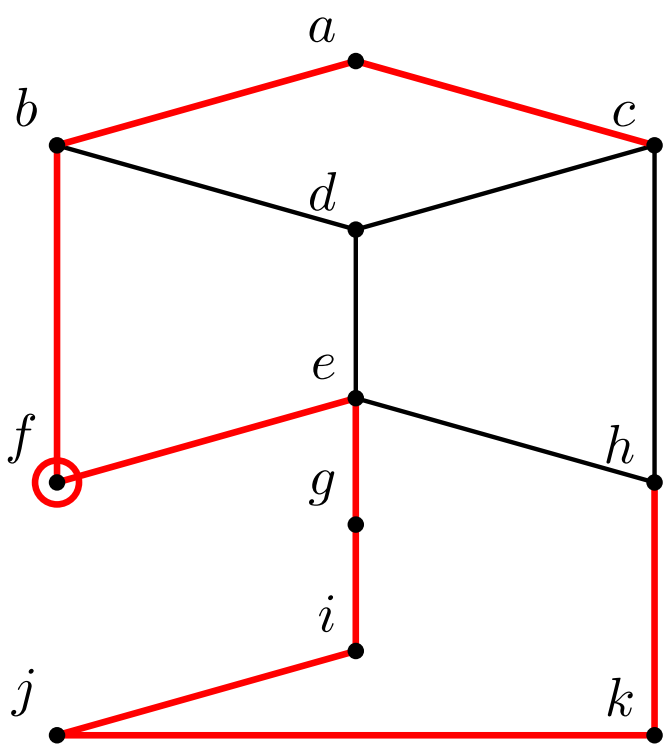
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

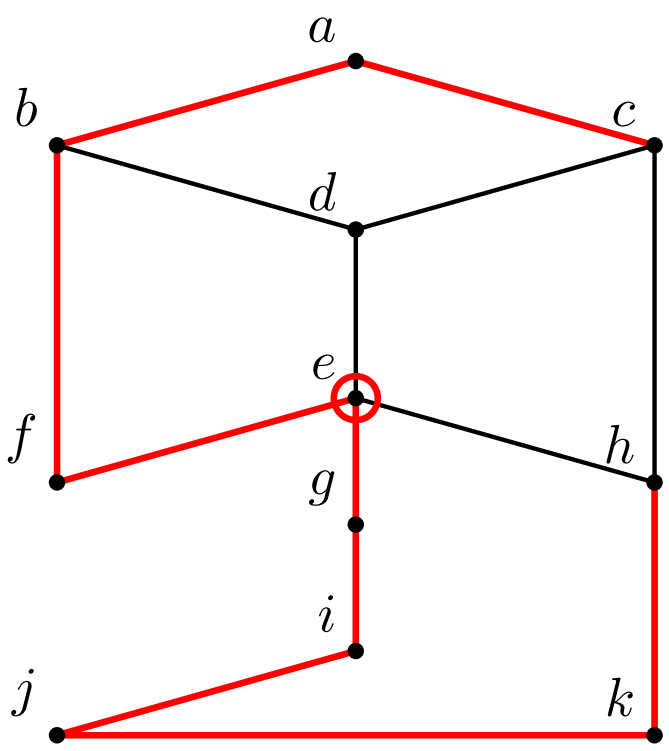
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

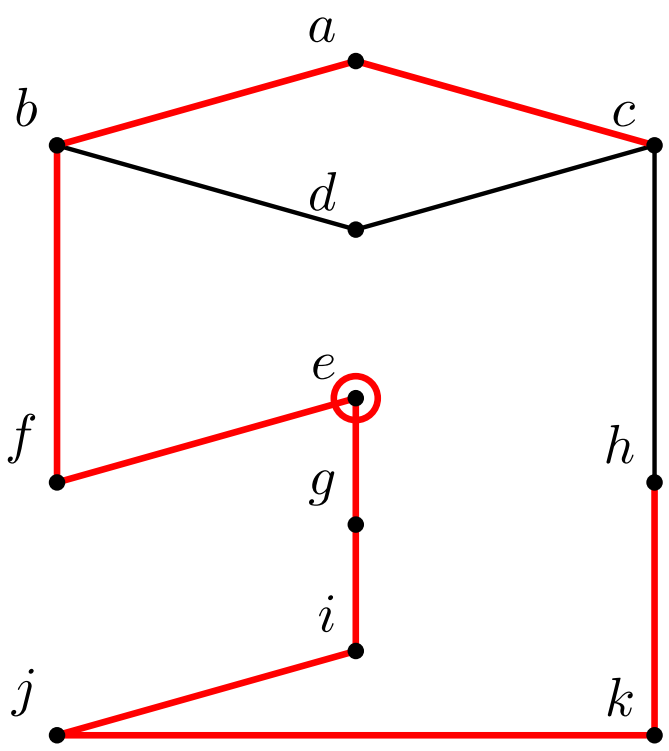
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

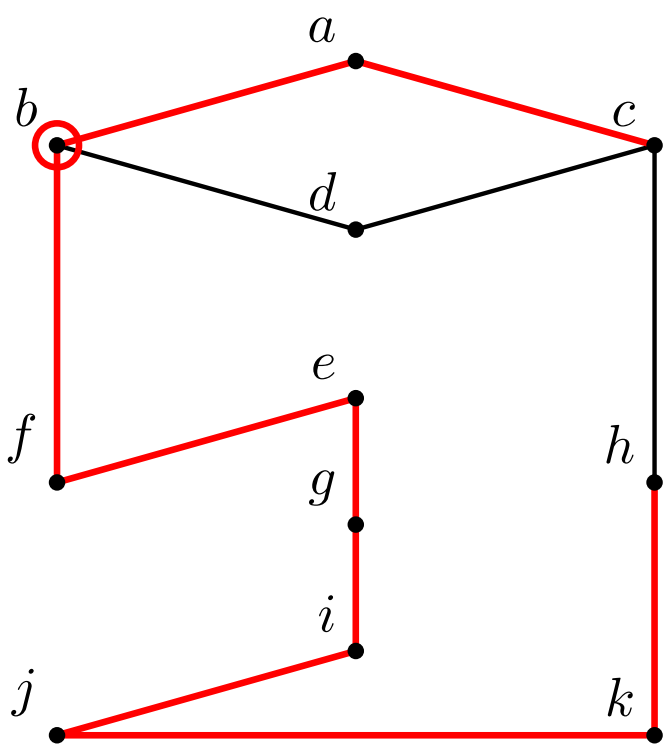
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

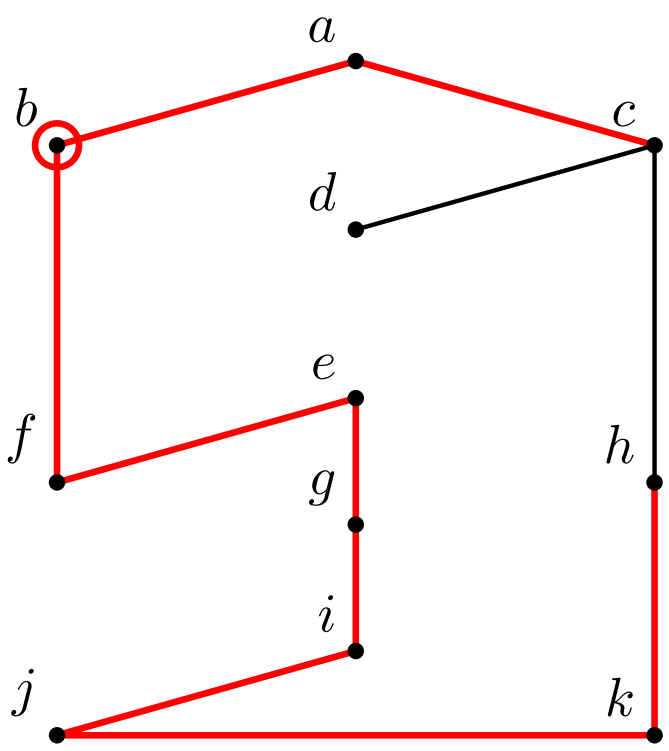
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

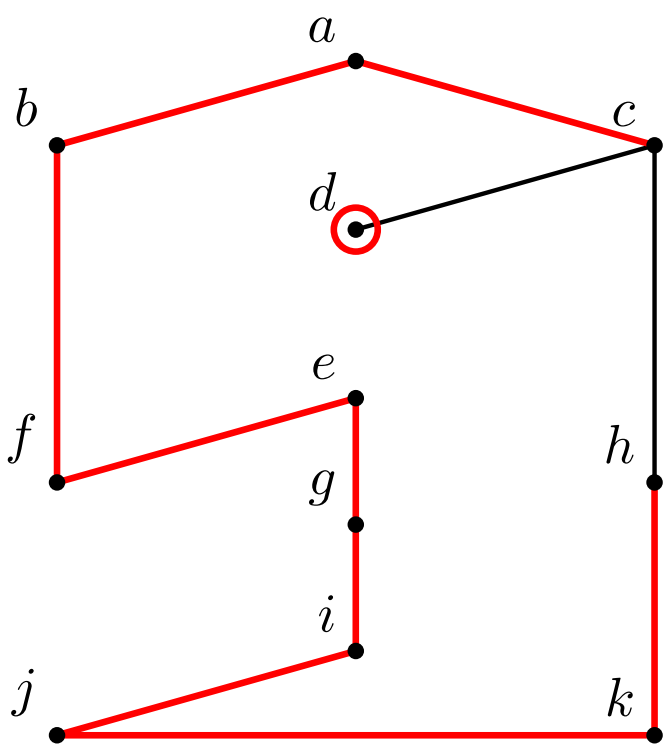
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

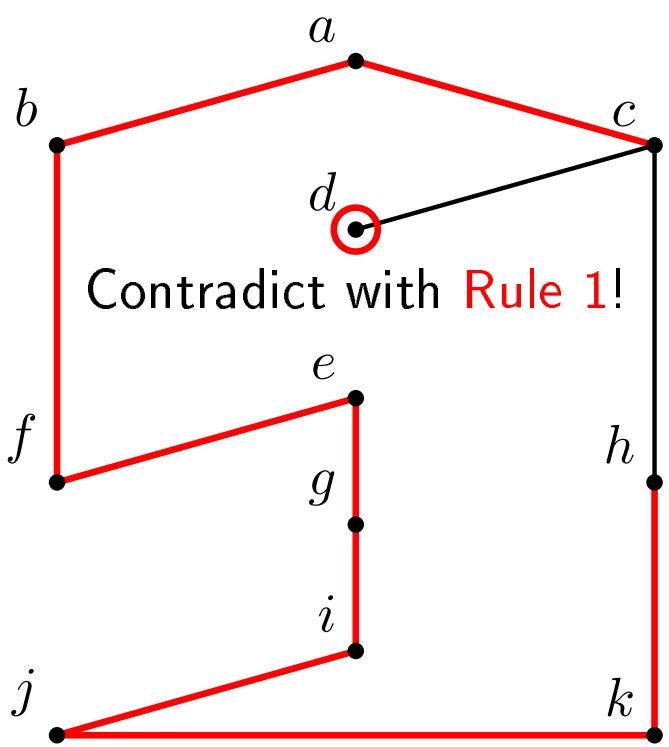
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.






Contents

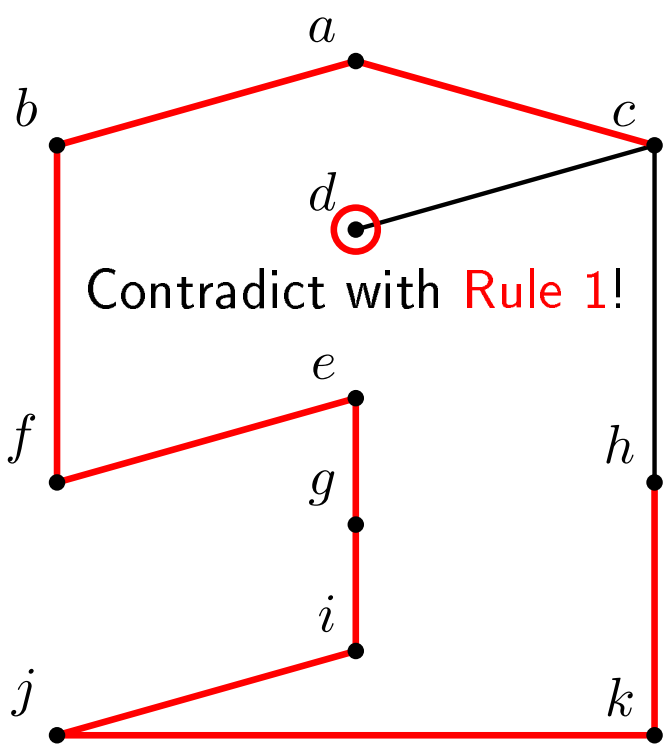
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

We can verify nonexistence of the graph during find Hamilton circuit.




Hamilton circuit doesn't exist!



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Application – Gray Code

**Definition**

The binary sequence that express consecutive numbers by differing just one position of sequence.

| Decimal number | | Binary number | Gray code |
|---|---|---|---|
| 1 | = | 001 | 000 |
| 2 | = | 010 | 100 |
| 3 | = | 011 | 110 |
| 4 | = | 100 | 010 |
| 5 | = | 101 | 011 |
| ⋮ | | ⋮ | ⋮ |

Used at digital communication for reduce the effect of noise; it prevents serious changes of information by noise.



Contents

Connectivity
Paths and Circuits

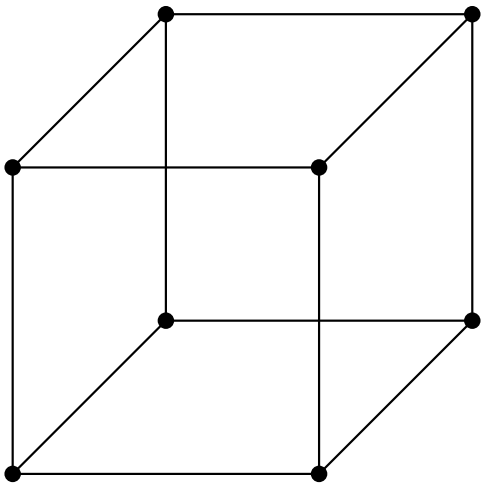
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.





Contents

Connectivity
Paths and Circuits

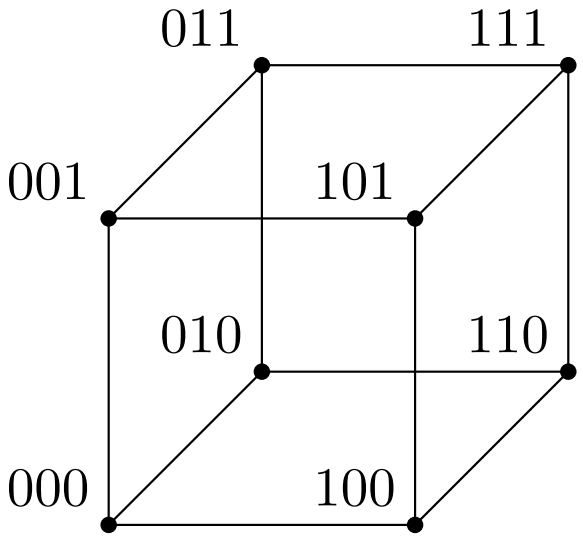
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.






Contents

Connectivity
Paths and Circuits

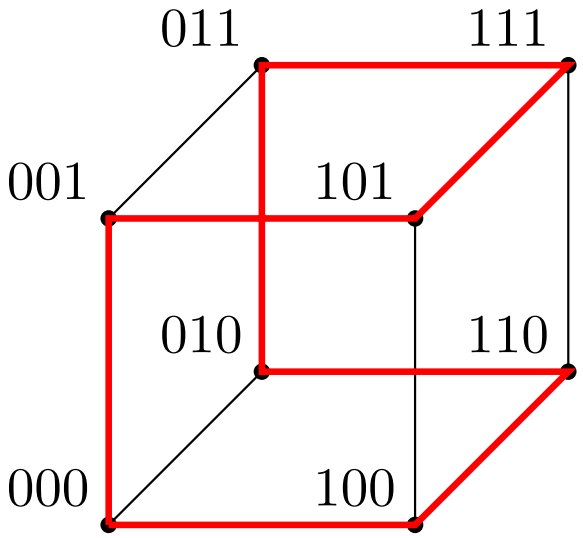
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.






Contents

Connectivity
Paths and Circuits

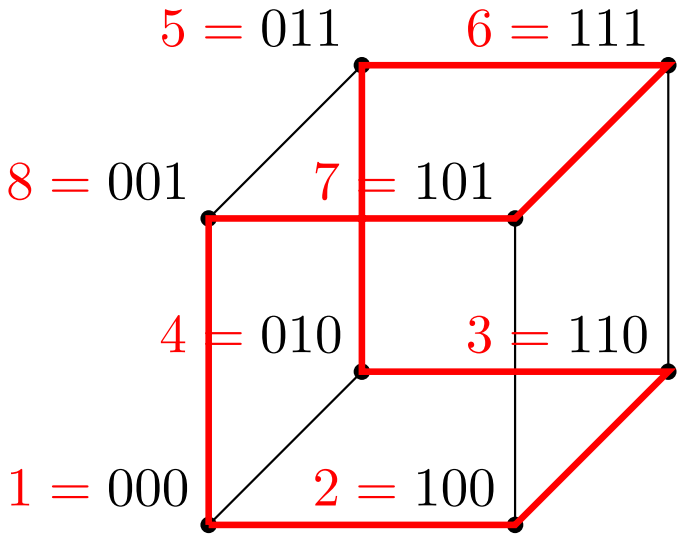
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.







Contents

Connectivity
Paths and Circuits

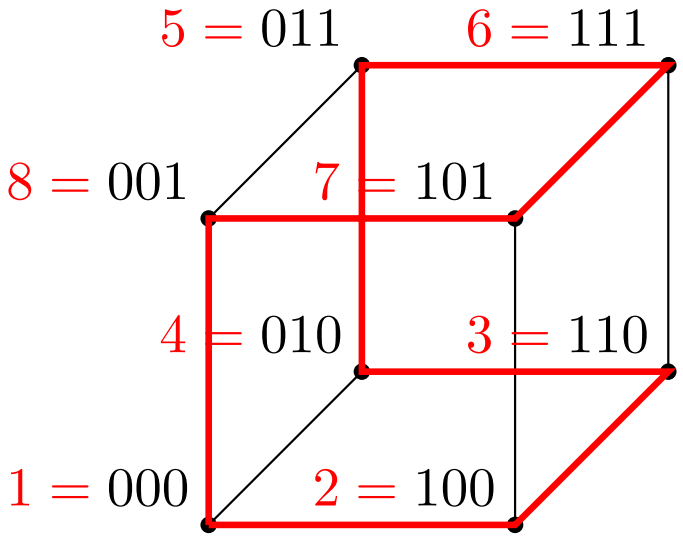
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.







Contents

Connectivity
Paths and Circuits

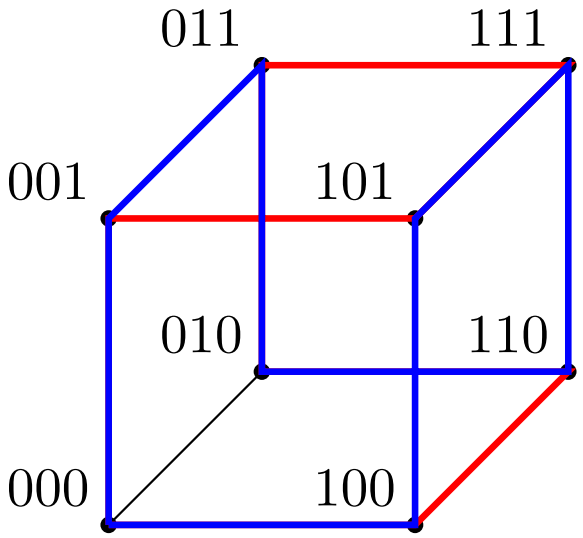
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.






Contents

Connectivity
Paths and Circuits

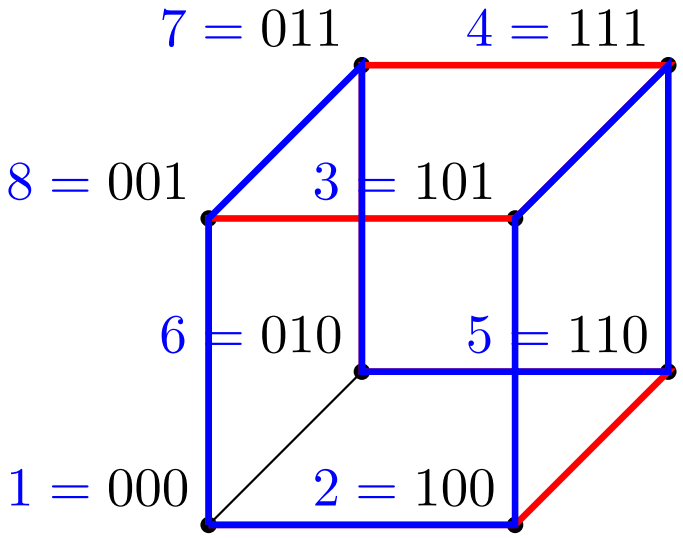
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.







Contents

Connectivity
Paths and Circuits

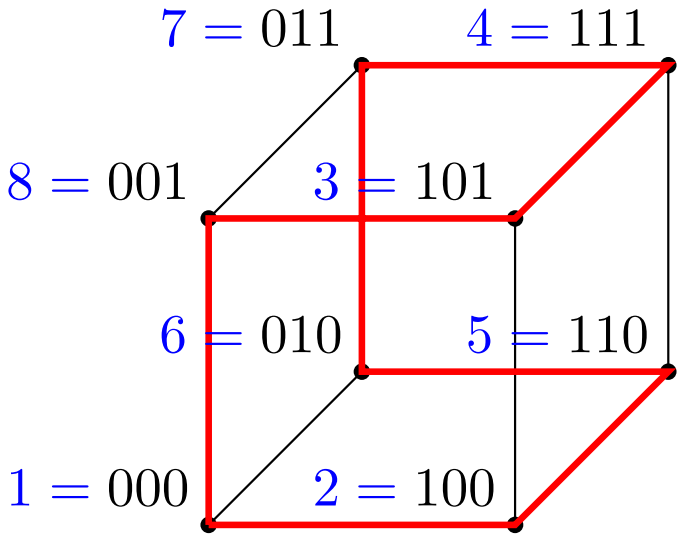
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.







Contents

Connectivity
Paths and Circuits

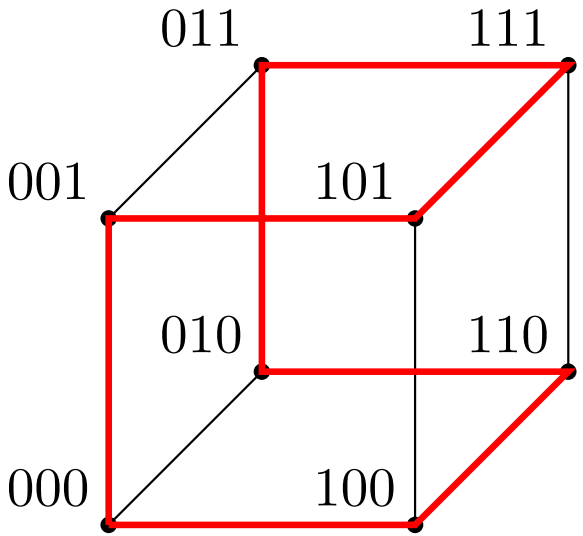
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Gray Code

$n$-digit gray code can be generated by finding Hamilton circuits of $n$-dimensional hypercube!
Consider the case $n = 3$.




Coordinate of each vertex is 3-digit binary sequences.
Coordinates of adjacent vertices differ in just on place.
Hamilton circuits of a cubic graph makes the order of binary sequences!

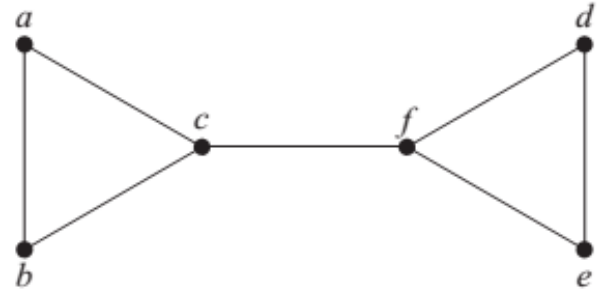


Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

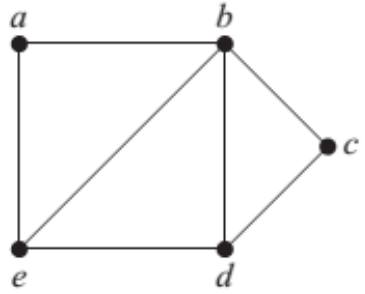
Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Exercise - Hamilton path & circuit

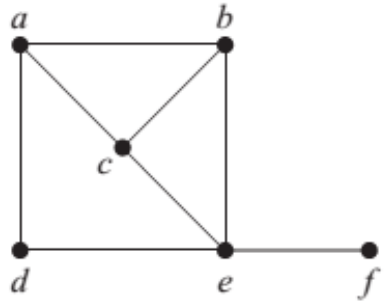
a)



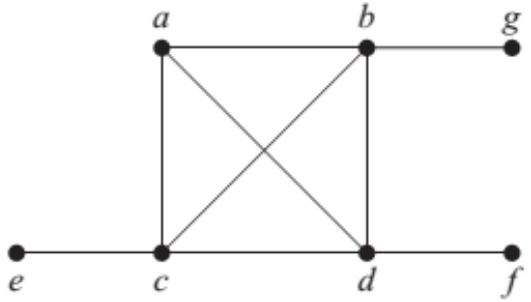

b)



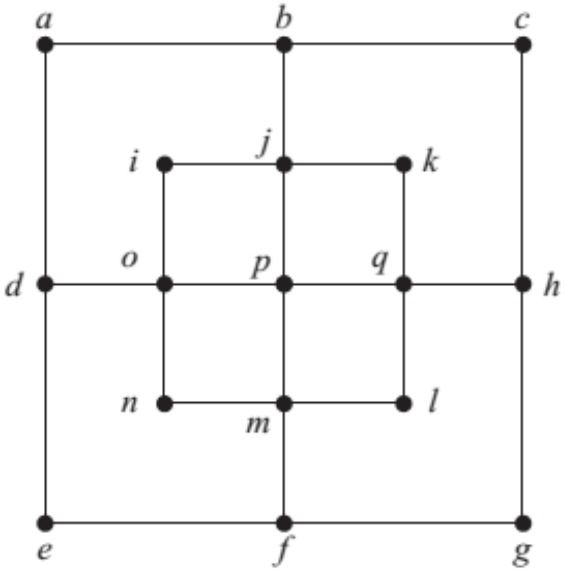

c)



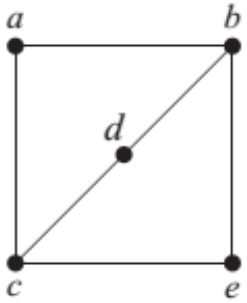

d)




e)




f)






Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

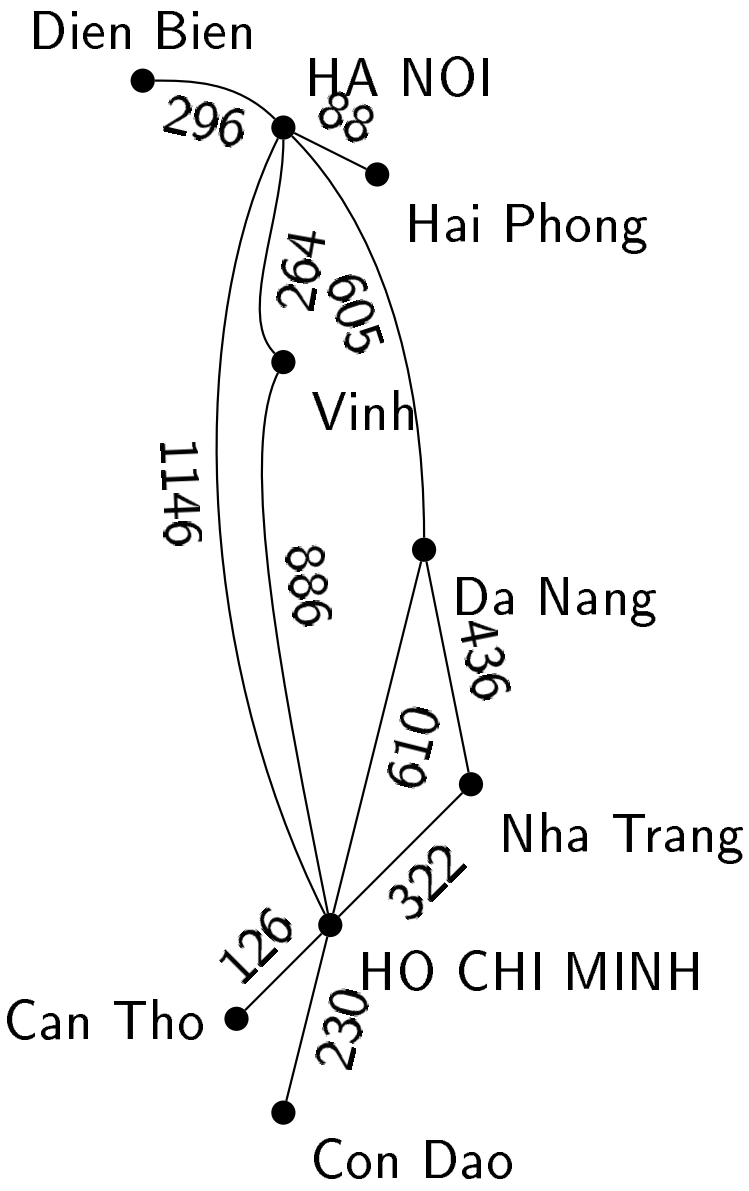
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Weighted Graphs






Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Problem

The problem is also sometimes called the single-pair shortest path problem, to distinguish it from the following generalizations:

- The single-source shortest path problem, in which we have to find shortest paths from a source vertex $v$ to all other vertices in the graph.
- The single-destination shortest path problem, in which we have to find shortest paths from all vertices in the graph to a single destination vertex $v$. This can be reduced to the single-source shortest path problem by reversing the edges in the graph.
- The all-pairs shortest path problem, in which we have to find shortest paths between every pair of vertices $v$, $v'$ in the graph.

These generalizations have significantly more efficient algorithms than the simplistic approach of running a single-pair shortest path algorithm on all relevant pairs of vertices.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Dijkstra's Algorithm

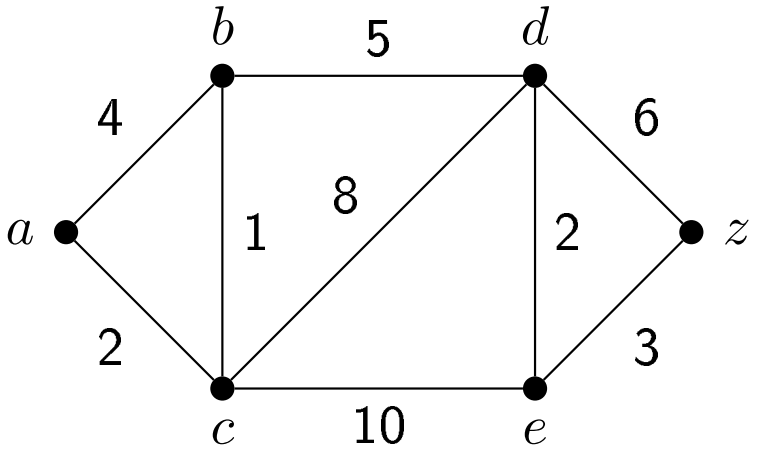
```
procedure Dijkstra(G,a)
// Initialization Step
  forall vertices v
     Label[v] := ∞
     Prev[v] := -1
  endfor
  Label(a) := 0 // a is the source node
  S := ∅

// Iteration Step
  while z ∉ S
     u := a vertex not in S with minimal Label
     S := S ∪ {u}
     forall vertices v not in S
       if (Label[u] + Wt(u,v)) < Label(v)
         then begin
               Label[v] := Label[u] + Wt(u,v)
               Pred[v] := u
             end
  endwhile
```



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents
Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

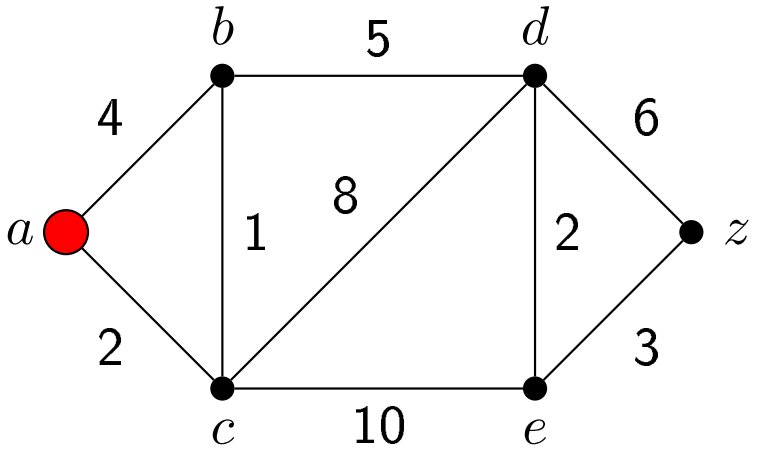
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

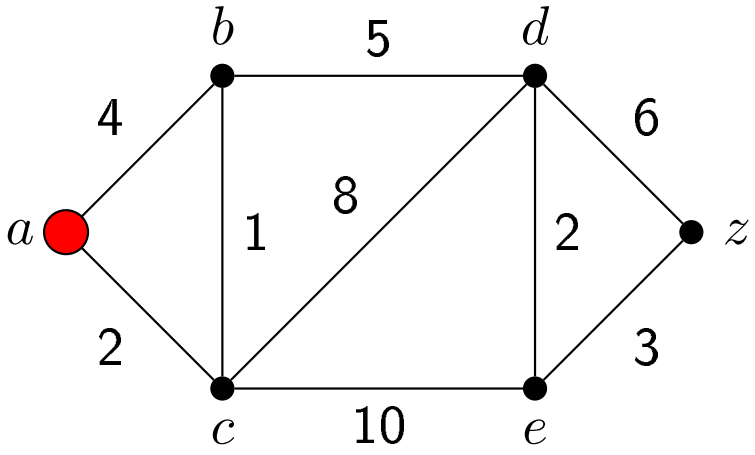
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | | | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

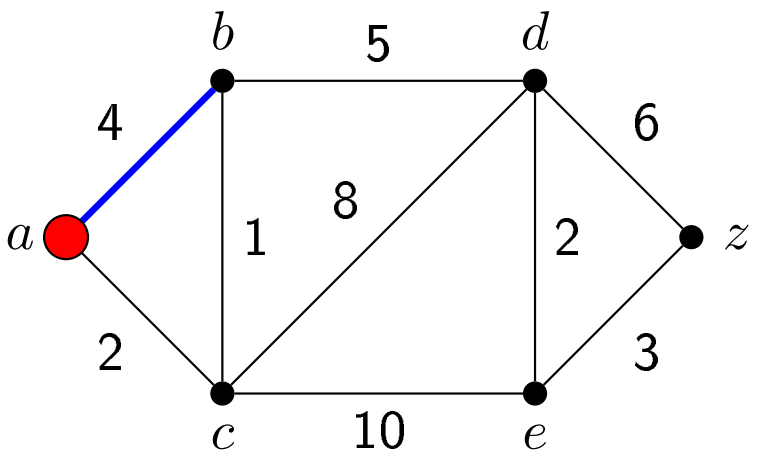
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | | | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

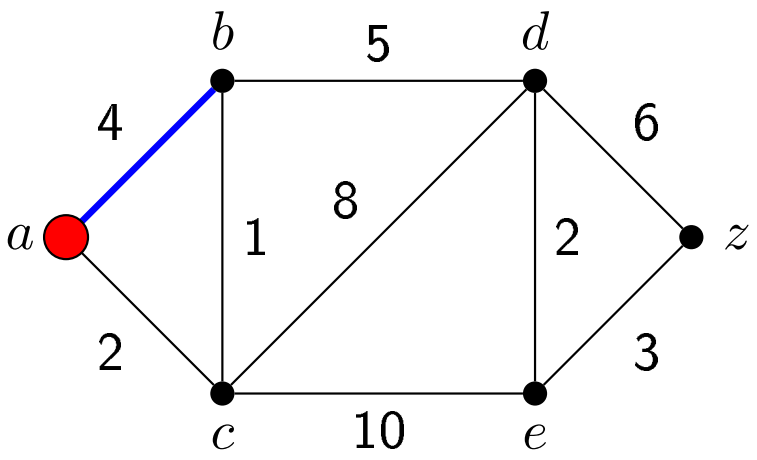
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

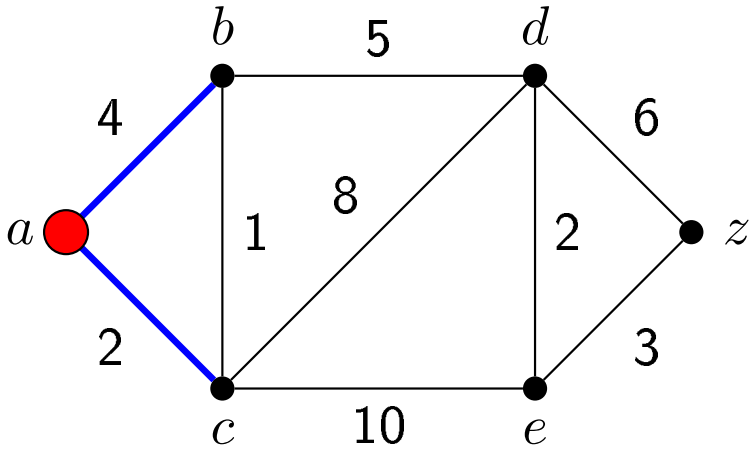
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

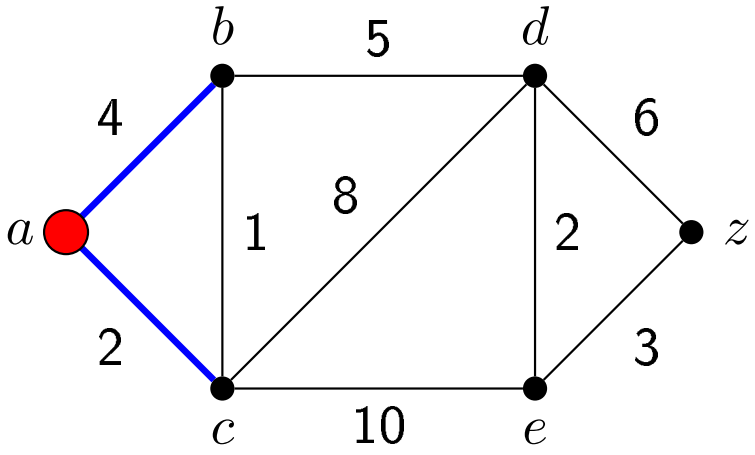
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

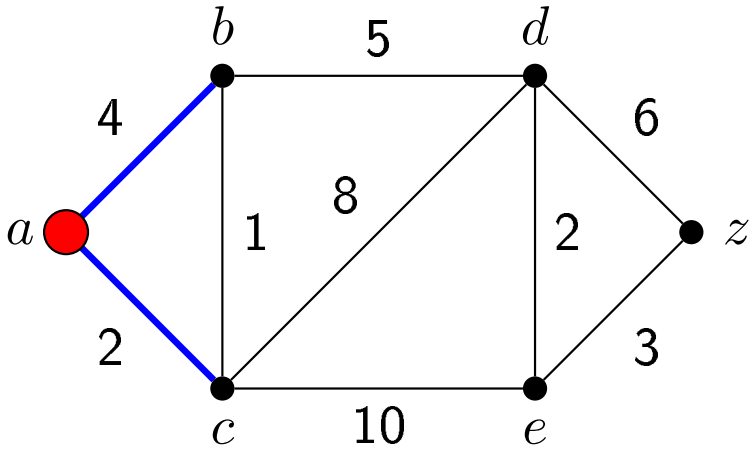
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | 2 | $\infty$ | $\infty$ | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

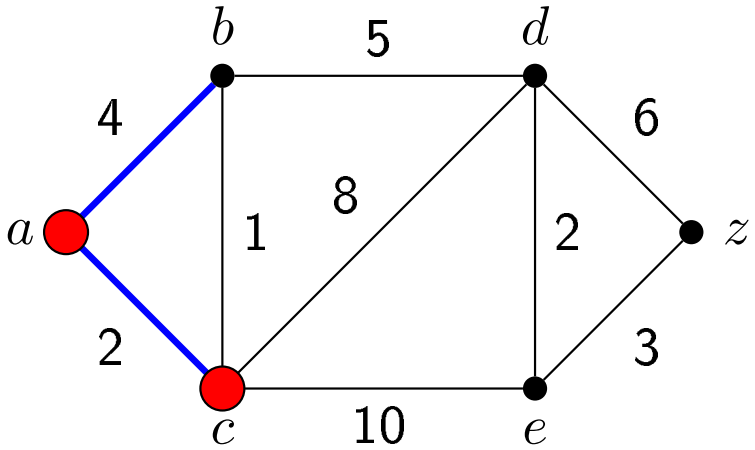
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

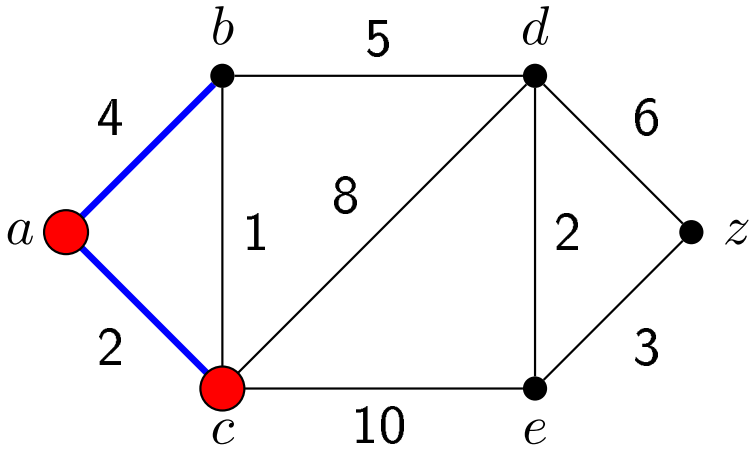
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

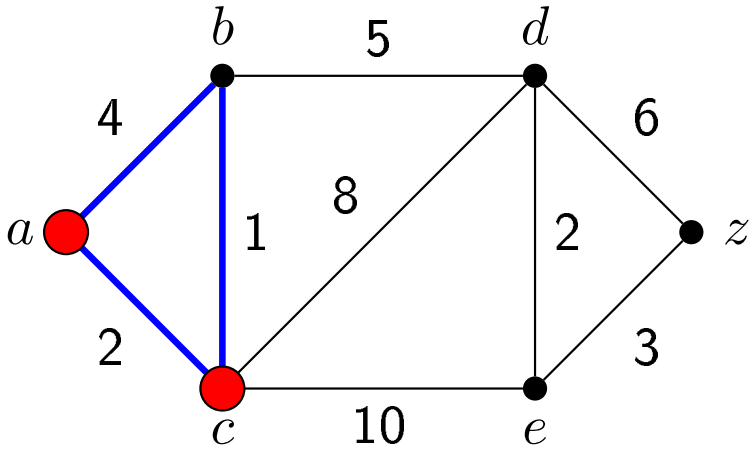
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

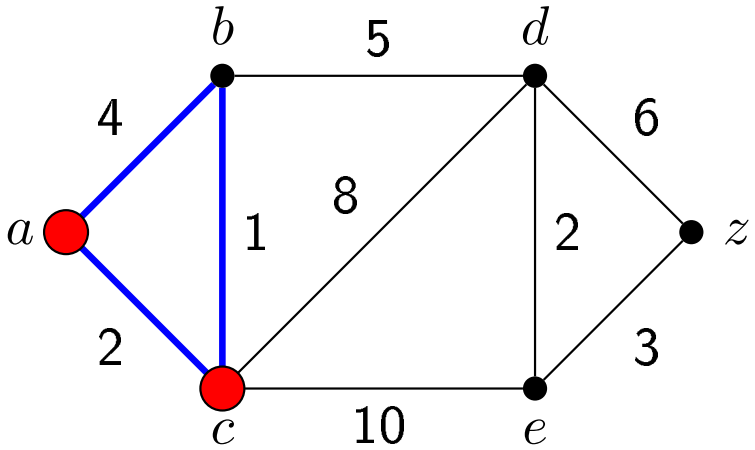
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

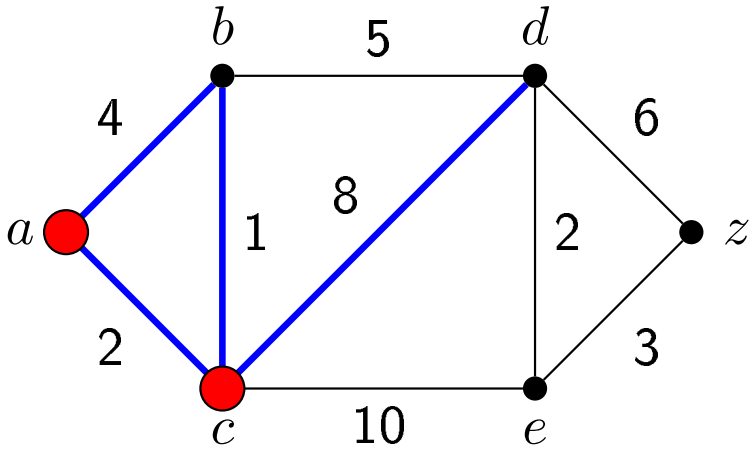
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

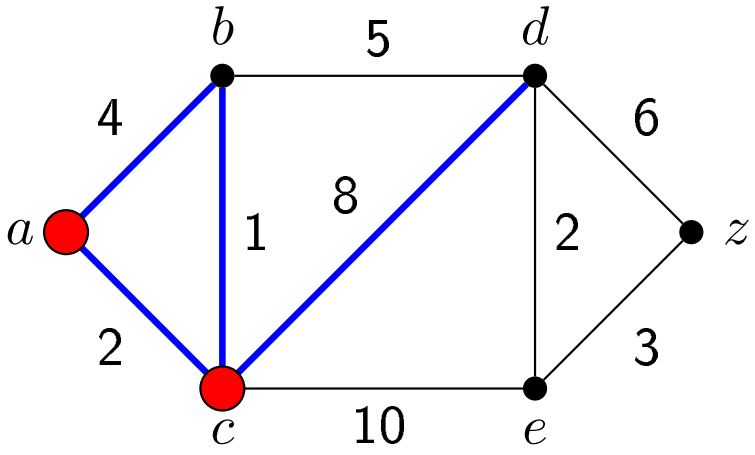
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | 10 | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

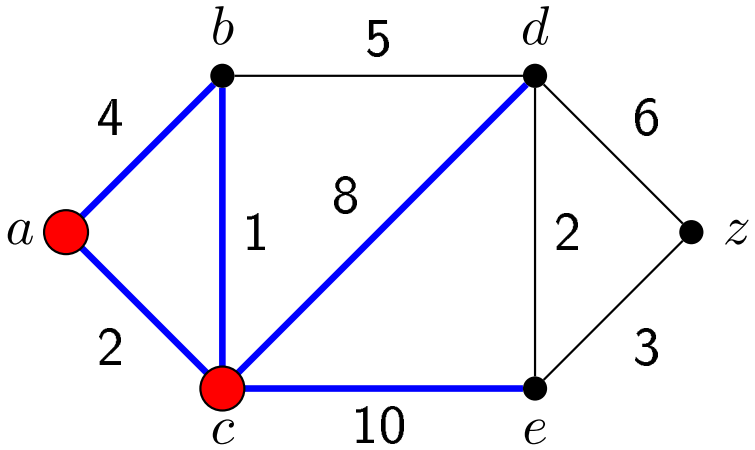
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | 10 | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

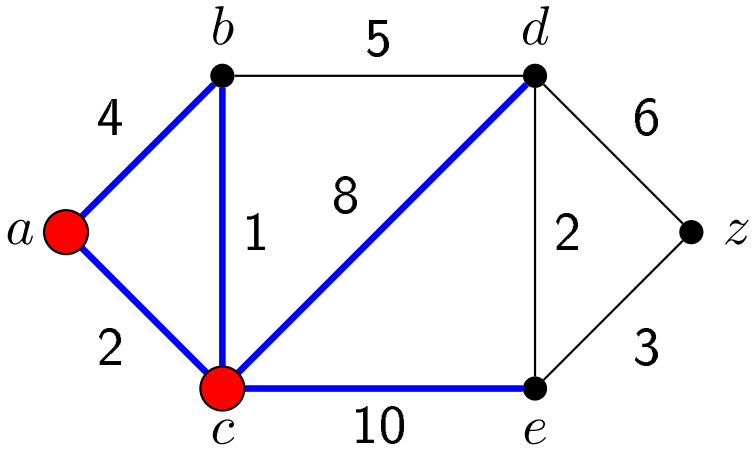
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | 10 | 12 | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

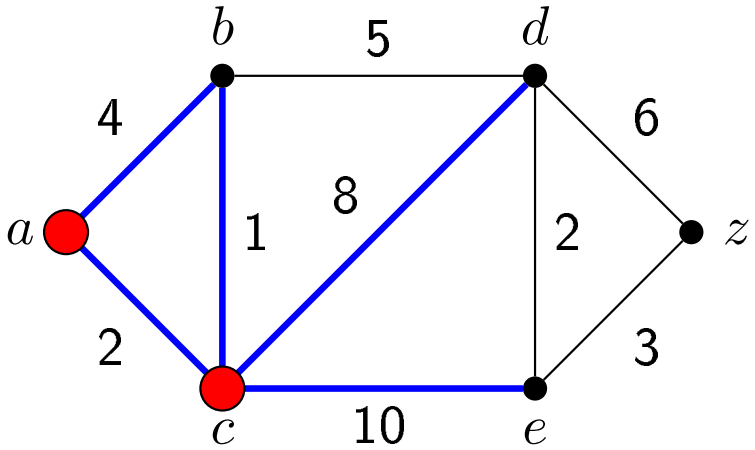
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | 3 | 2 | 10 | 12 | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

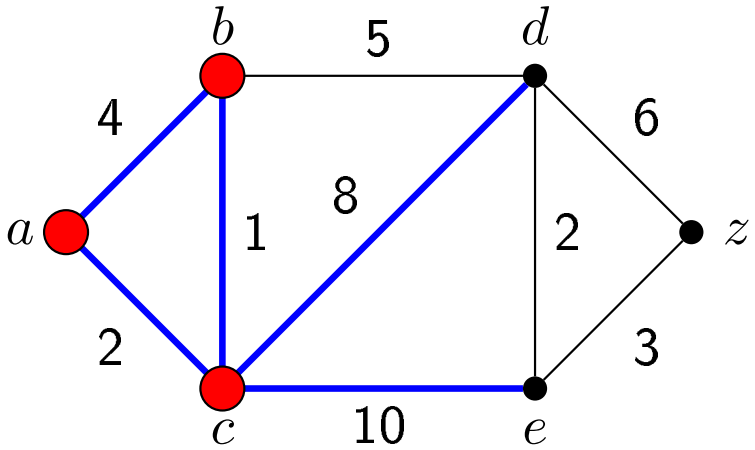
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

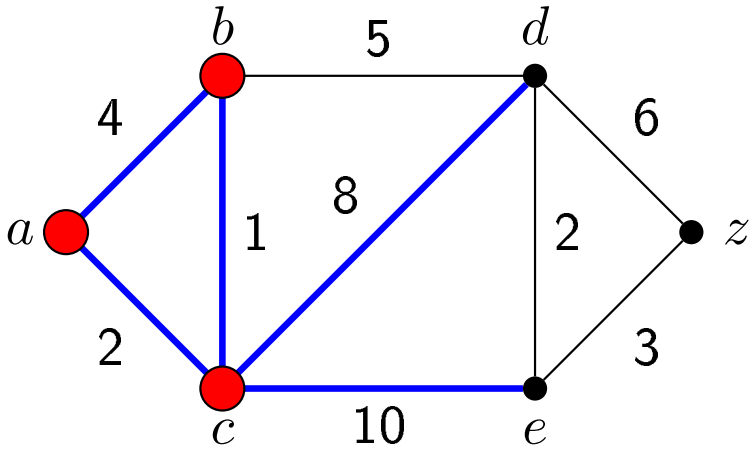
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

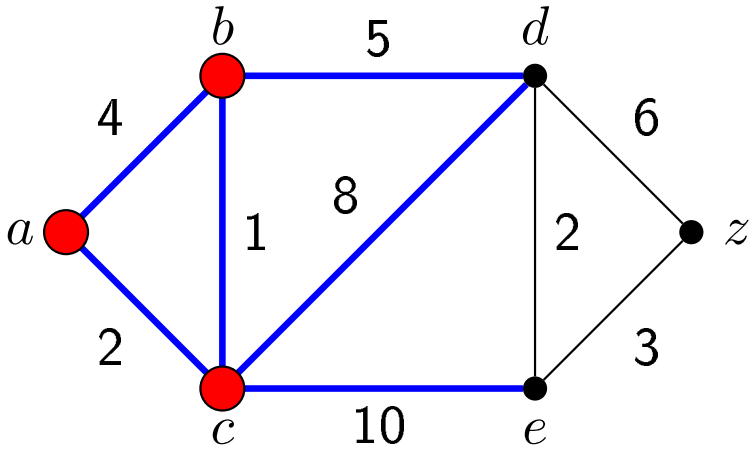
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

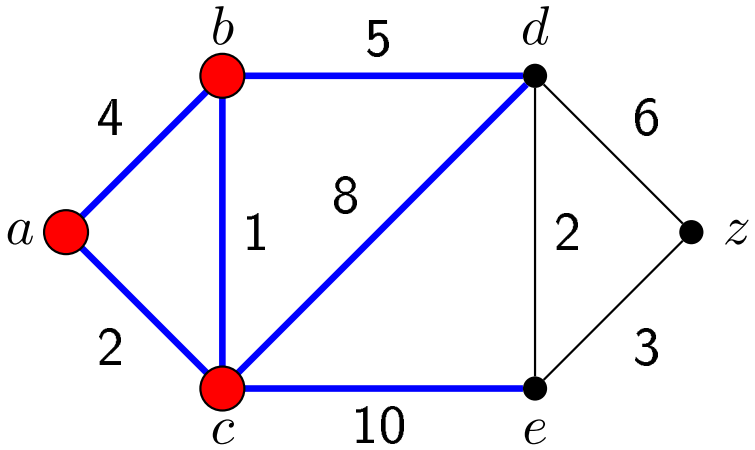
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | 8 | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

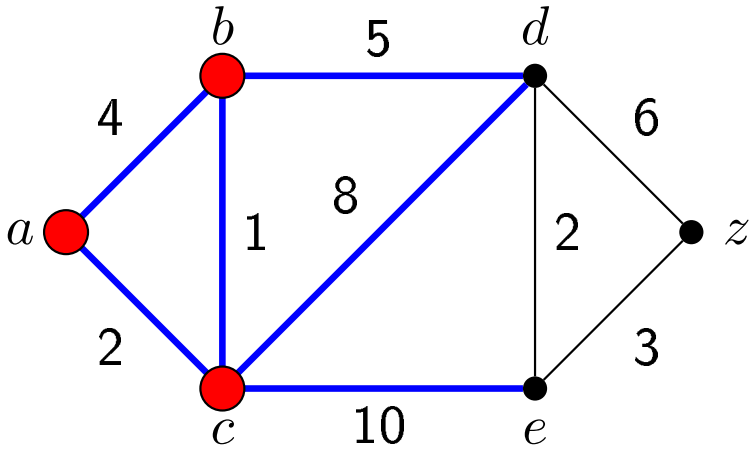
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | 8 | 12 | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

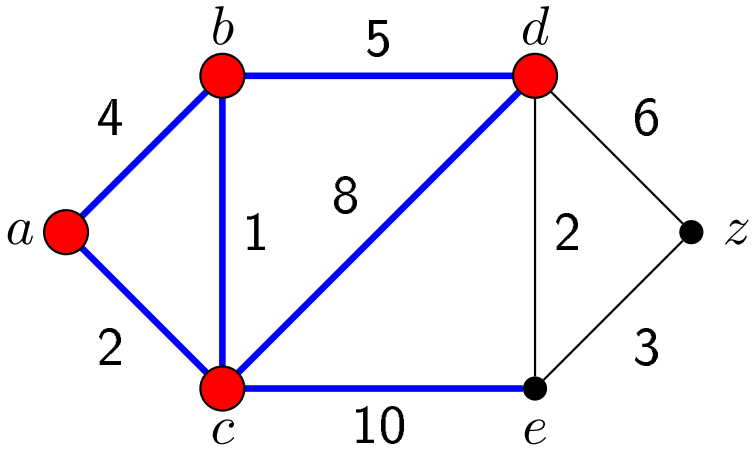
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

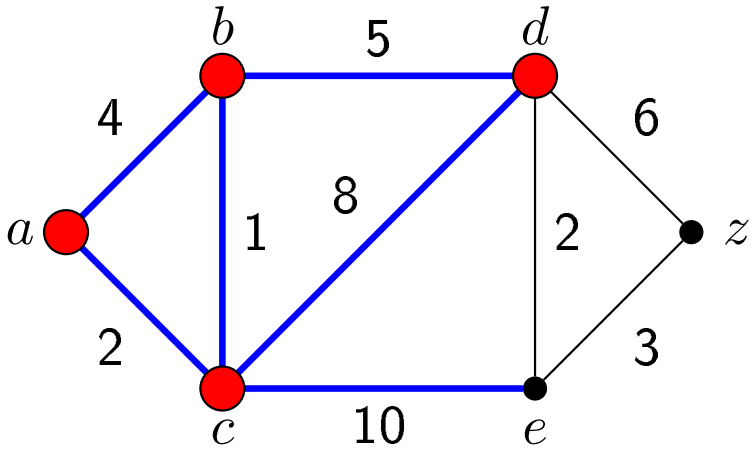
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

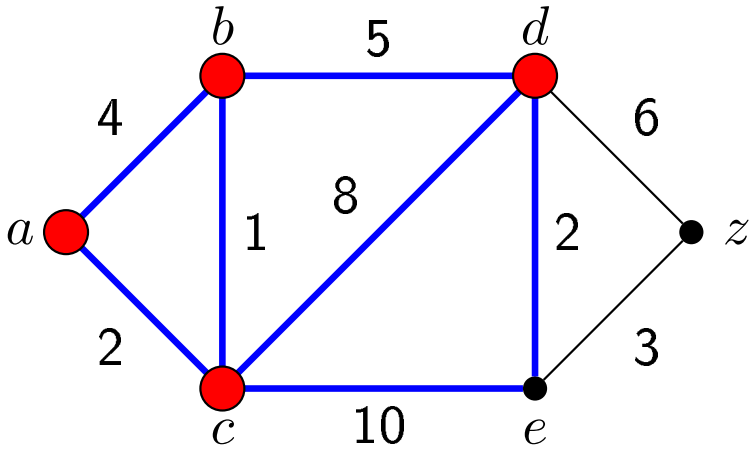
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

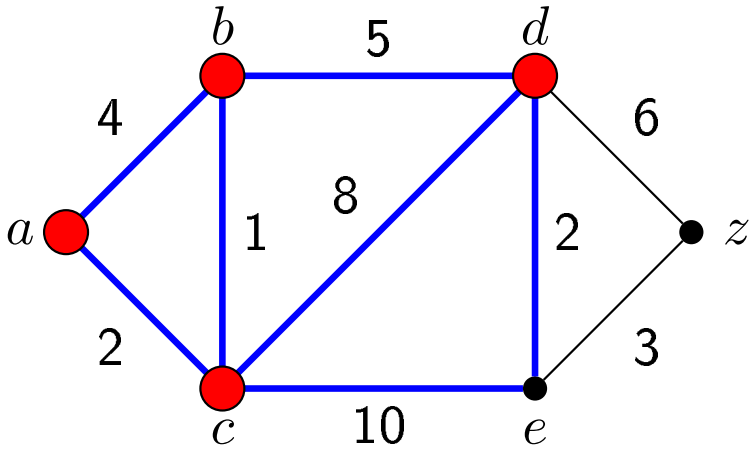
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | 10 | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

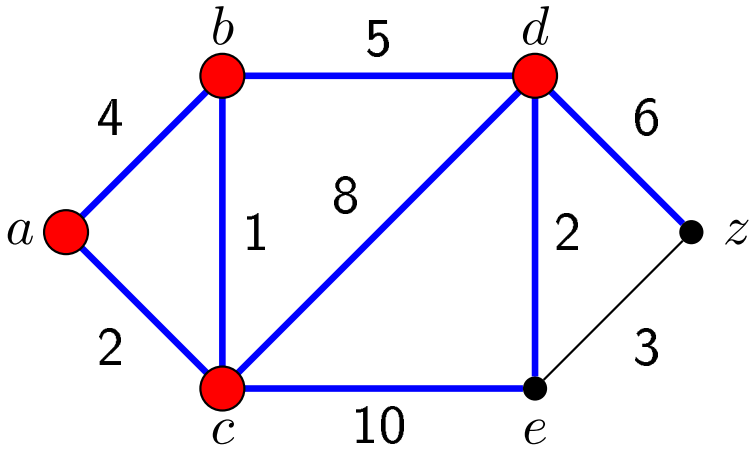
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | 10 | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

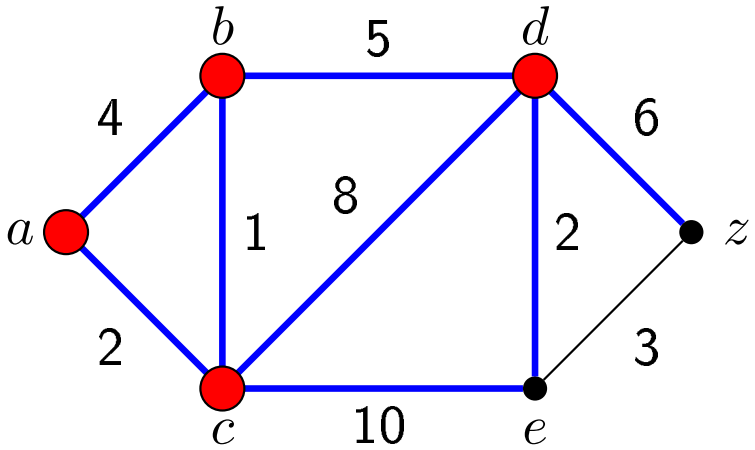
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | 10 | 14 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

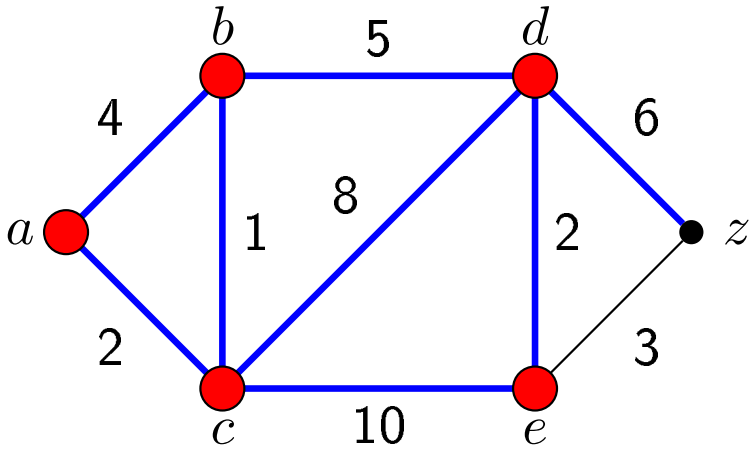
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | **10** | 14 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

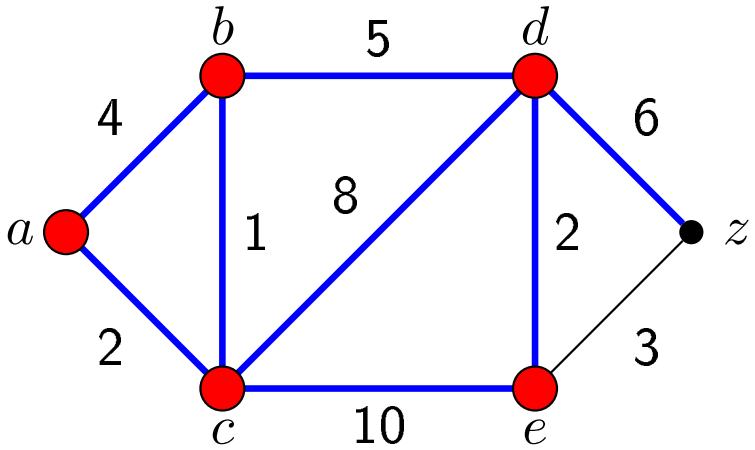
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | **10** | 14 |
| $e$ | 0 | 3 | 2 | 8 | 10 | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

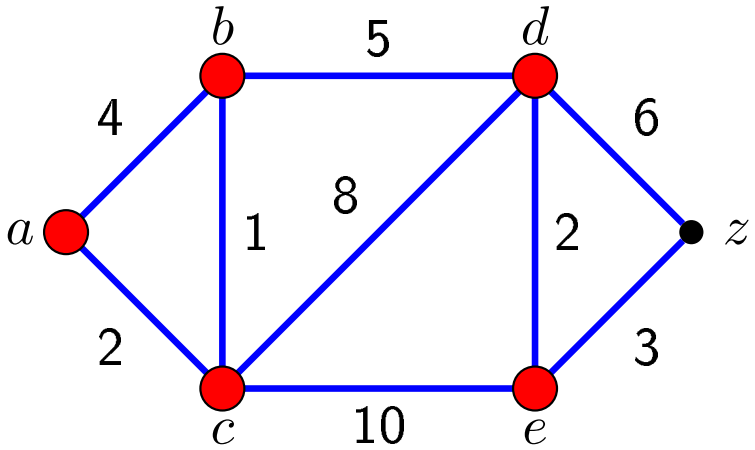
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | **10** | 14 |
| $e$ | 0 | 3 | 2 | 8 | 10 | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

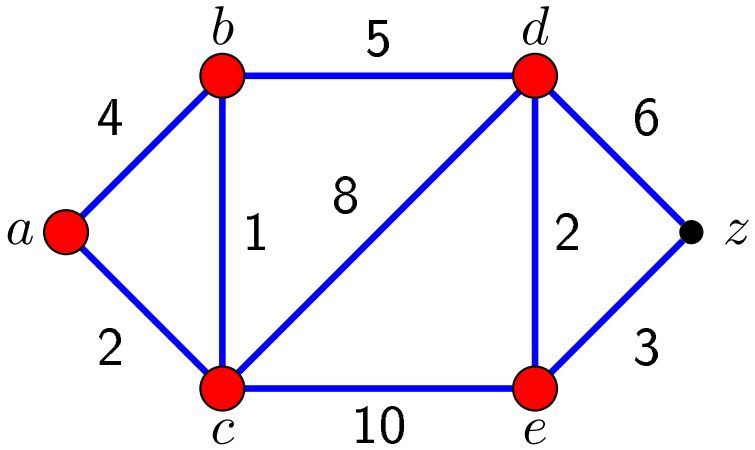
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | **10** | 14 |
| $e$ | 0 | 3 | 2 | 8 | 10 | 13 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

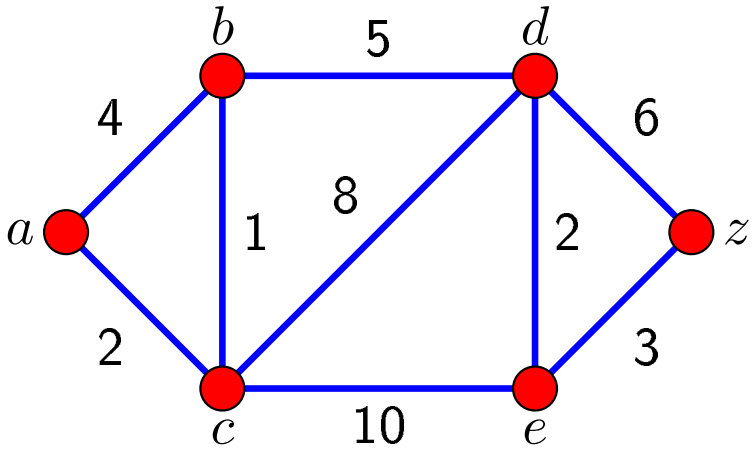
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | **10** | 14 |
| $e$ | 0 | 3 | 2 | 8 | 10 | **13** |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

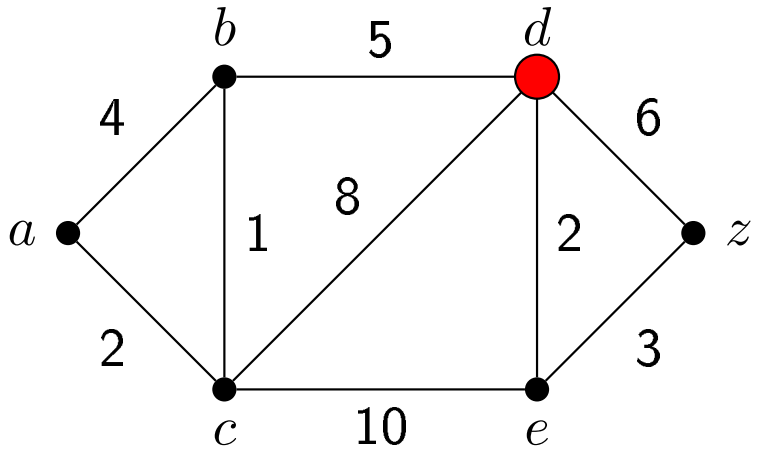
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | $\underline{8}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | 8 | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

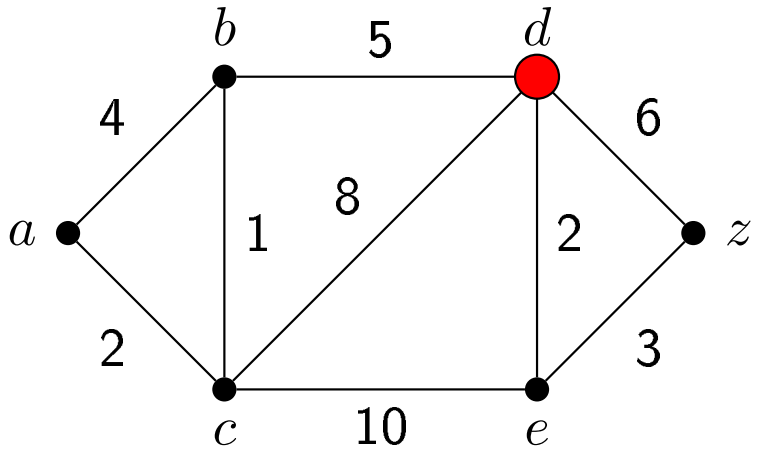
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | $\underline{8}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | 8 | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

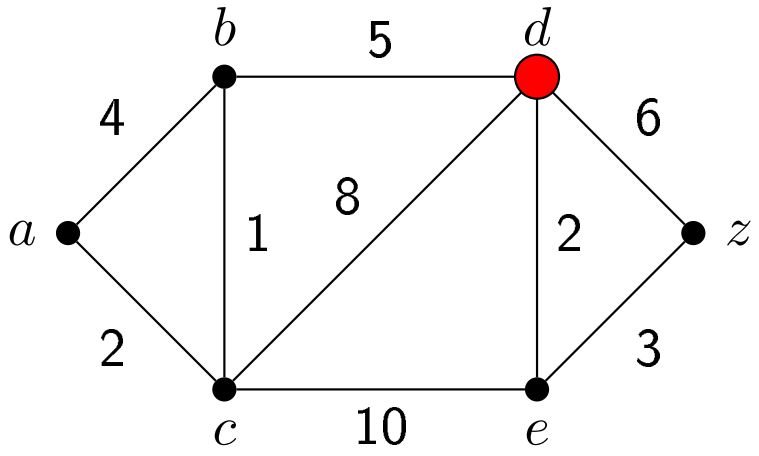
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | $\underline{8}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

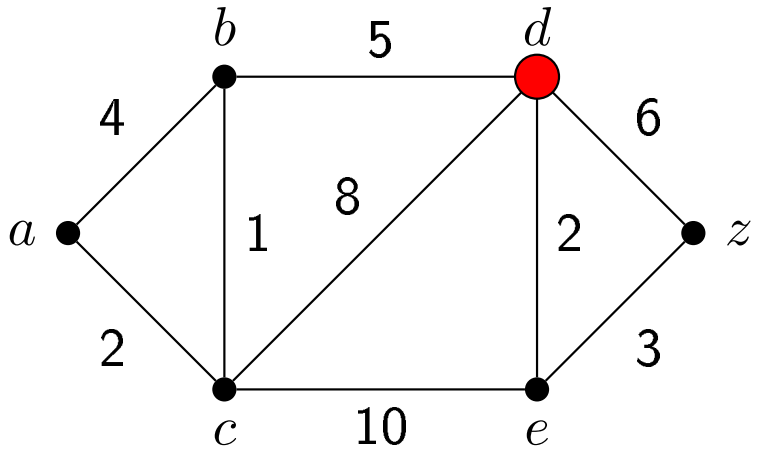
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

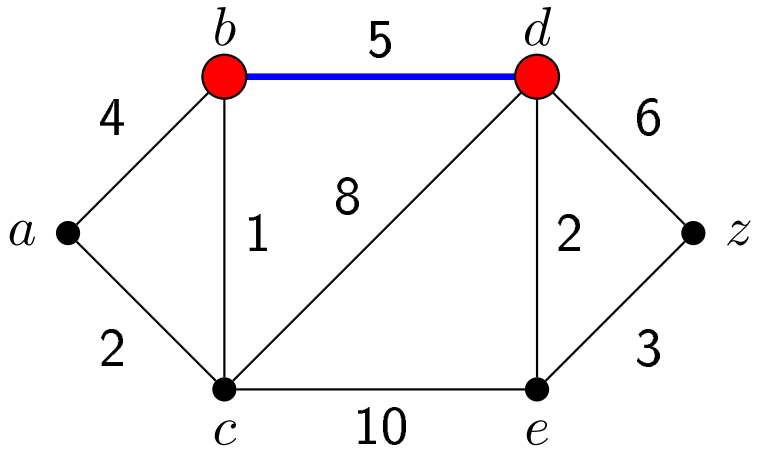
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | 3 | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

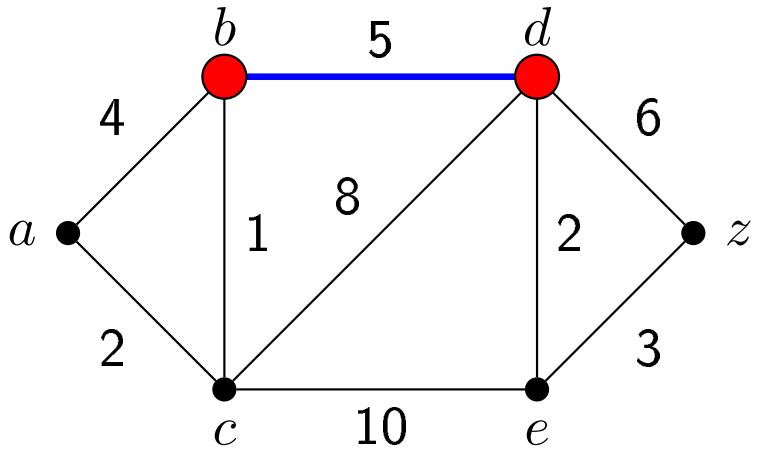
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{3}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | **3** | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

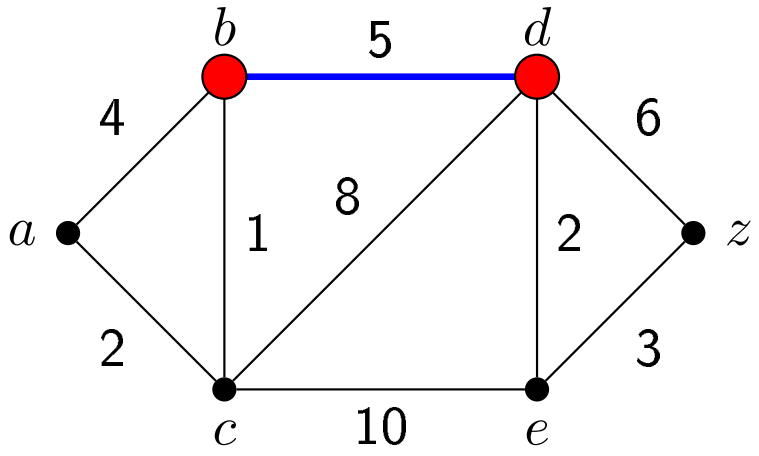
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{\mathbf{3}}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | **3** | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

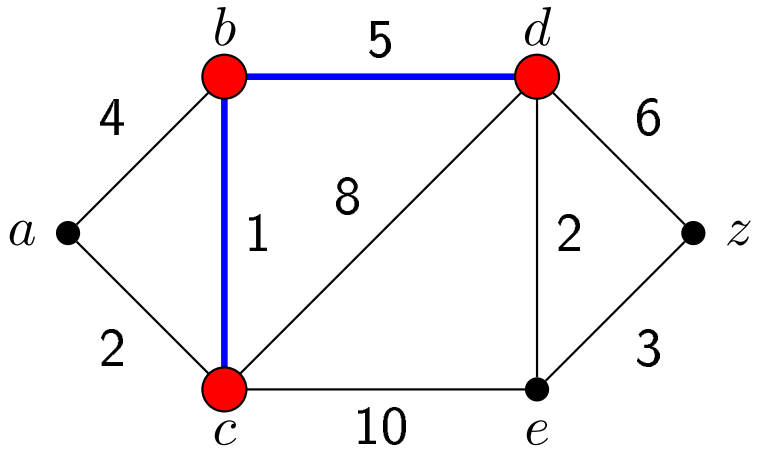
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{2}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{\mathbf{3}}$ | 2 | 10 | 12 | $\infty$ |
| $b$ | 0 | **3** | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

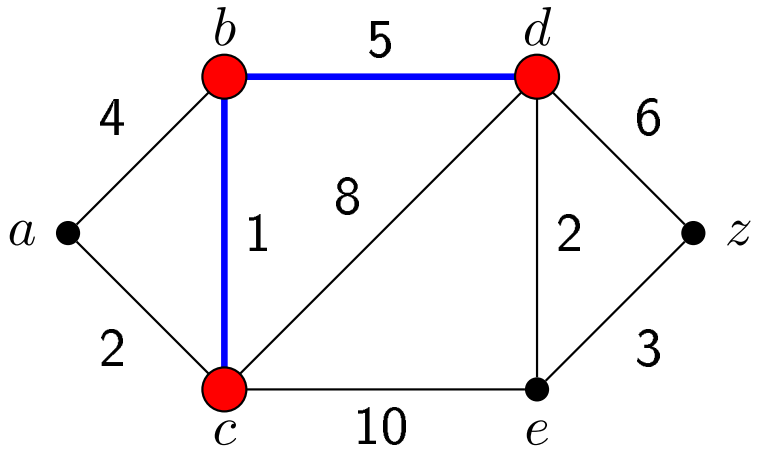
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | $\underline{0}$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | 0 | 4 | $\underline{\mathbf{2}}$ | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | $\underline{\mathbf{3}}$ | **2** | 10 | 12 | $\infty$ |
| $b$ | 0 | **3** | 2 | $\underline{\mathbf{8}}$ | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | $\underline{10}$ | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | $\underline{13}$ |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

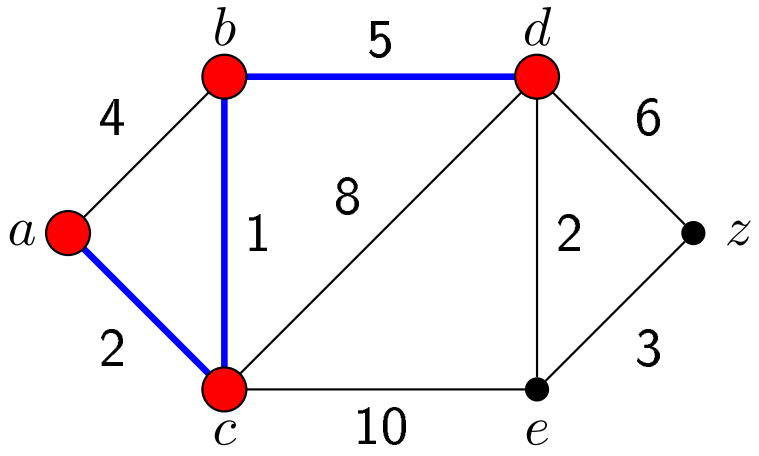
Ford's algorithm

Others

Graph Coloring

# Back tracking procedure

How to determine shortest path from $a$ to $d$ according to Dijkstra's algorithm?




| $S$ | $a$ | $b$ | $c$ | $d$ | $e$ | $z$ |
|---|---|---|---|---|---|---|
| $\emptyset$ | **0** | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| $a$ | **0** | 4 | **2** | $\infty$ | $\infty$ | $\infty$ |
| $c$ | 0 | **3** | **2** | 10 | 12 | $\infty$ |
| $b$ | 0 | **3** | 2 | **8** | 12 | $\infty$ |
| $d$ | 0 | 3 | 2 | **8** | 10 | 14 |
| $e$ | 0 | 3 | 2 | **8** | 10 | 13 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Dijkstra's Algorithm

**Property**

Applicable for any $G$, any length $\ell(v_i) \geq 0$, $\forall i$; one-to-all; complexity $O(|V|^2)$.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

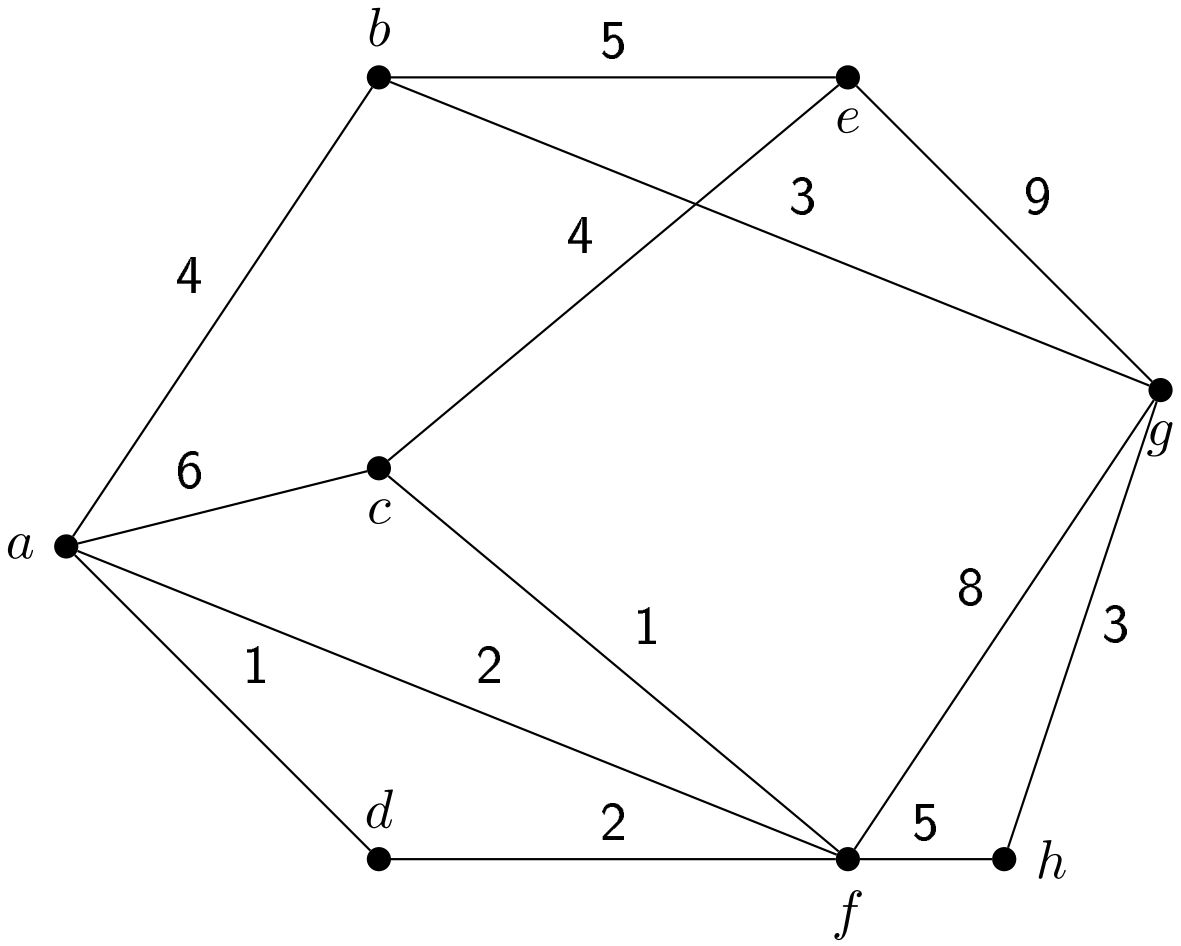
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Exercise

## Example

Find the shortest path from $a$ to other vertices using Dijkstra's algorithm.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

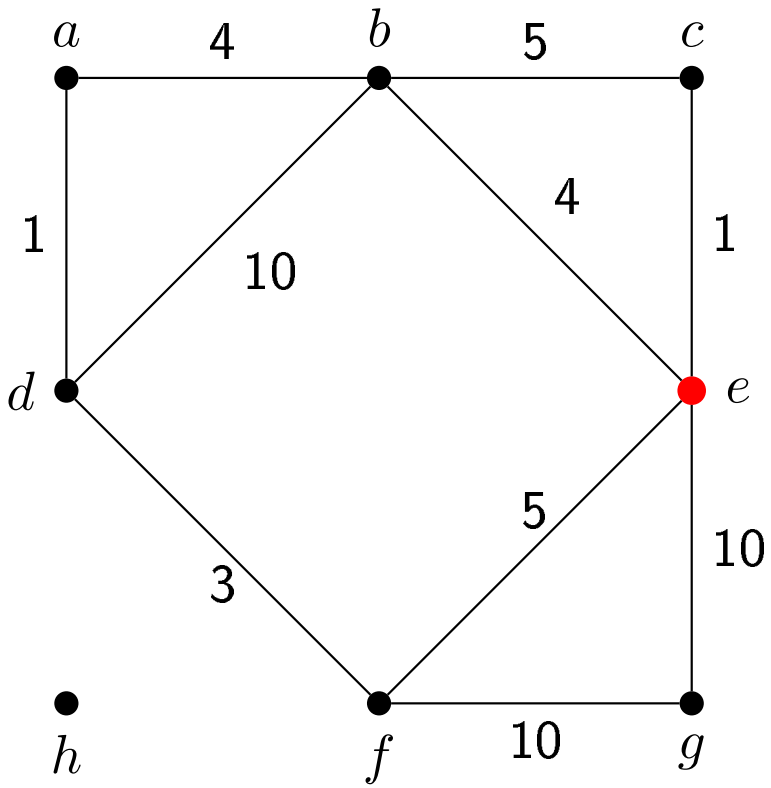
Others

Graph Coloring

# Exercise

## Example

Find the shortest path from $e$ to other vertices using Dijkstra's algorithm.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

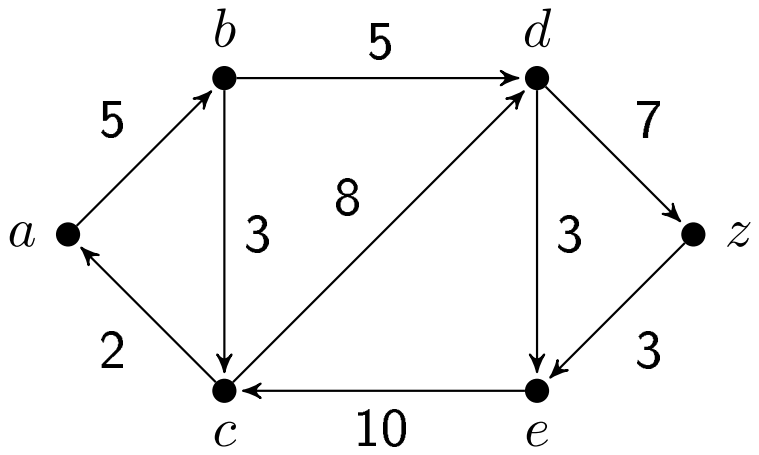
Others

Graph Coloring

# Exercise

## Example

Find the shortest path from $a$ to other vertices using Dijkstra's algorithm.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

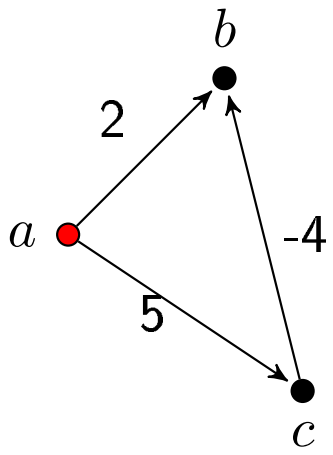
Others

Graph Coloring

# Dijkstra's Algorithm Flaw

## Can Dijkstra's Algorithm be used on...

- ...digraph?
  - Yes!
- ...negative weighted graph?
  - No! Why?







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Bellman-Ford Algorithm

```
procedure BellmanFord(G,a)
// Initialization Step
  forall vertices v
     Label[v] := ∞
     Prev[v] := -1
  Label(a) := 0 // a is the source node

// Iteration Step
  for i from 1 to size(vertices)-1
    forall vertices v
      if (Label[u] + Wt(u,v)) < Label[v]
        then
          Label[v] := Label[u] + Wt(u,v)
          Prev[v] := u

// Check circuit of negative weight
     forall vertices v
       if (Label[u] + Wt(u,v)) < Label(v)
         error "Contains circuit of negative weight"
```

**Property**

any $G$, any weighted; one-to-all; detect whether there exists a circle of negative length; complexity $O(|V| \times |E|)$.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

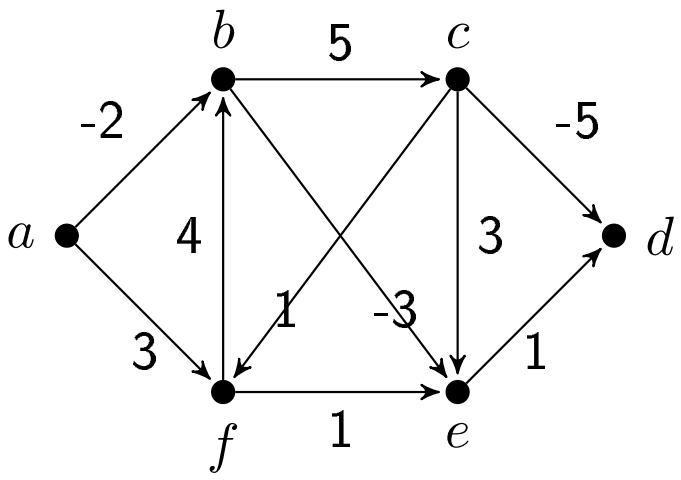
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

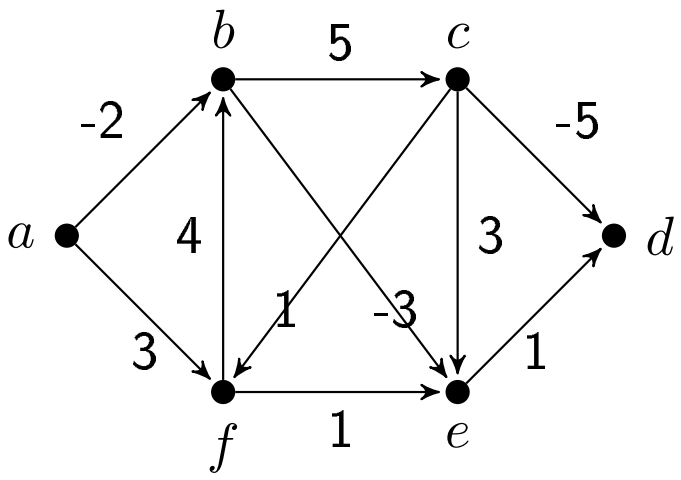
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |





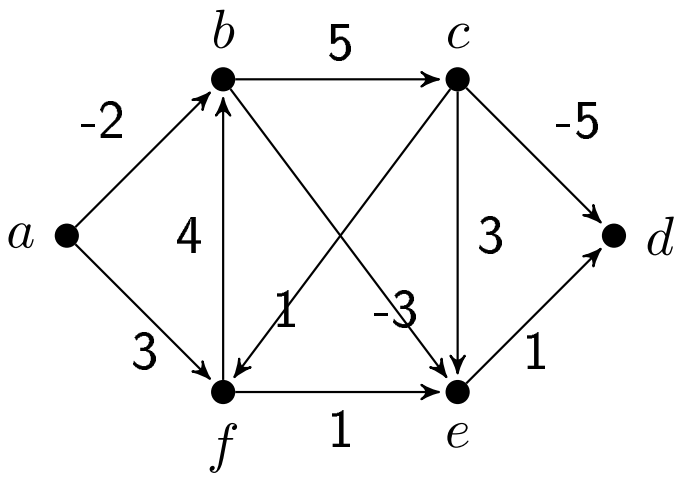

Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

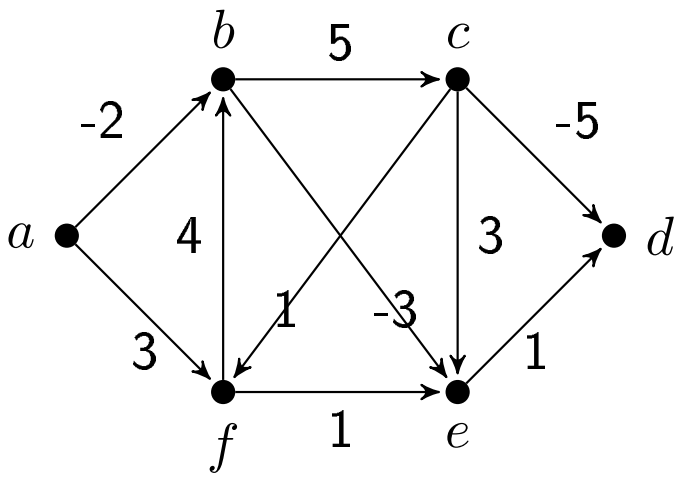
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

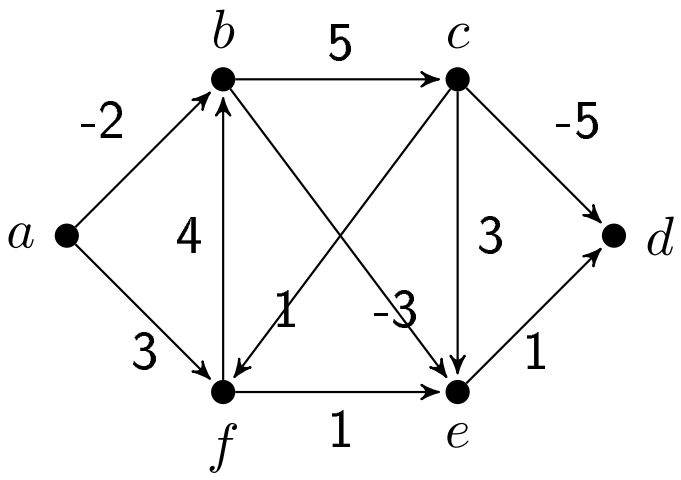
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | -4e | -5 | 3 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

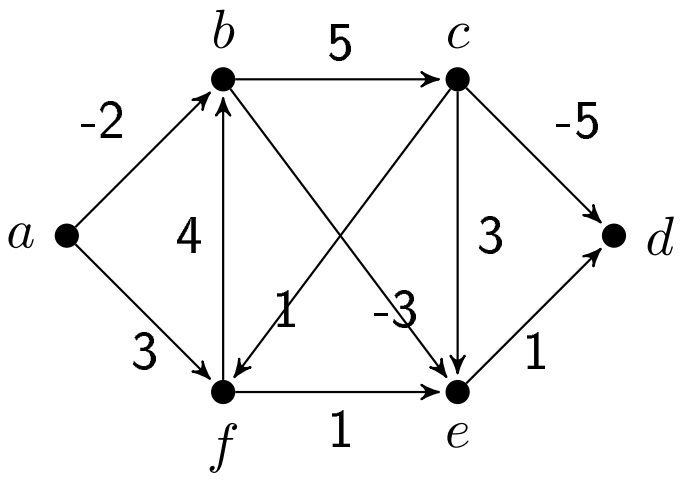
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | -4e | -5 | 3 |
| 4 | 0 | -2 | 3 | -4 | -5 | 3 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

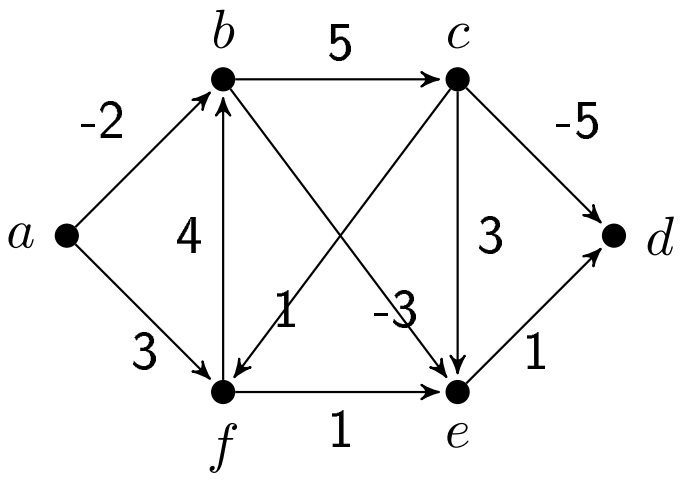
Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | -4e | -5 | 3 |
| 4 | 0 | -2 | 3 | -4 | -5 | 3 |

Stop since Step 4 = Step 3.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

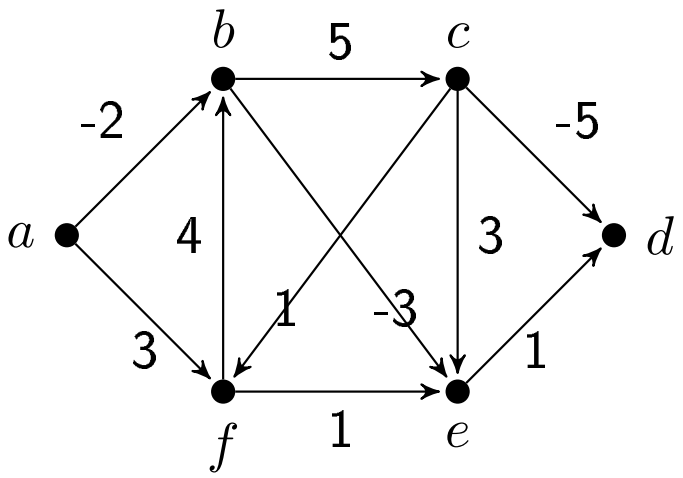
Graph Coloring

# Backtracking procedure

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | -4e | -5 | 3 |
| 4 | 0 | -2 | 3 | -4 | -5 | 3 |

Stop since Step 4 = Step 3.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

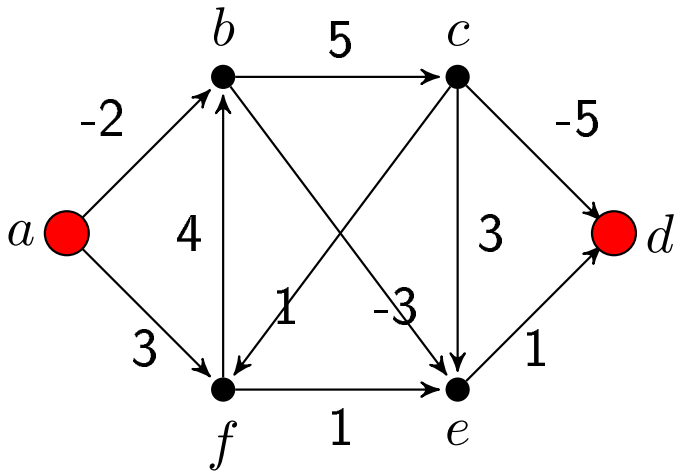
# Backtracking procedure

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | -4e | -5 | 3 |
| 4 | 0 | -2 | 3 | -4 | -5 | 3 |

Stop since Step 4 = Step 3.

How to find shortest path from $a$ to $d$? $a \rightarrow \qquad\qquad d$







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

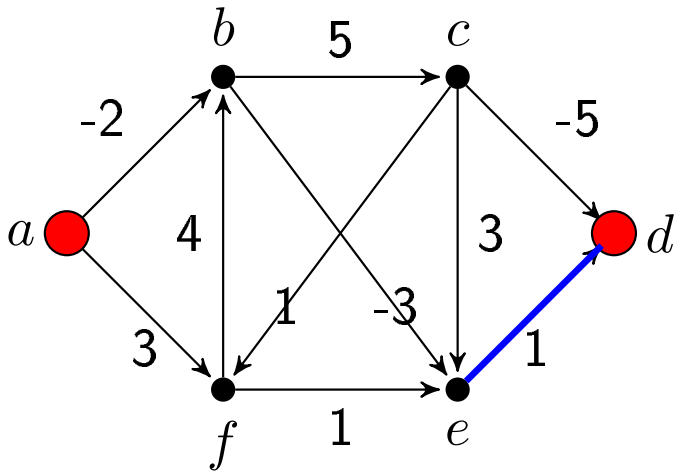
# Backtracking procedure

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | -5b | 3 |
| 3 | 0 | -2 | 3 | **-4e** | -5 | 3 |
| 4 | 0 | -2 | 3 | **-4** | -5 | 3 |

Stop since Step 4 = Step 3.

How to find shortest path from $a$ to $d$? $a \rightarrow \qquad e \rightarrow d$







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

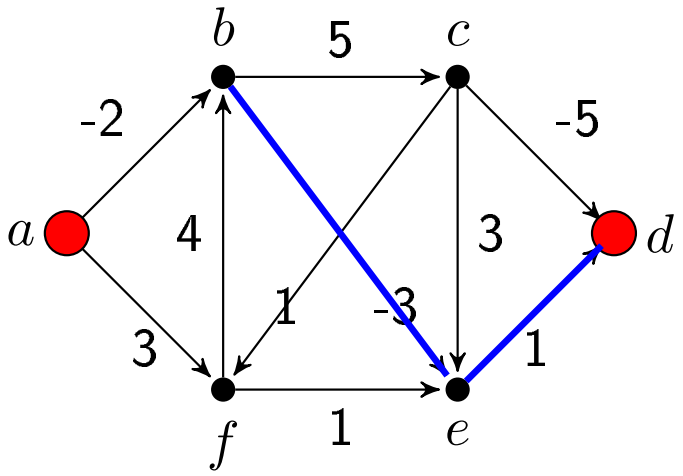
# Backtracking procedure

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | -2 | 3b | $\infty$ | **-5b** | 3 |
| 3 | 0 | -2 | 3 | **-4e** | **-5** | 3 |
| 4 | 0 | -2 | 3 | **-4** | -5 | 3 |

Stop since Step 4 = Step 3.

How to find shortest path from $a$ to $d$? $a \to b \to e \to d$







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

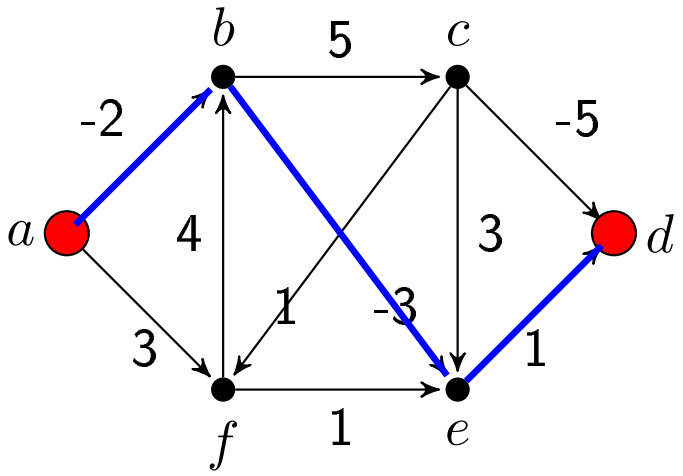
# Backtracking procedure

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | **-2a** | $\infty$ | $\infty$ | $\infty$ | 3a |
| 2 | 0 | **-2** | 3b | $\infty$ | **-5b** | 3 |
| 3 | 0 | -2 | 3 | **-4e** | **-5** | 3 |
| 4 | 0 | -2 | 3 | **-4** | -5 | 3 |

Stop since Step 4 = Step 3.

How to find shortest path from $a$ to $d$? $a \to b \to e \to d$







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

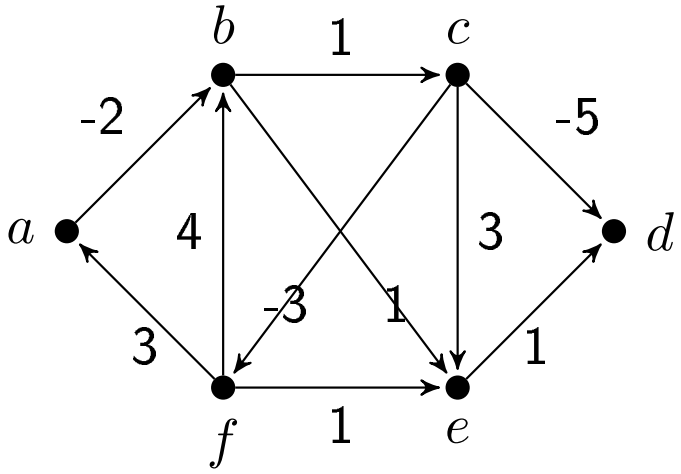
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|





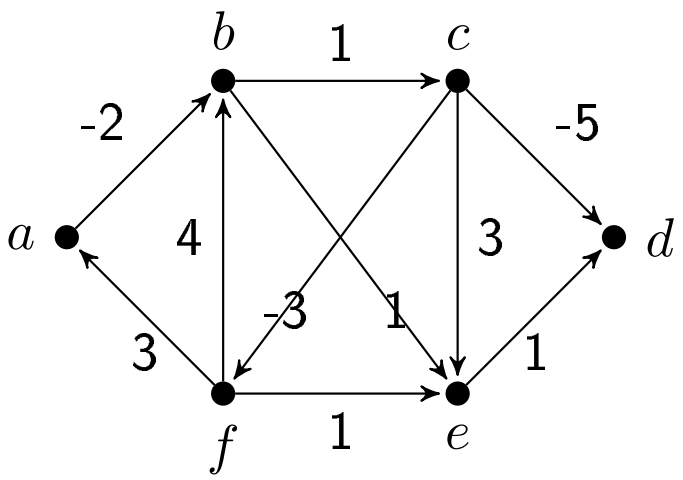

Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

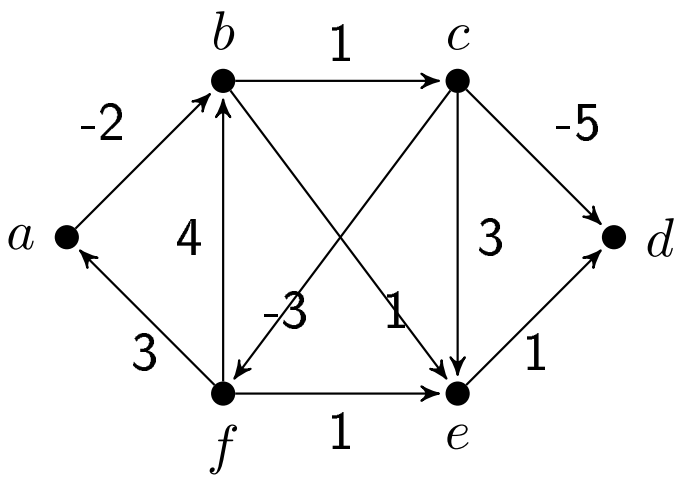
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

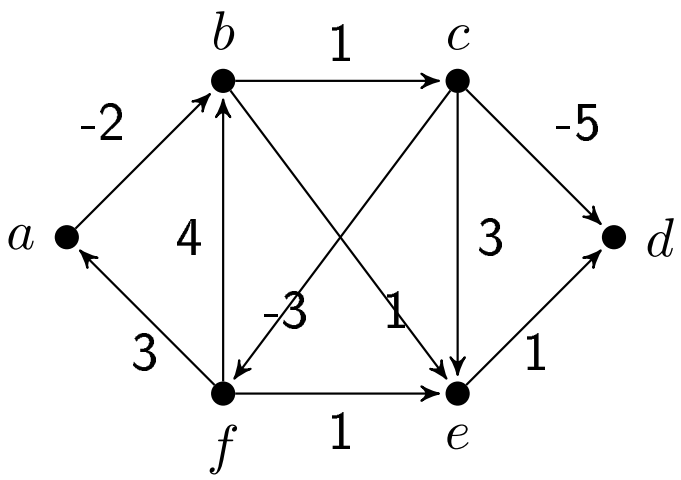
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

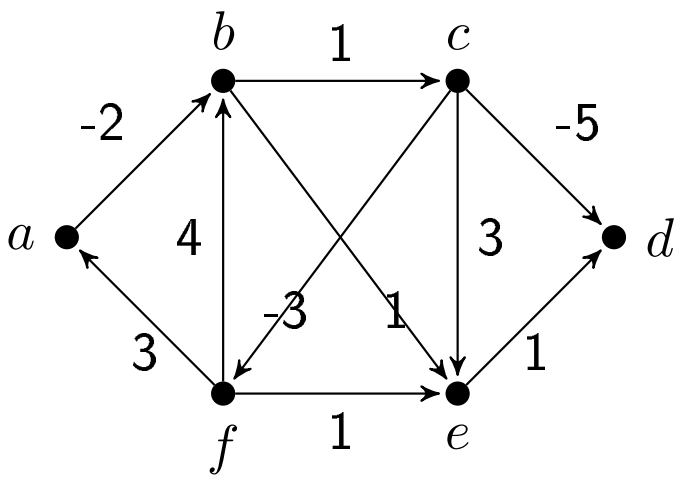
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

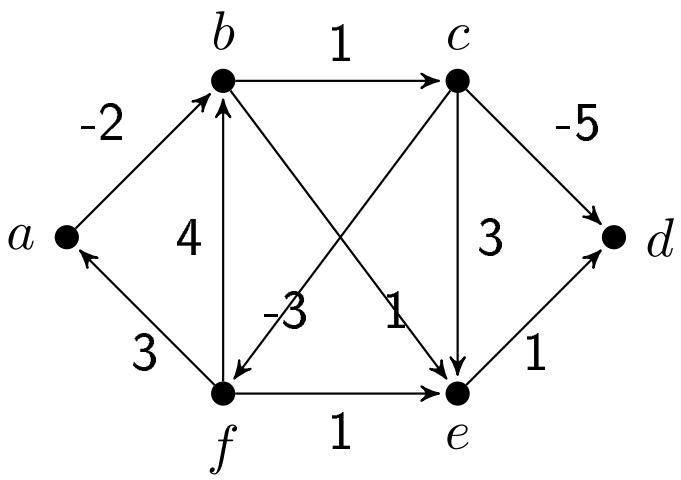
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |
| 4 | -1f | -2 | -1 | -6 | -3f | -4 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

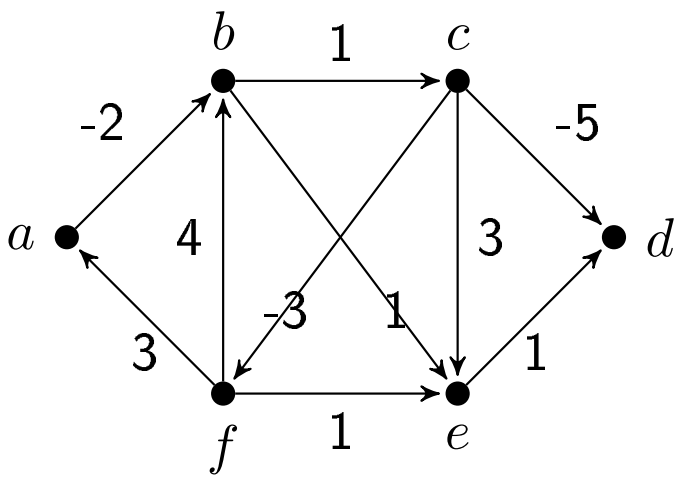
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |
| 4 | -1f | -2 | -1 | -6 | -3f | -4 |
| 5 | -1 | -3a | -1 | -6 | -3 | -4 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

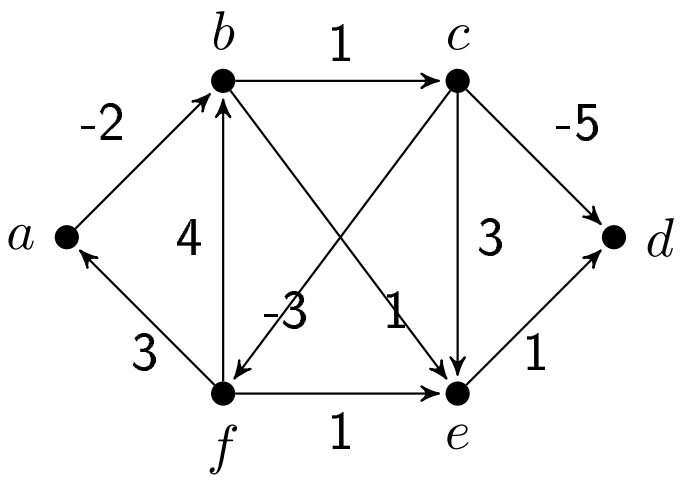
Others

Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |
| 4 | -1f | -2 | -1 | -6 | -3f | -4 |
| 5 | -1 | -3a | -1 | -6 | -3 | -4 |
| 6 | -1 | -3 | -2b | -6 | -3 | -4 |







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

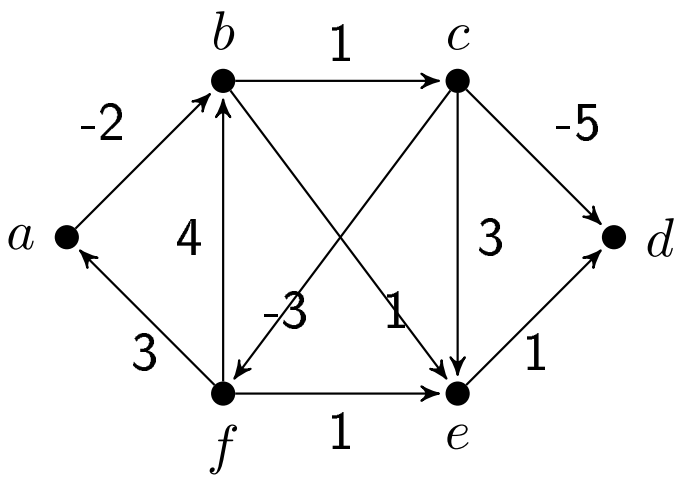
Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |
| 4 | -1f | -2 | -1 | -6 | -3f | -4 |
| 5 | -1 | -3a | -1 | -6 | -3 | -4 |
| 6 | -1 | -3 | -2b | -6 | -3 | -4 |

There exists a circle of negative length since Step 6 $\neq$ Step 5.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

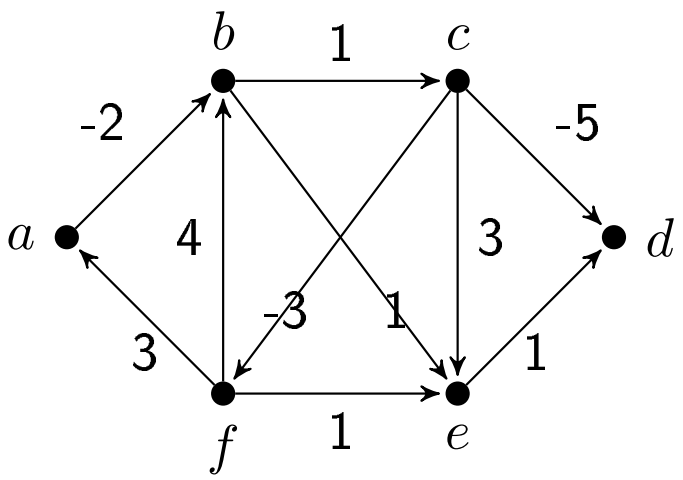
Graph Coloring

# Example

## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | -2a | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 2 | 0 | -2 | -1b | $\infty$ | -1b | $\infty$ |
| 3 | 0 | -2 | -1 | -6c | -1 | -4c |
| 4 | -1f | -2 | -1 | -6 | -3f | -4 |
| 5 | -1 | -3a | -1 | -6 | -3 | -4 |
| 6 | -1 | -3 | -2b | -6 | -3 | -4 |
| 7 | -1 | -3 | -2 | -7c | -3 | -4 |

There exists a circle of negative length since Step 6 $\neq$ Step 5.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

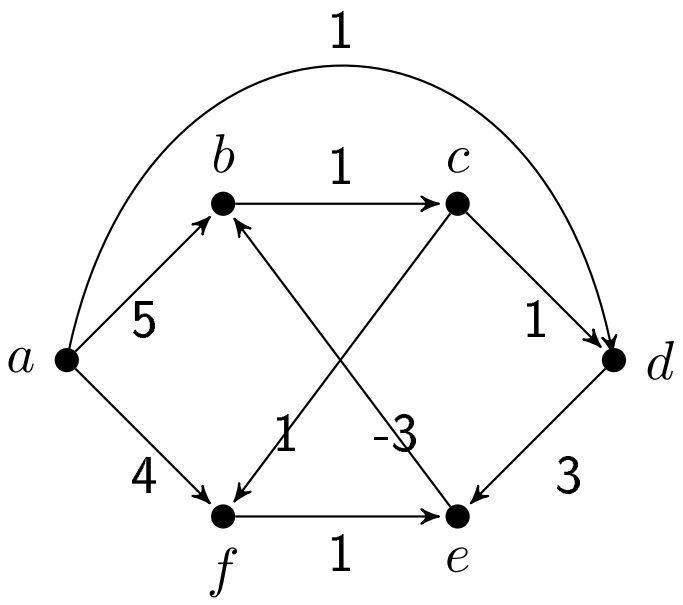
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




## Example

| Step | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|---|---|---|---|---|---|---|
| 0 | 0 | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ |
| 1 | 0 | 5a | $\infty$ | 1a | $\infty$ | 4a |
| 2 | 0 | 5a | 6b | 1a | 4d | 4a |
| 3 | 0 | 1e | 6b | 1a | 4d | 4a |
| 4 | 0 | 1e | 2b | 1a | 4d | 4a |
| 5 | 0 | 1e | 2b | 1a | 4d | 3c |
| 6 | 0 | 1e | 2b | 1a | 4d | 3c |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

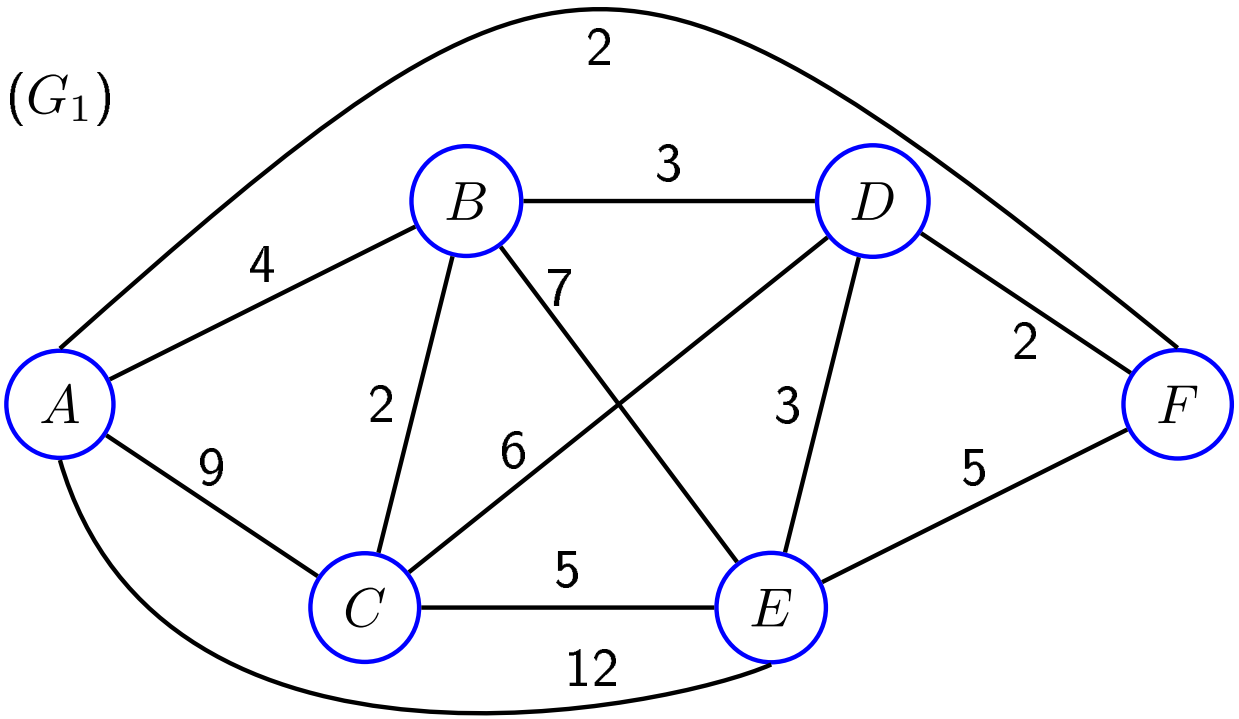
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Exercise

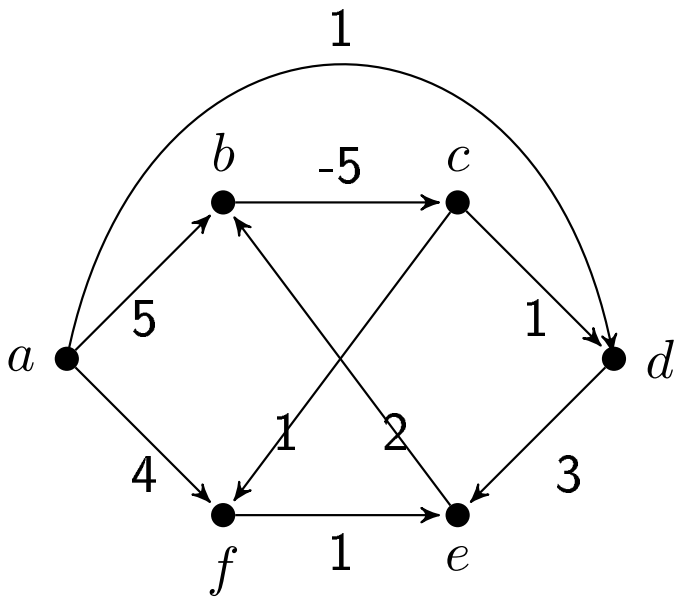





Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Exercise






Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Floyd-Warshall Algorithm [1962]

```
procedure FloydWarshall ()
   for k := 1 to n
      for i := 1 to n
         for j := 1 to n
            path[i,j] = min (path[i,j],
                             path[i,k]+path[k,j]);
```

## Property

any $G$, any weighted; all-to-all; this is an software algorithm; complexity $O(|V|^3)$.



Contents

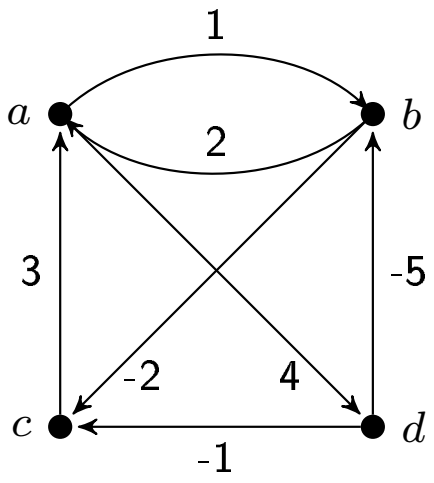
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

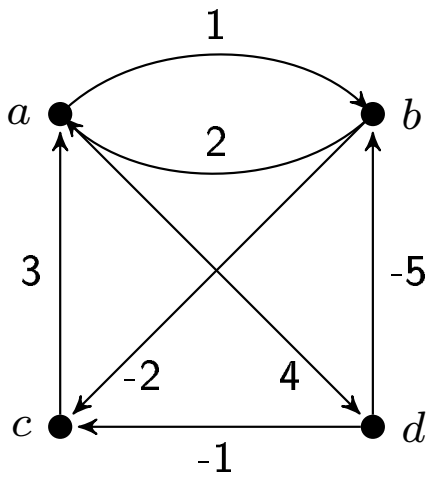
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & & & \\ 3_0 & & & \\ \infty_0 & & & \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

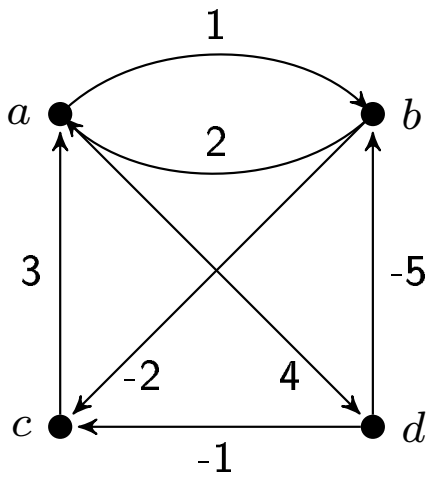
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

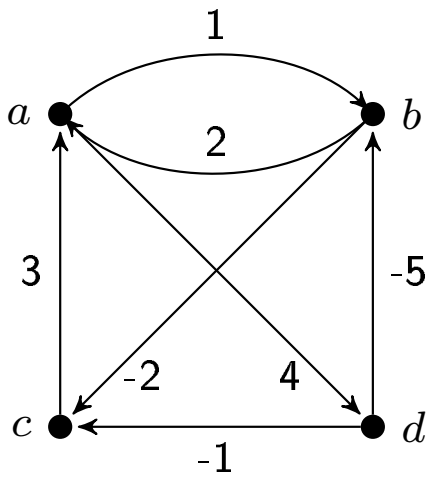
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} & 1_0 & & \\ 2_0 & 0_0 & -2_0 & 6_1 \\ & 4_1 & & \\ & -5_0 & & \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

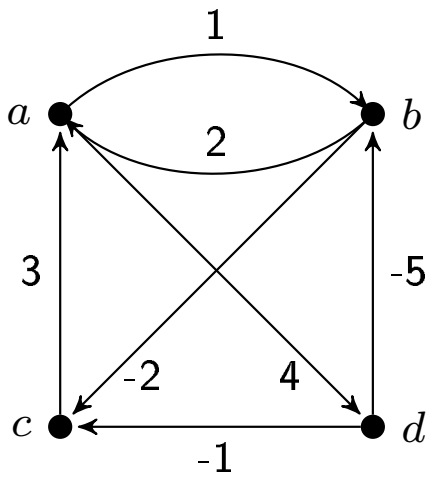
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

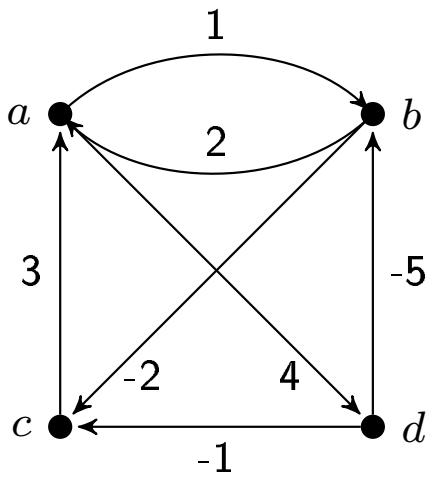
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} & & -1_2 & \\ & & -2_0 & \\ 3 & 4_1 & 0_0 & 7_1 \\ & & -7_2 & \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

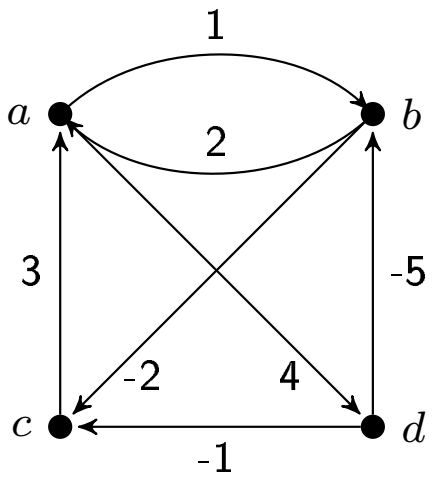
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

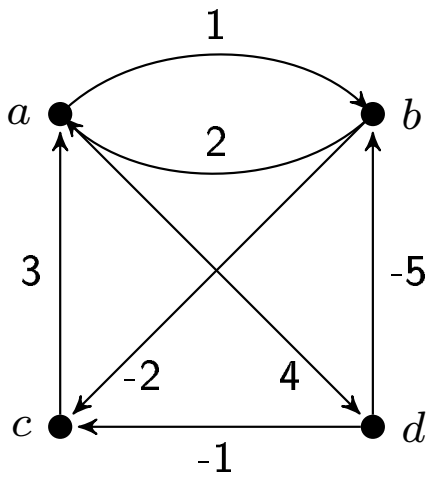
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} & & & 4_0 \\ & & & 5_3 \\ & & & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

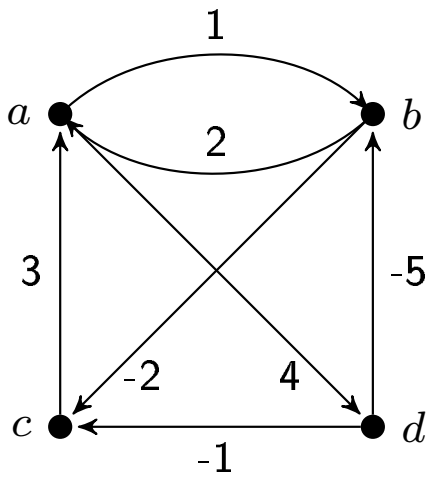
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} 0_0 & -1_4 & -3_4 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3_0 & 2_4 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

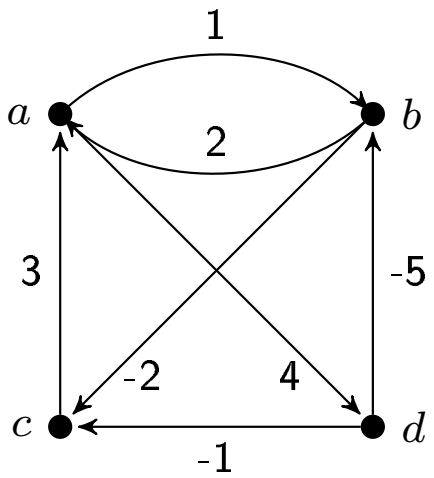
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




Shortest path from $b$ to $d$ ($5_3$ from $L^{(4)}$):

$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} 0_0 & -1_4 & -3_4 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3_0 & 2_4 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

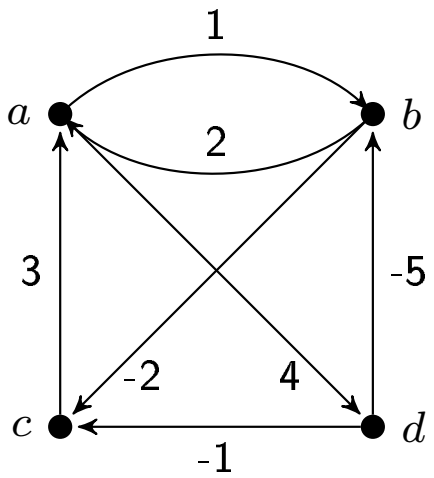
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




Shortest path from $b$ to $d$
($5_3$ from $L^{(4)}$):
$bd = bc + cd$
($5_3 = -2_0 + 7_1$ from $L^{(3)}$)

$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} 0_0 & -1_4 & -3_4 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3_0 & 2_4 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

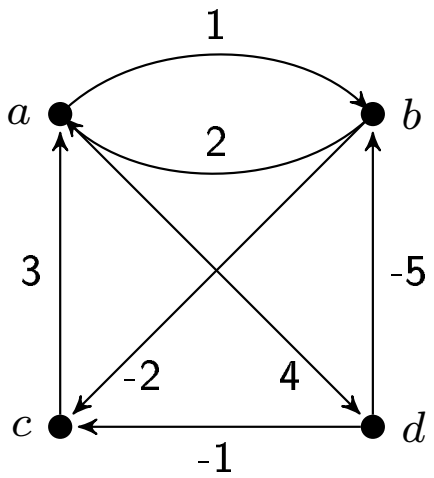
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example




Shortest path from $b$ to $d$
($5_3$ from $L^{(4)}$):
$bd = bc + cd$
($5_3 = -2_0 + 7_1$ from $L^{(3)}$)
$cd = ca + ad$
($7_1 = 3_0 + 4_0$ from $L^{(1)}$)

$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} 0_0 & -1_4 & -3_4 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3_0 & 2_4 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

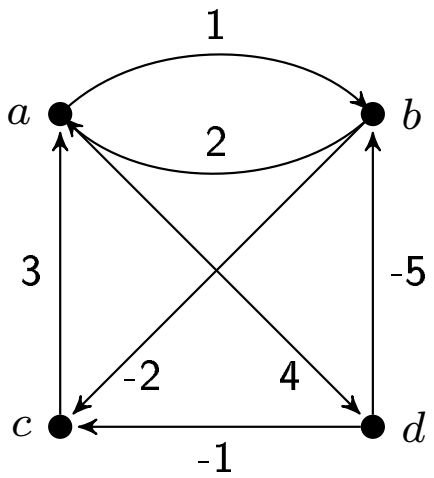
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example




Shortest path from $b$ to $d$
($5_3$ from $L^{(4)}$):
$bd = bc + cd$
($5_3 = -2_0 + 7_1$ from $L^{(3)}$)
$cd = ca + ad$
($7_1 = 3_0 + 4_0$ from $L^{(1)}$)
$\Rightarrow bd = bc + ca + ad$

$$L^{(0)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & \infty_0 \\ 3_0 & \infty_0 & 0_0 & \infty_0 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(1)} = \begin{pmatrix} 0_0 & 1_0 & \infty_0 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ \infty_0 & -5_0 & -1_0 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 2_0 & 0_0 & -2_0 & 6_1 \\ 3_0 & 4_1 & 0_0 & 7_1 \\ -3_2 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(3)} = \begin{pmatrix} 0_0 & 1_0 & -1_2 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3 & 4_1 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$

$$L^{(4)} = \begin{pmatrix} 0_0 & -1_4 & -3_4 & 4_0 \\ 1_3 & 0_0 & -2_0 & 5_3 \\ 3_0 & 2_4 & 0_0 & 7_1 \\ -4_3 & -5_0 & -7_2 & 0_0 \end{pmatrix}$$




Contents

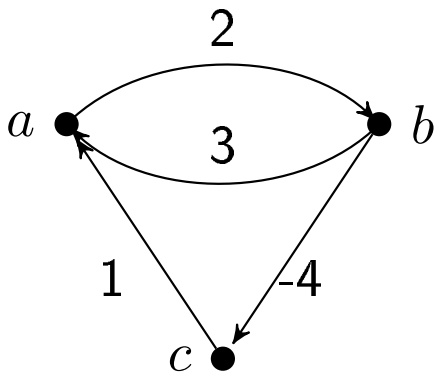
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example



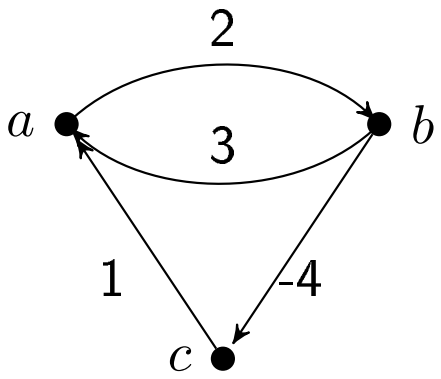

$$L^{(0)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & \infty_0 & 0_0 \end{pmatrix}$$



Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & \infty_0 & 0_0 \end{pmatrix} \quad L^{(1)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & 3_1 & 0_0 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

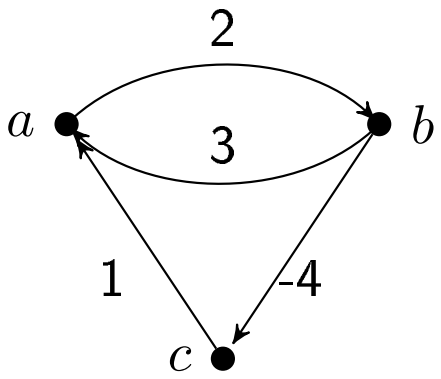
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & \infty_0 & 0_0 \end{pmatrix} \quad L^{(1)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & 3_1 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 2_0 & -2_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & 3_1 & -1_2 \end{pmatrix}$$




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

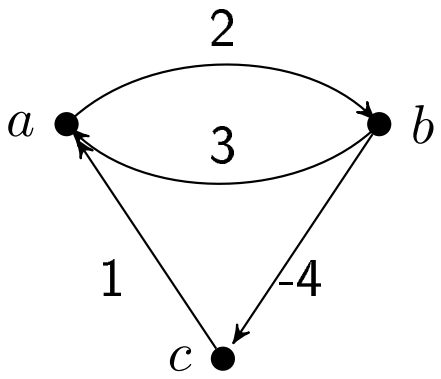
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




$$L^{(0)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & \infty_0 & 0_0 \end{pmatrix} \quad L^{(1)} = \begin{pmatrix} 0_0 & 2_0 & \infty_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & 3_1 & 0_0 \end{pmatrix}$$

$$L^{(2)} = \begin{pmatrix} 0_0 & 2_0 & -2_0 \\ 3_0 & 0_0 & -4_0 \\ 1_0 & 3_1 & -1_2 \end{pmatrix}$$

STOP, there exists a circuit of negative length.

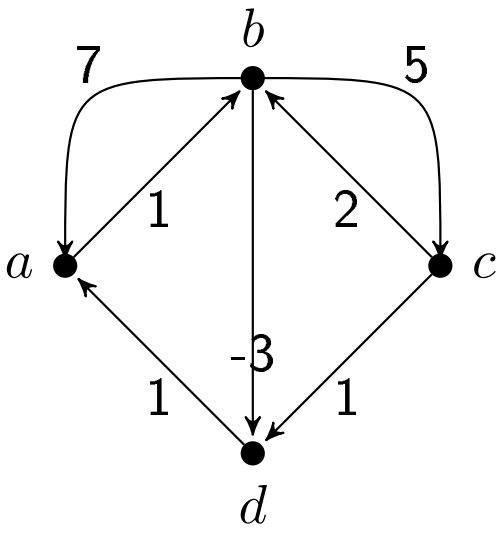


Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents

Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Exercise






Contents

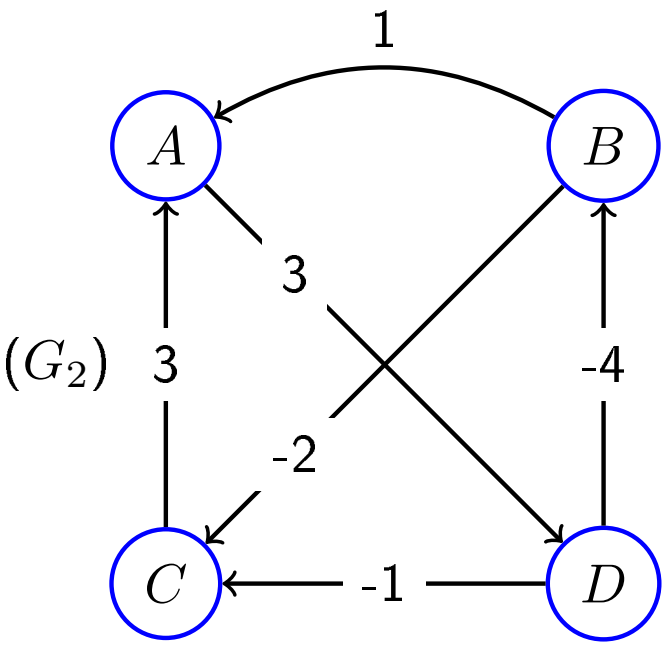
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Exercise






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Ford's algorithm

$\pi(1) = 0$

For each $j \in V$ do

$\pi(j) = \min_{i \in \rho_j^{-1}} (\pi(i) + \ell[i,j])$

End

## Property

$G$ without circle, positive length; one-to-all; rank table definition; complexity $O(|V|)$.



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents

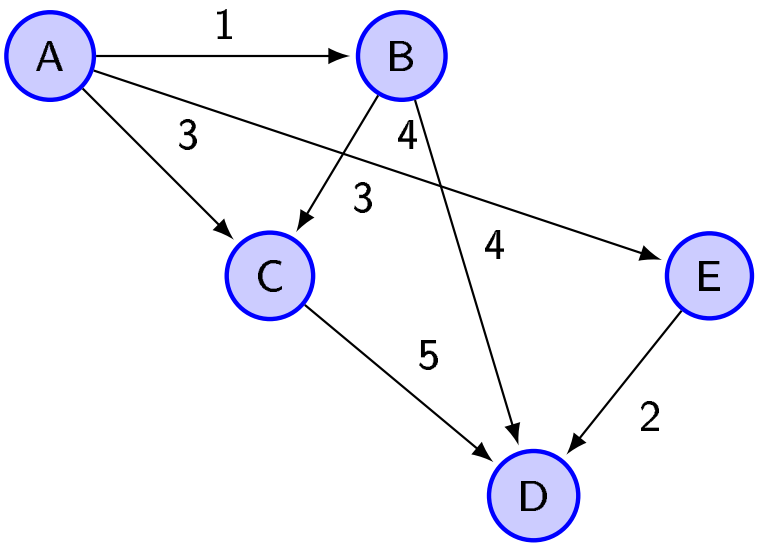
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example






Contents

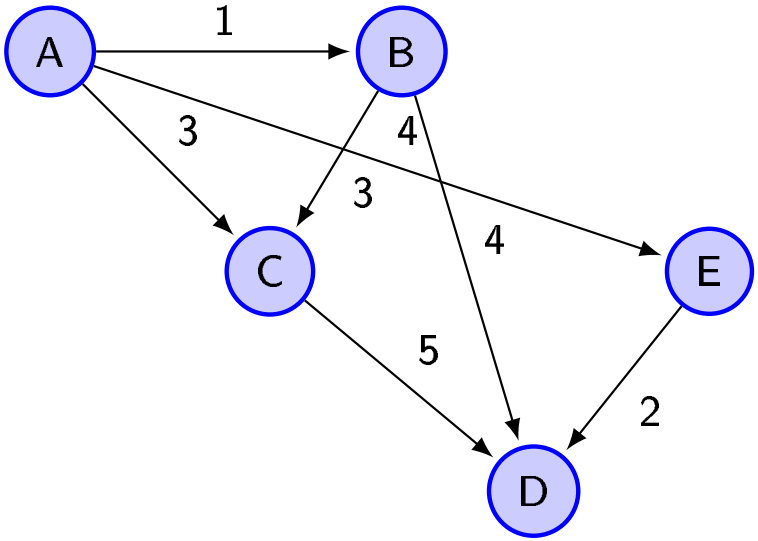
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | | |
| B | | |
| C | | |
| D | | |
| E | | |



Contents

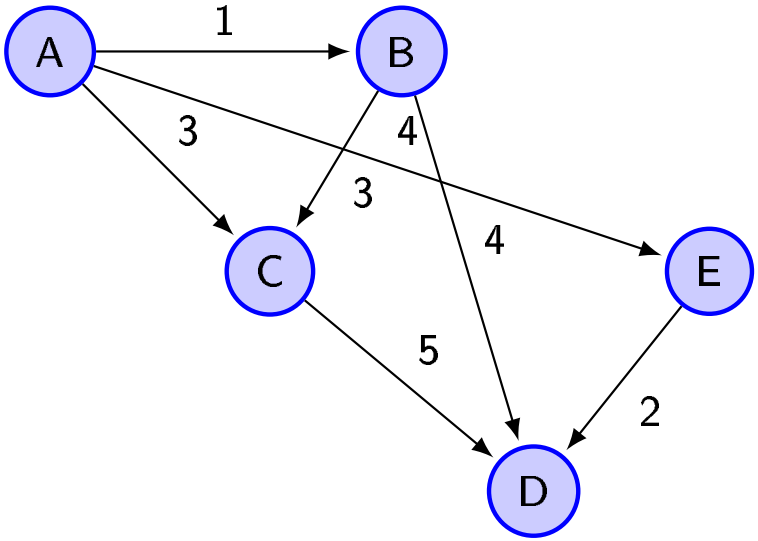
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | |
| B | A | |
| C | A, B | |
| D | B, C , E | |
| E | A | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

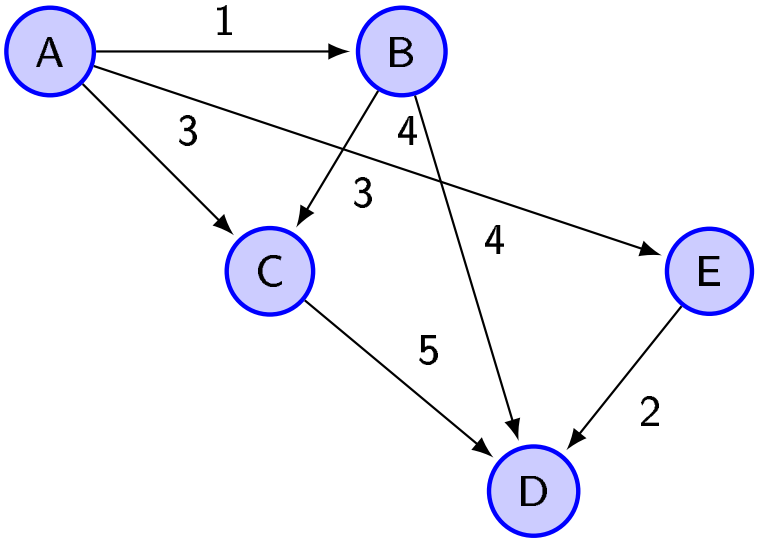
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | A | |
| C | A, B | |
| D | B, C , E | |
| E | A | |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

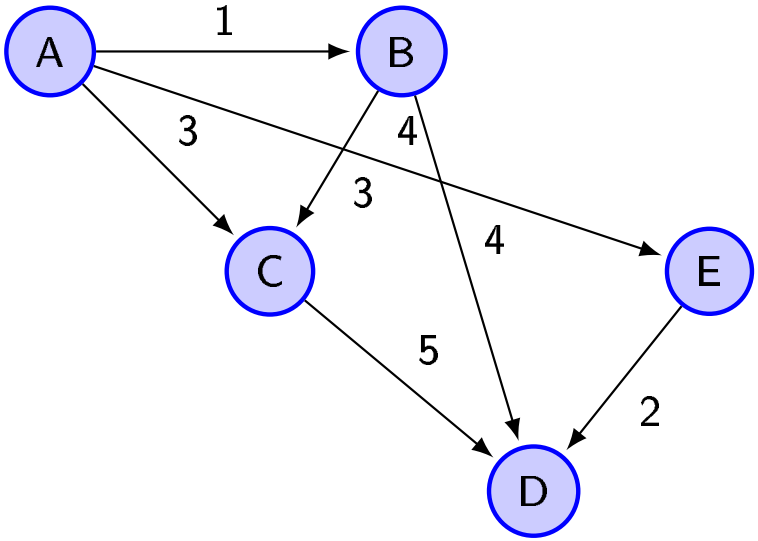
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | B | |
| D | B, C , E | |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

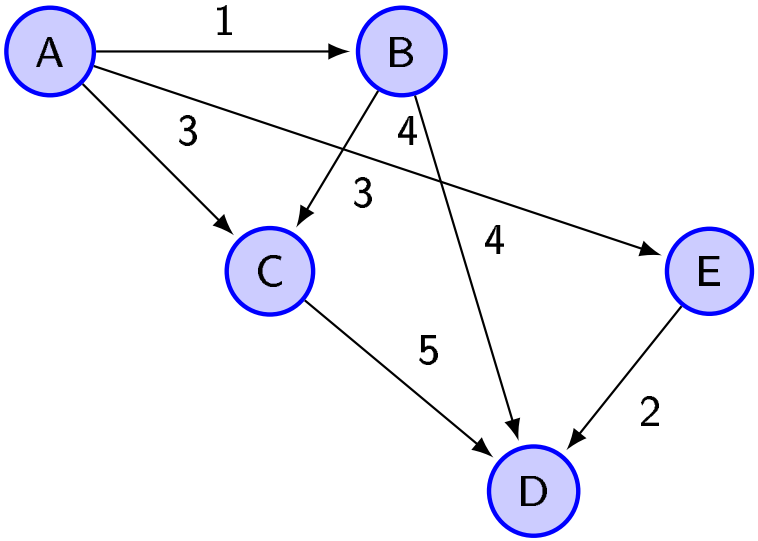
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | C | |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

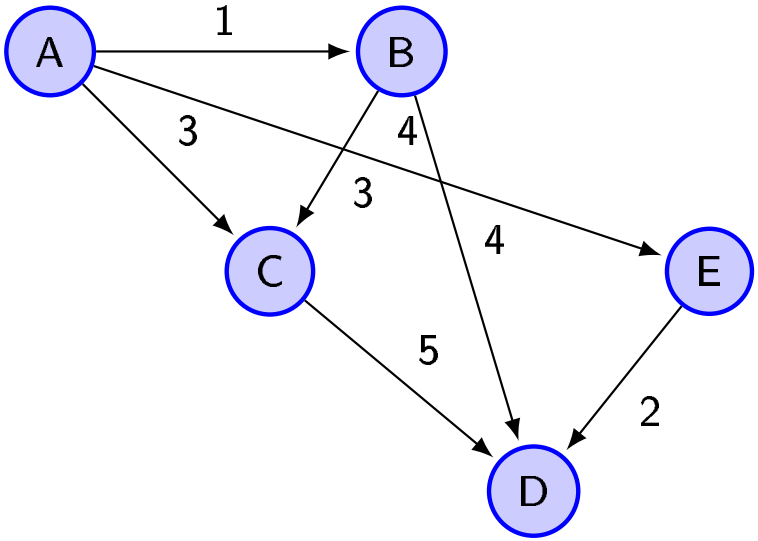
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

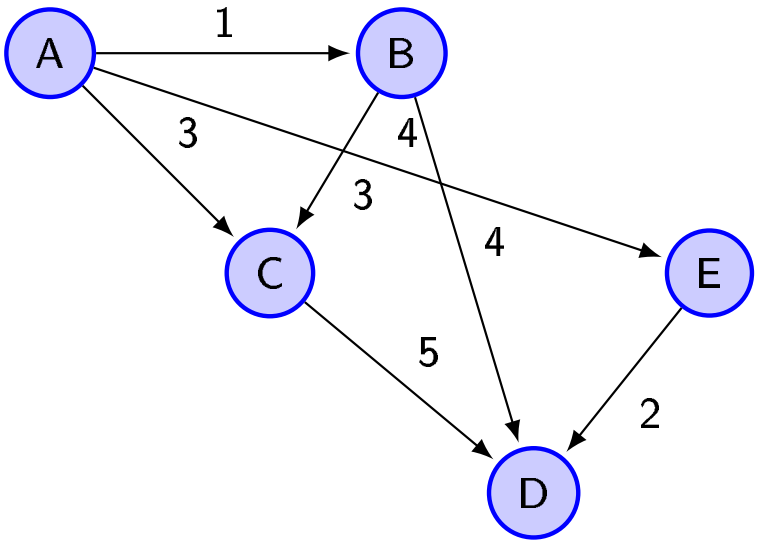
Floyd-Warshall Algorithm

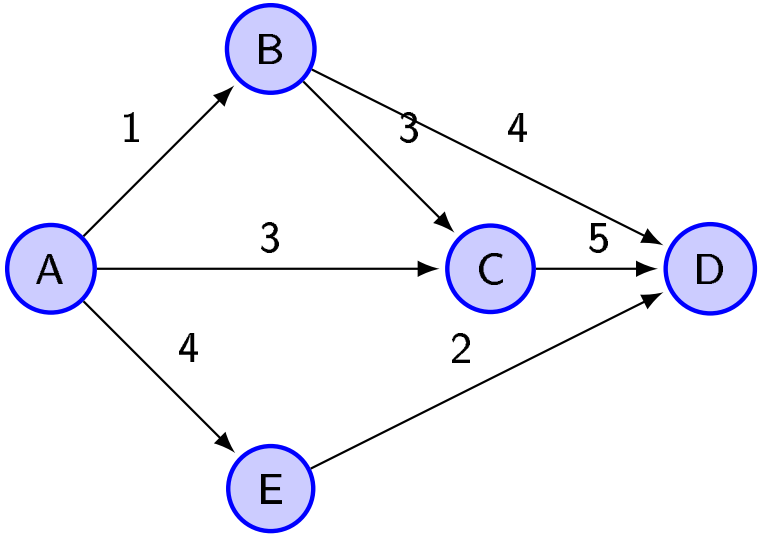
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

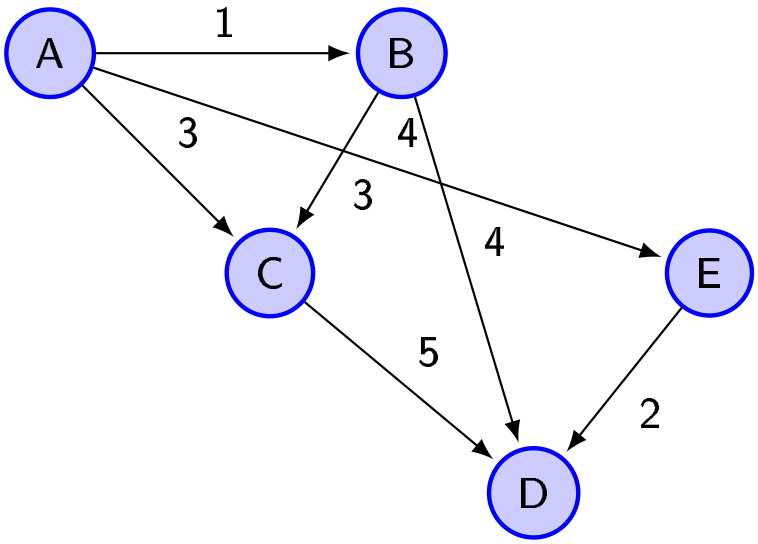
Floyd-Warshall Algorithm

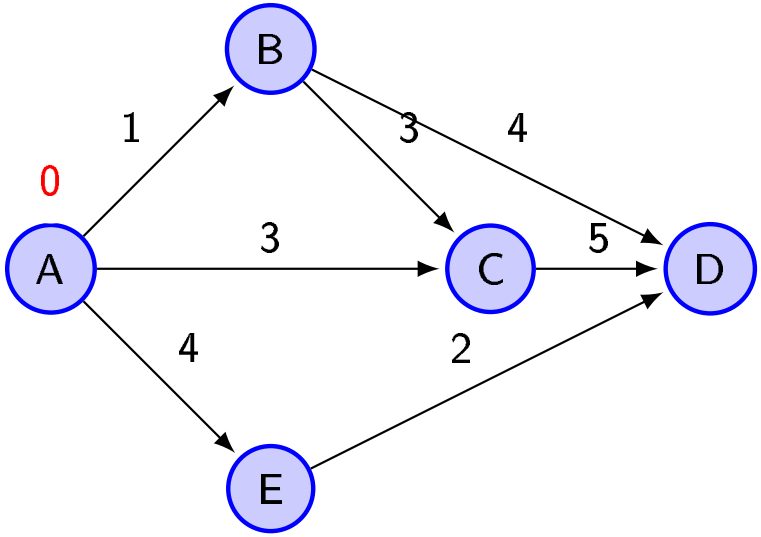
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

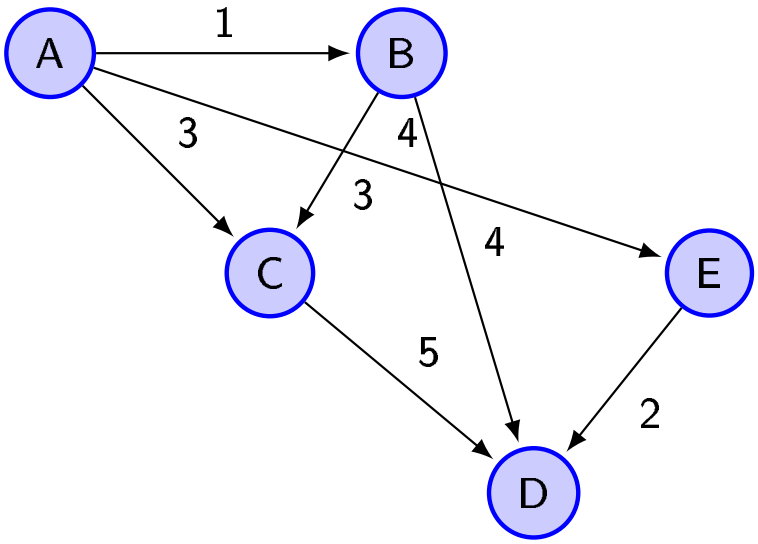
Floyd-Warshall Algorithm

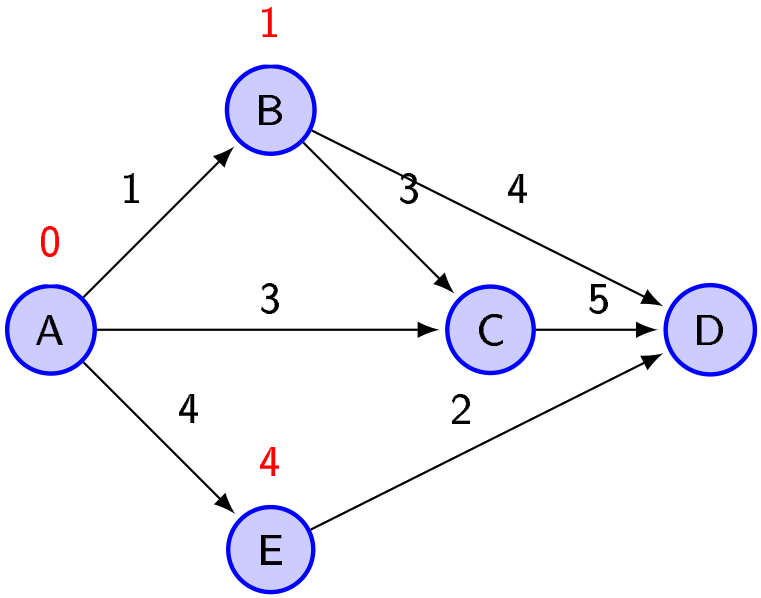
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

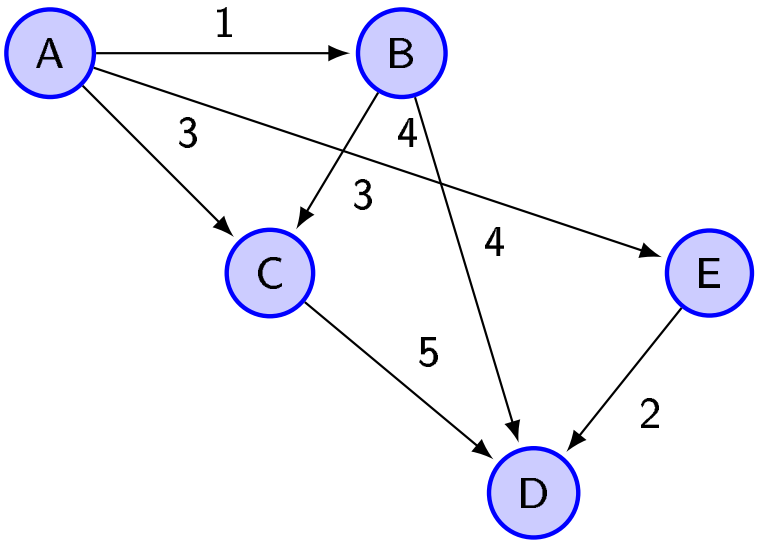
Floyd-Warshall Algorithm

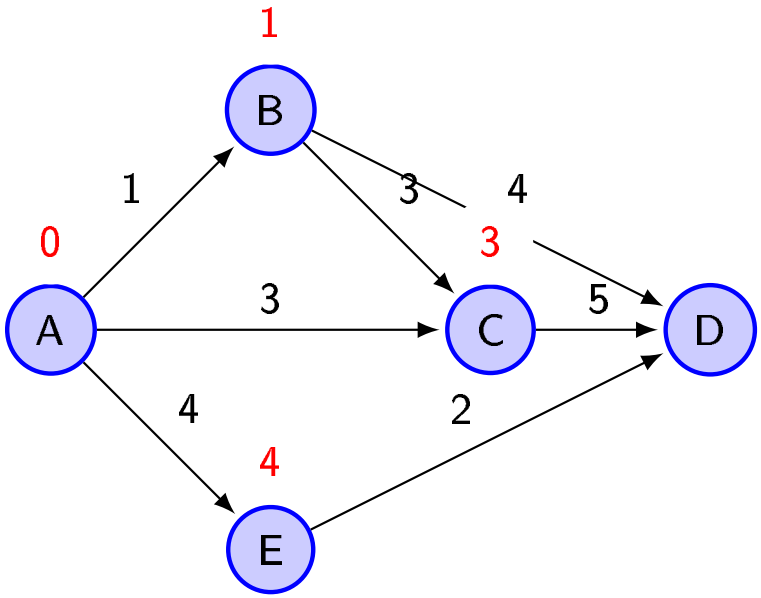
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

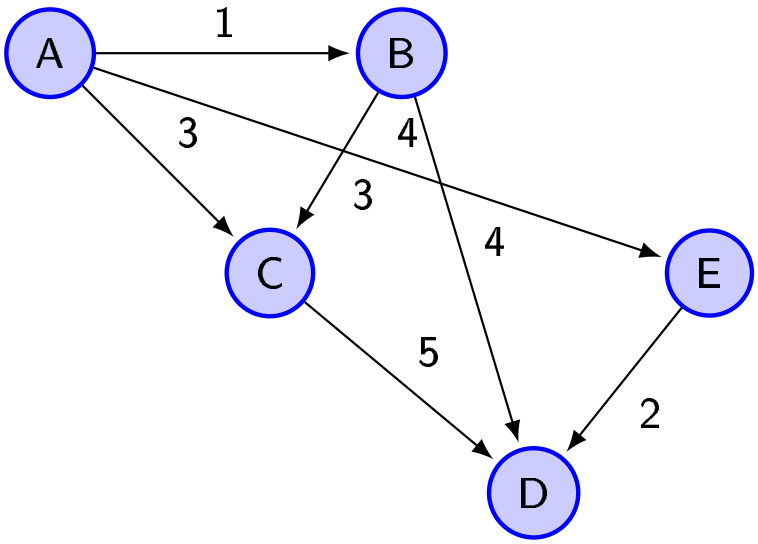
Floyd-Warshall Algorithm

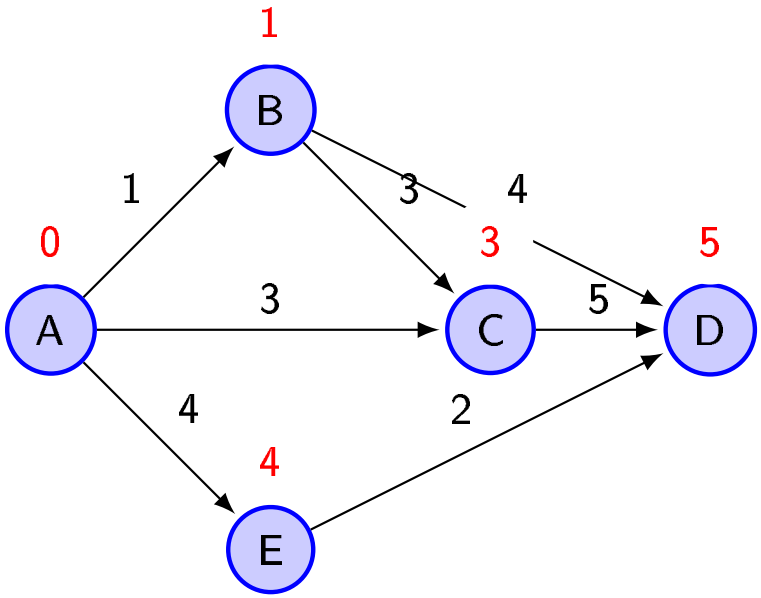
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

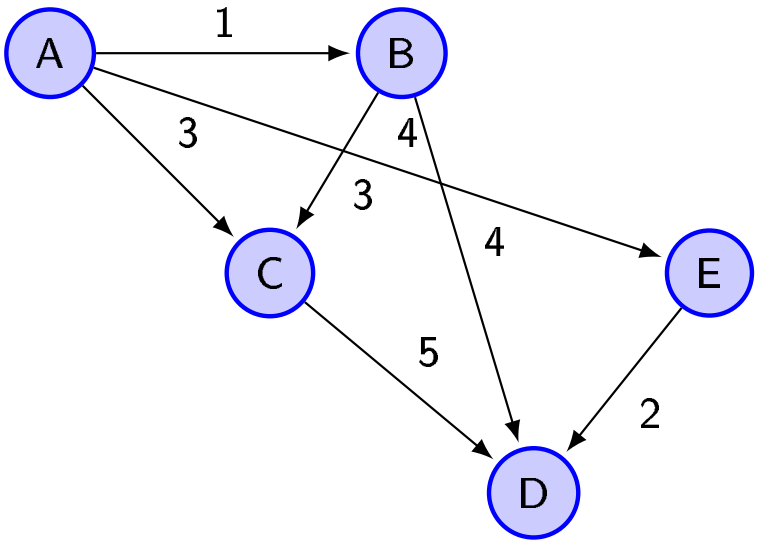
Floyd-Warshall Algorithm

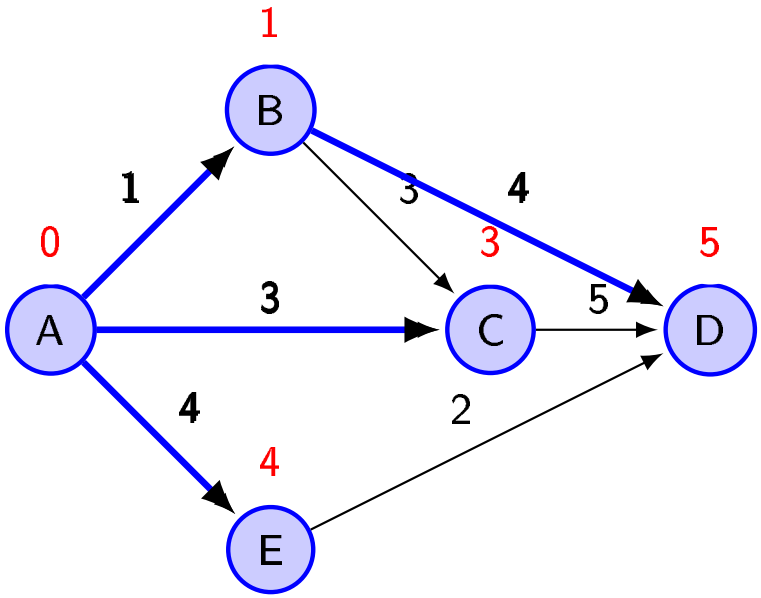
Ford's algorithm

Others

Graph Coloring

# Example




| i | $\Gamma_i^{-1}$ | rank(i) |
|---|---|---|
| A | - | 0 |
| B | | 1 |
| C | | 2 |
| D | | 3 |
| E | | 1 |




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

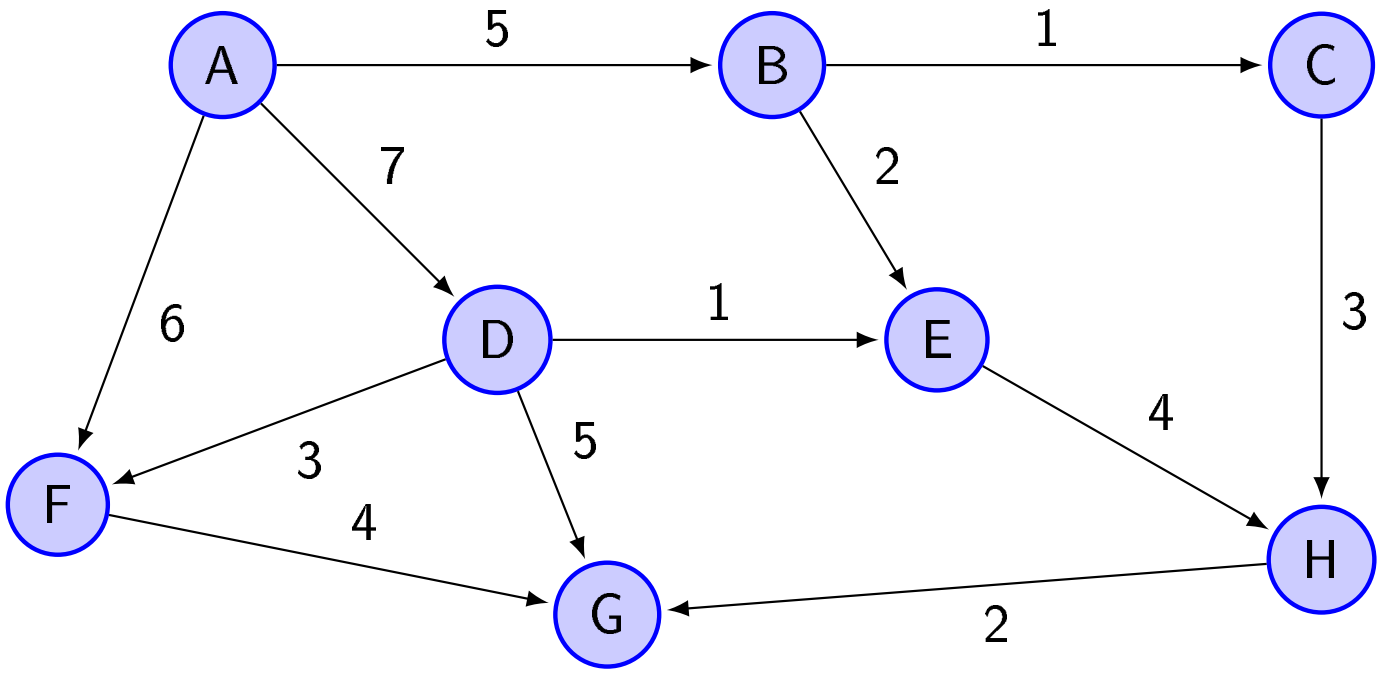
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Exercise






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Application

## Problem

A young professor in Hue is invited to teach some years in Ho Chi Minh university of technology. He decides to represent the diverse operations of his transfer by a graph and, in this purpose, establishes the list of following operations:

- A: Find a house in Ho Chi Minh city.
- B: Choose a removal man and sign a contract of move
- C: Make pack his furniture by the removal man
- D: Make transport his furniture towards Ho Chi Minh city
- E: Find an accommodation to HCM (from Hue)
- F: Transport his family to HCM
- G: Move into his new accommodation
- H: Register the children to their new school
- I: Look for a temporary work for his wife
- J: Fit out the new accommodation and pay this arrangement with the first treatment of his wife
- K: Find a small bar to celebrate in family the success of the move and express the enjoyment to live in a good accommodation arrangement



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Application

Considering constraint of posteriority following: $A < F$; $B < C$; $C < D \wedge F$; $D < G$; $E < F$; $F < G \wedge H \wedge I$; $G < K$; $H < K$; $I < J$; $J < K$.

Approximated job processing times :

| A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | 2 | 3 | 4 | 7 | 3 | 5 | 1 | 3 | 8 | 2 |



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Application

Considering constraint of posteriority following: $A < F$; $B < C$; $C < D \wedge F$; $D < G$; $E < F$; $F < G \wedge H \wedge I$; $G < K$; $H < K$; $I < J$; $J < K$.
Approximated job processing times :

| A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | 2 | 3 | 4 | 7 | 3 | 5 | 1 | 3 | 8 | 2 |

## Question

- Determine a schedule of the 'movement' with minimal duration.
- What happens if his new accommodation is not available before date 20? In that case, of what margin we have to make the task $J$?



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Question

How to determine a shortest path from $u$ to $v$ in graph $G$ which traverses at most $\leq$ a given constant number of intermediate vertices.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Other shortest path problems

- multicriteria shortest path problem
  - linear combinaison
  - $\varepsilon$-constraint approach
  - lexico-graphical order
- $k$ shortest paths problem
  - allowing loop
  - loppless
- multi-point shortest path
  - TSP, VRP



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Top $k$ shortest paths query

When the shortest path is not sufficient for application, top-k shortest paths are desired.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

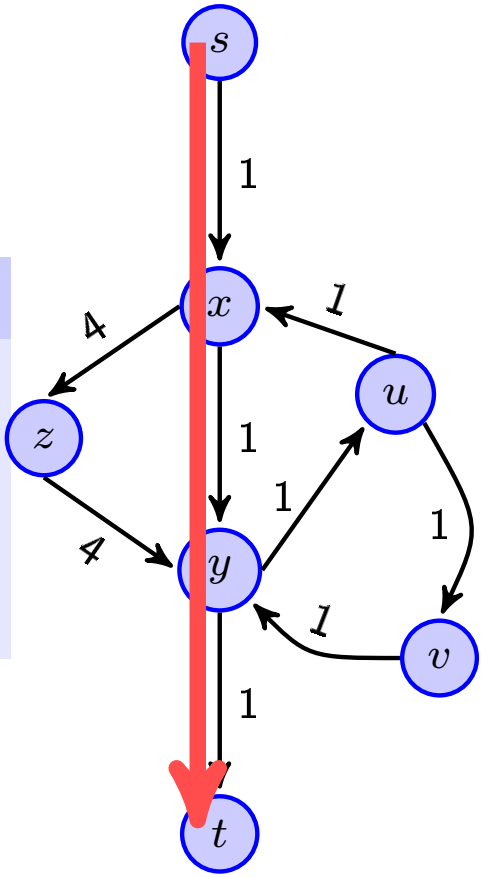
Graph Coloring

# Top $k$ shortest paths query

When the shortest path is not sufficient for application, top-k shortest paths are desired.

## Top 3 general shortest paths (allowing loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

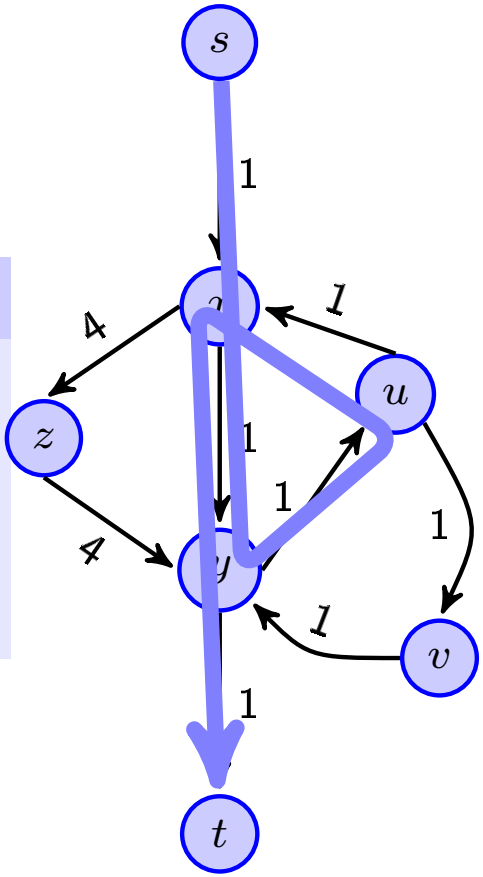
Graph Coloring

# Top $k$ shortest paths query

When the shortest path is not sufficient for application, top-k shortest paths are desired.

## Top 3 general shortest paths (allowing loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$
- $2^{nd}$:
  $s \to x \to y \to u \to x \to y \to t$, $l = 6$
  or $s \to x \to y \to u \to v \to y \to t$, $l = 6$







Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Top $k$ shortest paths query

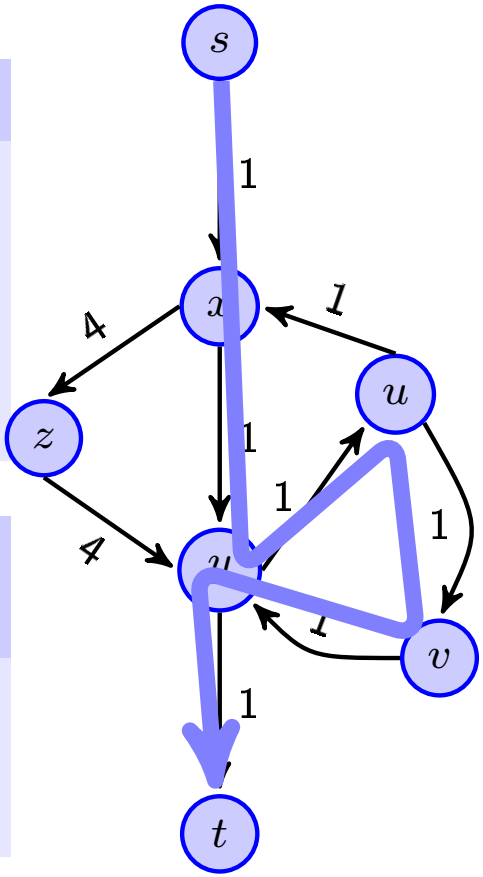
When the shortest path is not sufficient for application, top-k shortest paths are desired.

## Top 3 general shortest paths (allowing loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$
- $2^{nd}$:
  $s \to x \to y \to u \to x \to y \to t$, $l = 6$
  or $s \to x \to y \to u \to v \to y \to t$, $l = 6$

## Top 2 elementary shortest paths (without loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$
- $2^{nd}$: $s \to x \to z \to y \to t$, $l = 10$







Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Top $k$ shortest paths query

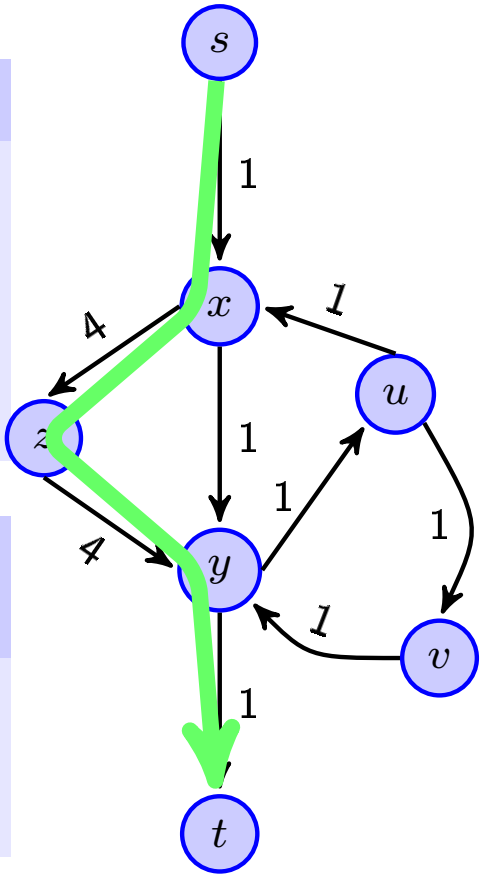
When the shortest path is not sufficient for application, top-k shortest paths are desired.

## Top 3 general shortest paths (allowing loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$
- $2^{nd}$:
  $s \to x \to y \to u \to x \to y \to t$, $l = 6$
  or $s \to x \to y \to u \to v \to y \to t$, $l = 6$

## Top 2 elementary shortest paths (without loops)

- $1^{st}$: $s \to x \to y \to t$, $l = 3$
- $2^{nd}$: $s \to x \to z \to y \to t$, $l = 10$







Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Traveling Salesman Problem (TSP)

## Problem

- Given a set of $n$ customers located in $n$ cities and distances for each pair of cities, the problem involves finding a round-trip with the minimum traveling cost.
- The vehicle must visit each customer exactly once and return to its point of origin also called depot.
- The objective function is the total cost of the tour.
- $\mathcal{NP}$-complete: all known techniques for obtaining an exact solution require an exponentially increasing number of steps (computing resources) as the problems become larger.
- **TSP is one of the most intensely studied problems in computational mathematics, yet no effective solution method.**



Contents

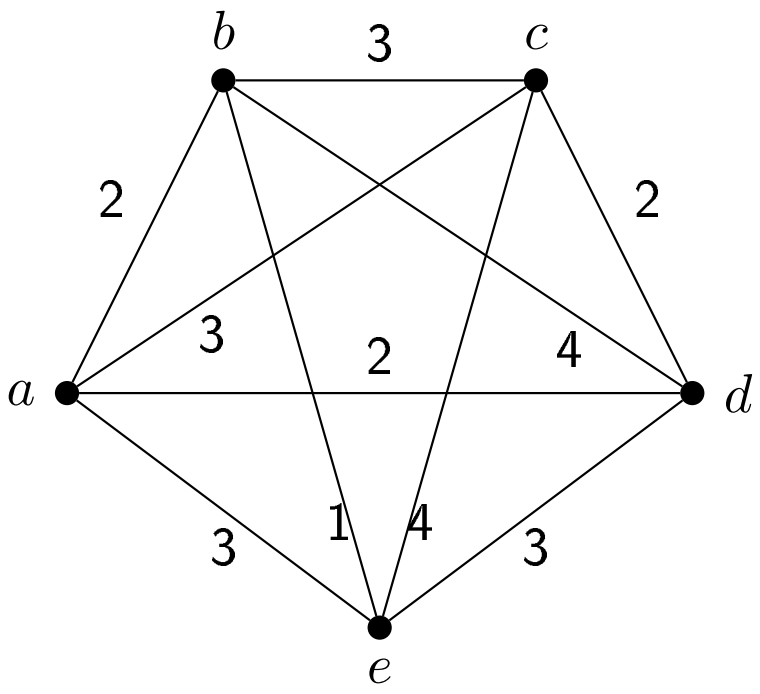
Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Traveling Salesman Problem




- The total number of possible Hamilton circuit is $(n-1)!/2$.
- For example, if there are $25$ customers to visit, the total number of solutions is $24!/2 = 3.1 \times 10^{23}$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

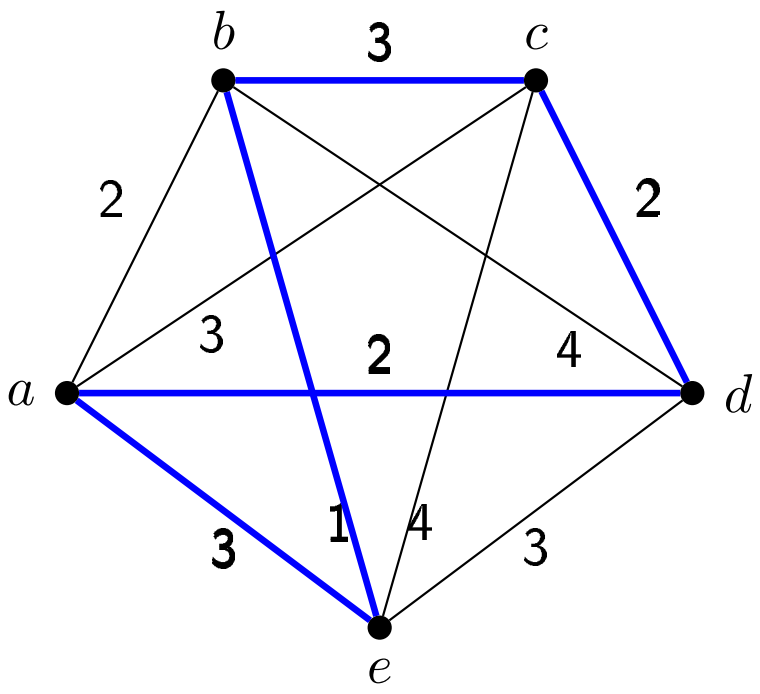
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Traveling Salesman Problem




- The total number of possible Hamilton circuit is $(n-1)!/2$.
- For example, if there are $25$ customers to visit, the total number of solutions is $24!/2 = 3.1 \times 10^{23}$.
- If the depot is located at node $1$, then the optimal tour is $1-5-2-3-4-1$ with total cost equal to $11$.




Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Vehicle Routing Problem (VRP)

## Problem

- The vehicle routing problem involves finding a set of trips, one for each vehicle, to deliver known quantities of goods to a set of customers.
- The objective is to minimize the travel costs of all trips combined.
- There may be upper bounds on the total load of each vehicle and the total duration of its trip.
- The most basic Vehicle Routing Problem (VRP) is the single-depot capacitate VRP.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Maps and Graphs

## Definition

- Every map can be represented by a graph. We call it dual graph.
- Problem of coloring the regions of a map $\rightarrow$ coloring the vertices of the dual graph so that no two adjacent vertices have the same color.



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

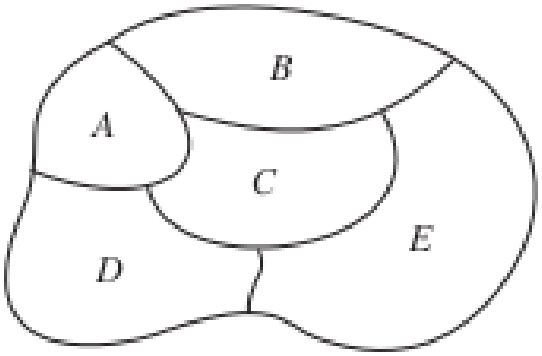
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

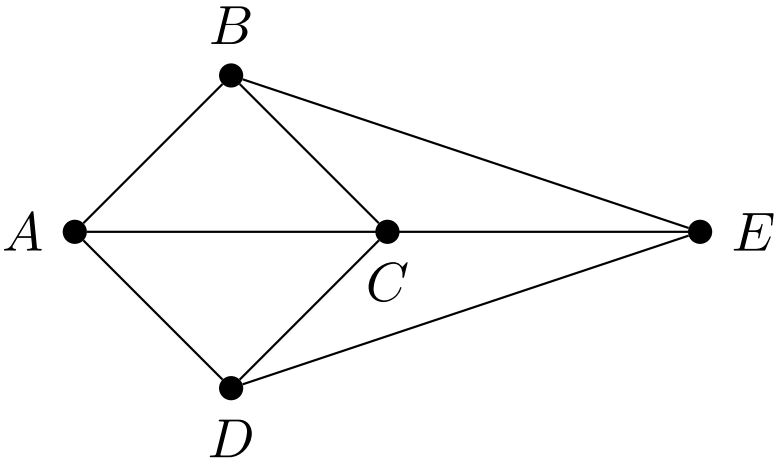
Graph Coloring

# Maps and Graphs

## Definition

- Every map can be represented by a graph. We call it dual graph.
- Problem of coloring the regions of a map $\rightarrow$ coloring the vertices of the dual graph so that no two adjacent vertices have the same color.






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

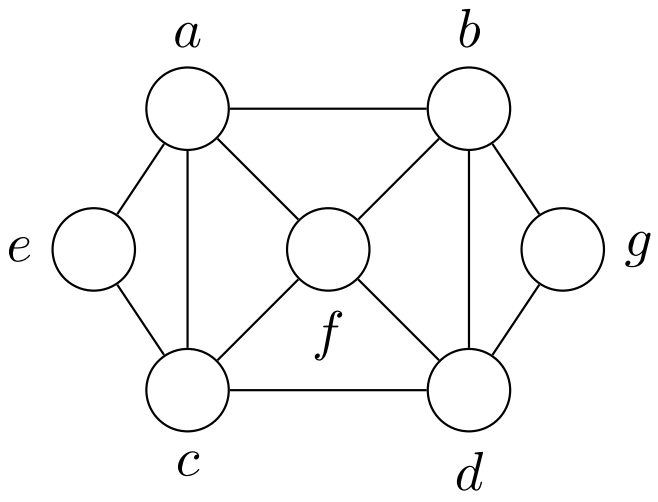
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Graph coloring

## Definition

- A coloring (*tô màu*) of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

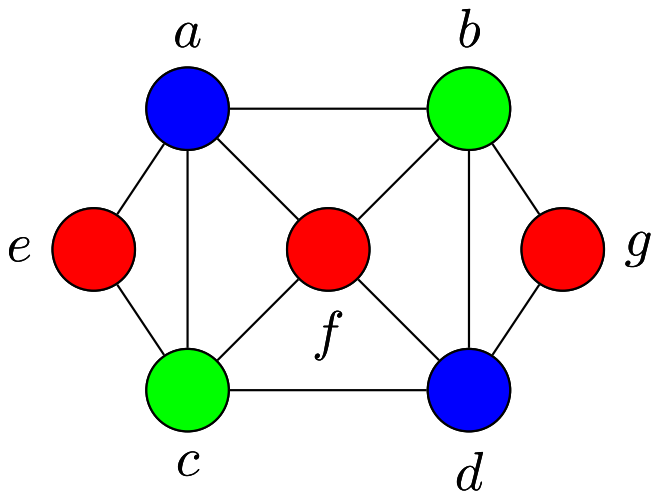
Others

Graph Coloring

# Graph coloring

## Definition

- A coloring (*tô màu*) of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
- The chromatic number (*số màu*) of a graph, denoted by $\chi(G)$, is the least number of colors needed for a coloring of this graph.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

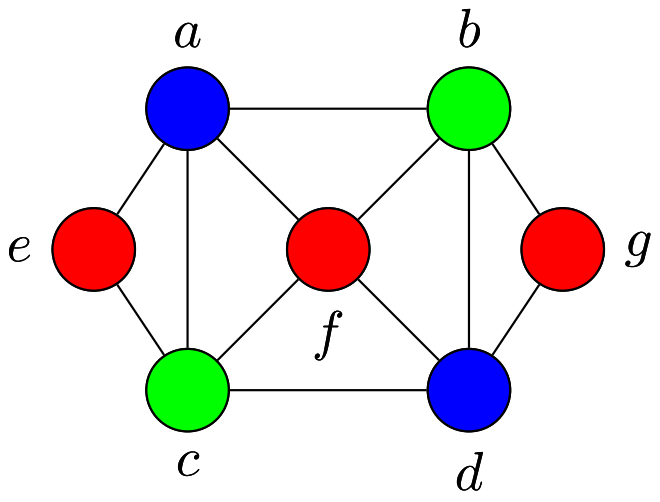
Others

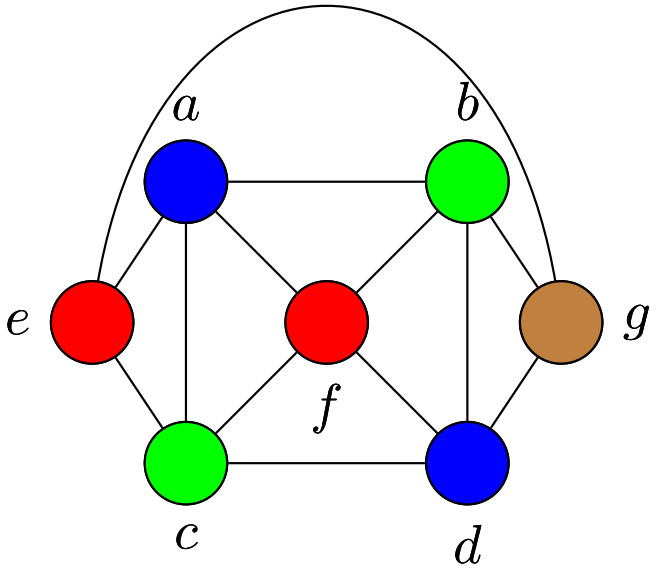
Graph Coloring

# Graph coloring

## Definition

- A coloring (*tô màu*) of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
- The chromatic number (*số màu*) of a graph, denoted by $\chi(G)$, is the least number of colors needed for a coloring of this graph.







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Four color theorem

**Theorem (Four color theorem)**

*The chromatic number of a planar graph is no greater than four.*

- Was a conjecture in the 1850s
- Was not proved completely until 1976 by Kenneth Appel and Wolfgang Haken, using computer
- No proof not relying on a computer has yet been found



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

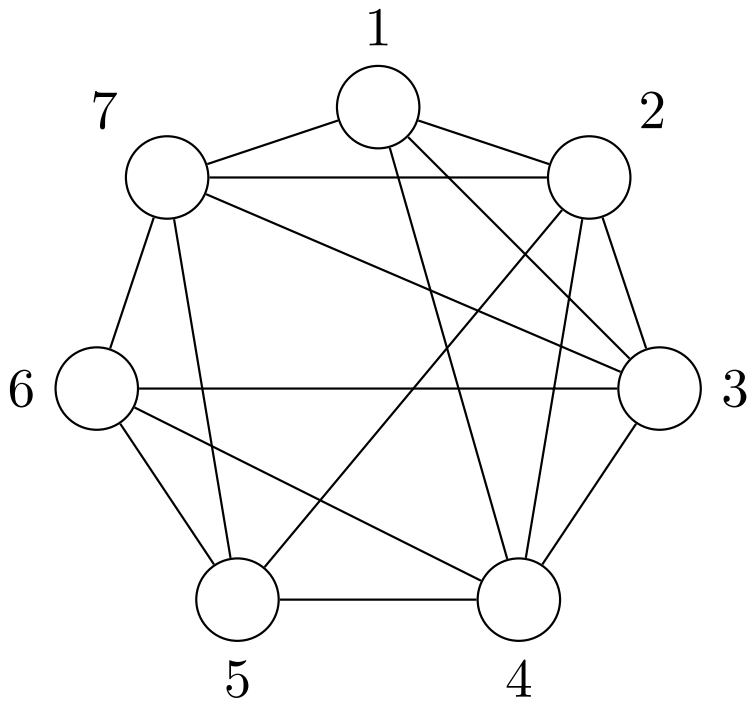
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Applications of Graph coloring

## Scheduling Final Exam

- How can the final exams at a university be scheduled so that no student has two exams at the same time?
- Suppose we have 7 finals, numbered 1 through 7.
- The pairs of courses have common students are depicted in the following graph






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

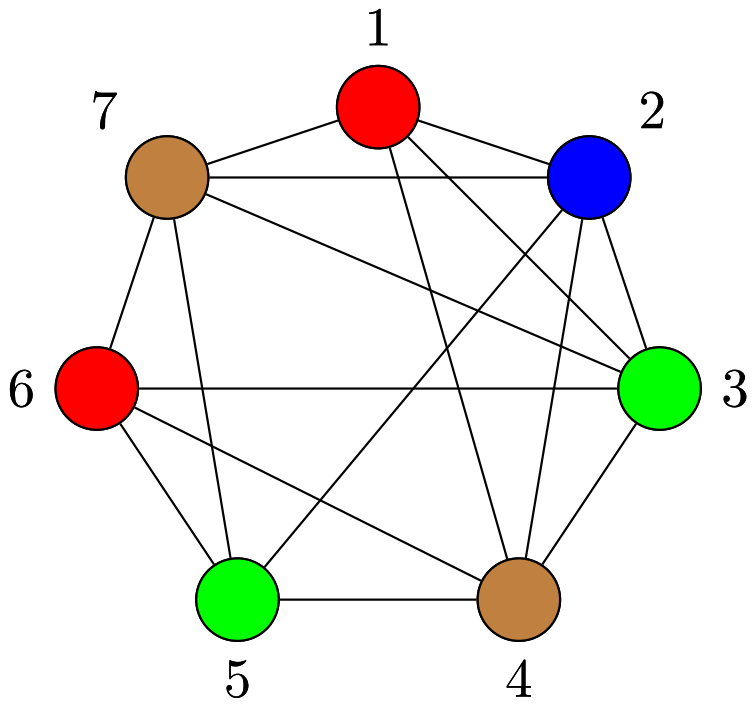
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Applications of Graph coloring

## Scheduling Final Exam

- How can the final exams at a university be scheduled so that no student has two exams at the same time?
- Suppose we have 7 finals, numbered 1 through 7.
- The pairs of courses have common students are depicted in the following graph






Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Applications of Graph Coloring

## Other Applications

- Frequency Assignments: Television channels 2 through 12 are assigned to stations in North America so that no two stations within 150 miles can operate on the same channel. How can the assignment of channels be modeled by graph coloring?



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Applications of Graph Coloring

## Other Applications

- Frequency Assignments: Television channels 2 through 12 are assigned to stations in North America so that no two stations within 150 miles can operate on the same channel. How can the assignment of channels be modeled by graph coloring?
- Index Registers: In an execution of loop, the frequently used variables should be stored in index registers to speed up. How many index registers are needed?



Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Revision

Given $H = (V, E)$ is a simple undirected graph. Which of the following assessment is/are true?

A) $H$ and $H^c$ are connected graphs.

B) $H$ and $H^c$ exist Euler path.

C) $H$ or $H^c$ is connected graphs.

D) $H$ or $H^c$ exists Hamilton path.



Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

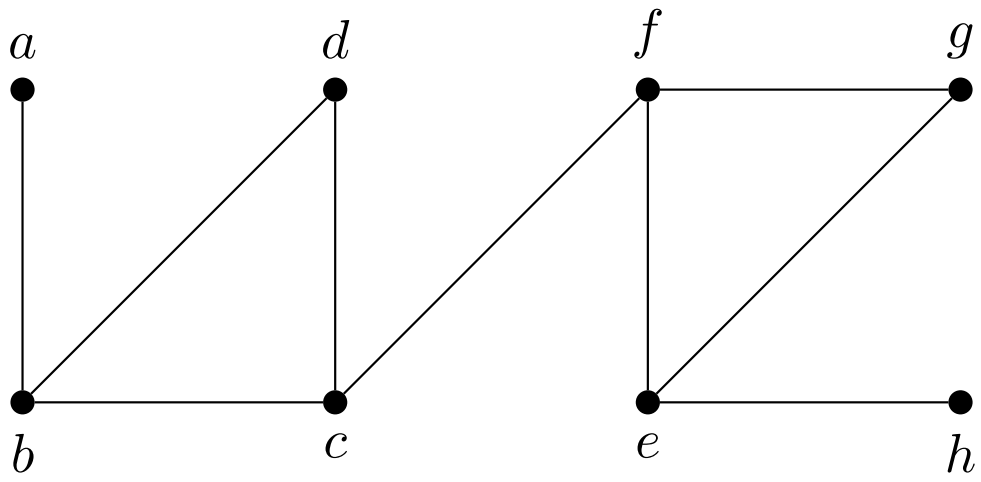
Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring

# Revision




Which of the following assessments is/are true about the cut vertex and cut edge of the graph above?

A) The graph has 2 cut vertex.

B) The graph has 4 cut vertex.

C) The graph has 2 cut edge.

D) The graph has 2 cut vertex and 1 bridge.



Nguyen An Khuong, Tran Tuan Anh, Mai Xuan Toan, Tran Hong Tai

Contents

Connectivity
Paths and Circuits

Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Revision

Choose the correct answer about the reliation of cut edge and cut vertex

A) Two endpoints of a cut edge must be cut vertices.

B) Two endpoints of a cut edge maybe not the cut vertices.

C) One of two endpoints of a cut edge must be a cut vertex.

D) two endpoints of a cut edge are not cut vertices.



Contents

Connectivity
Paths and Circuits

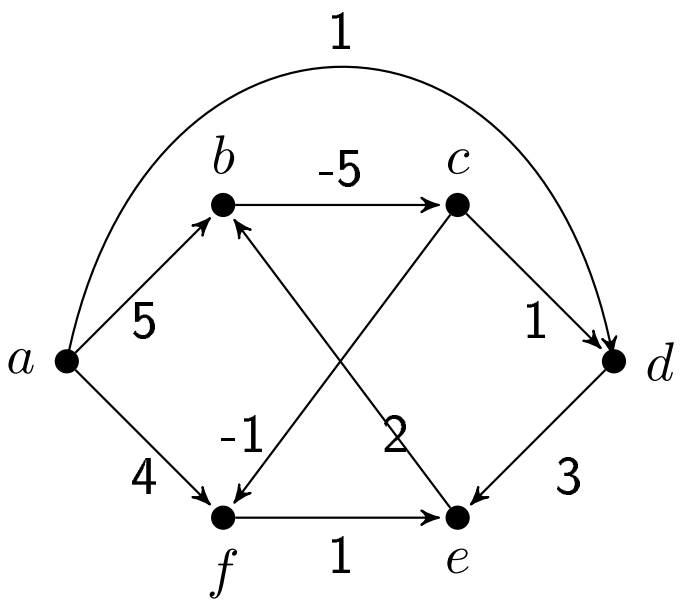
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits

Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others

Graph Coloring

# Revision

Determine a shortest path from $a$ to other vertices in the following graph.



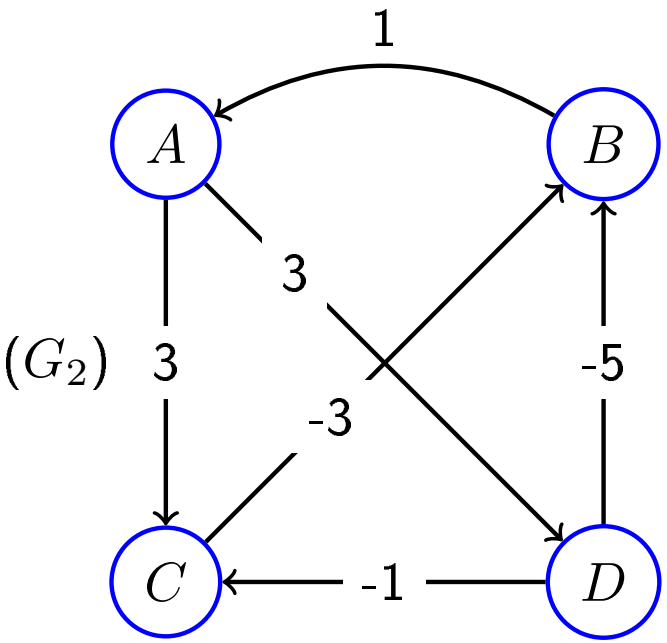



Contents


Connectivity
Paths and Circuits
Euler and Hamilton Paths
Euler Paths and Circuits
Hamilton Paths and Circuits
Shortest Path Problem
Dijkstra's Algorithm
Bellman-Ford Algorithm
Floyd-Warshall Algorithm
Ford's algorithm
Others
Graph Coloring

# Revision

Determine a shortest path from any vertex to other vertex in the following graph (using Floyd-Warshall algorithm).







Contents

Connectivity

Paths and Circuits

Euler and Hamilton Paths

Euler Paths and Circuits

Hamilton Paths and Circuits

Shortest Path Problem

Dijkstra's Algorithm

Bellman-Ford Algorithm

Floyd-Warshall Algorithm

Ford's algorithm

Others

Graph Coloring